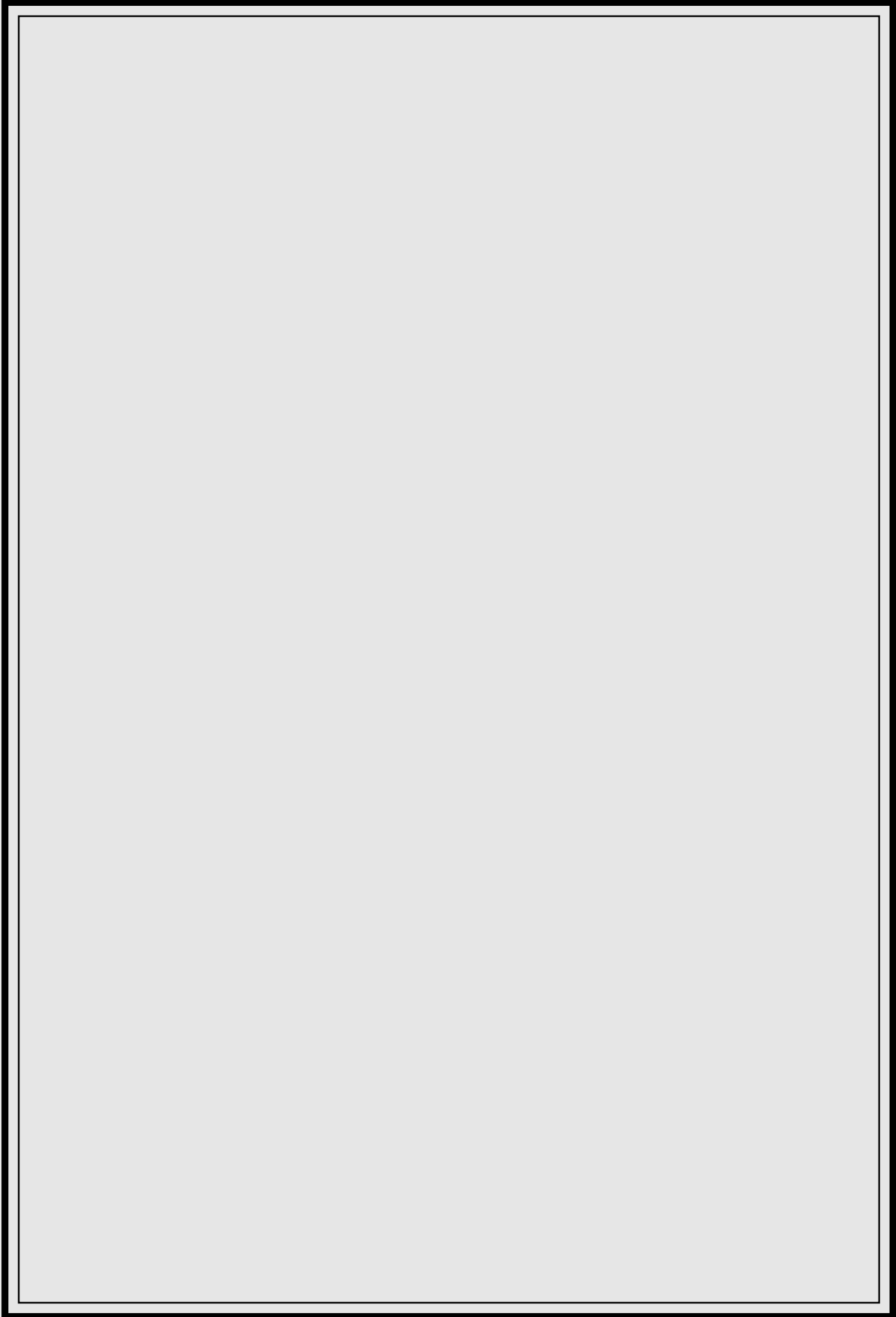




Railway applications — Fixed installations —

Part 1: Protective provisions relating to electrical safety and earthing

The European Standard EN 50122-1:1997 has the status of a
British Standard

ICS 29.120.50; 29.280,45.020

National foreword

This British Standard is the English language version of EN 50122-1:1997 published by the European Committee for Electrotechnical Standardization (CENELEC).

The UK participation in its preparation was entrusted to Technical Committee PEL/9, Railway electrotechnical applications, to Subcommittee PEL/9/3, Fixed equipment, which has the responsibility to:

- aid enquirers to understand the text;
- present to the responsible European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;
- monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

Cross-references

The British Standards which implement international or European publications referred to in this document may be found in the BSI Standards Catalogue under the section entitled “International Standards Correspondence Index”, or by using the “Find” facility of the BSI Standards Electronic Catalogue.

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Summary of pages

This document comprises a front cover, an inside front cover, pages i and ii, the EN title page, pages 2 to 63 and a back cover.

This standard has been updated (see copyright date) and may have had amendments incorporated. This will be indicated in the amendment table on the inside front cover.

Amendments issued since publication

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ICS 29.120.50; 45.020

Descriptors: Railway fixed equipment, safety, accident prevention, earthing, safety measures, safety devices, protection against live parts, protection against electric shocks

English version

Railway applications — Fixed installations Part 1: Protective provisions relating to electrical safety and earthing

Applications ferroviaires — Installations fixes
Partie 1: Mesures de protection relatives à la
sécurité électrique et à la mise à la terre

Bahnanwendungen — Ortsfeste Anlagen
Teil 1: Schutzmaßnahmen in Bezug auf
elektrische Sicherheit und Erdung

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CENELEC

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Foreword

This European Standard was prepared by SC 9XC, Electrical supply and earthing systems the Technical Committee CENELEC TC 9X, Electrical and electronic applications for railways.

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The following dates were fixed:

- latest date by which the EN has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 1997-12-01
- latest date by which the national standards conflicting with the EN have to be withdrawn (dow) 1997-12-01

Annexes designated “normative” are part of the body of the standard.

Annexes designated “informative” are given for information only.

In this standard, Annex B, Annex E and Annex G are normative and Annex A, Annex C, Annex D, Annex F and Annex H are informative.

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1 Scope

This standard specifies requirements for the protective provisions relating to electrical safety in fixed installations associated with a.c.- and d.c.-traction systems and to any installations that may be endangered by the traction power supply system.

It also applies to all fixed installations that are necessary to ensure electrical safety during maintenance work within electric traction systems.

NOTE Other provisions may be required to protect work sites for maintenance purposes which are not included in this standard.

This standard applies to all new lines and to all major revisions to existing lines for the following electric traction systems:

- railways;
- guided mass transport systems such as: tramways, elevated and underground railways, mountain railways, trolleybus systems and magnetic levitated systems;
- material transportation systems.

This standard does not apply to:

- mine traction systems in underground mines;
- cranes, transportable platforms and similar transportation equipment on rails, temporary structures (e.g. exhibition structures) in so far as these are not supplied directly or via transformers from the contact line system and are not endangered by the traction power supply system;
- suspended cable cars;
- funicular railways;
- maintenance work.

2 Normative references

This European Standard incorporates by dated and undated reference provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

EN 50122-2, *Railway applications — Fixed installations — Part 2: Protective provisions against the effects of stray currents caused by d.c. traction systems*¹⁾.

EN 50124-1, *Railway applications — Insulation co-ordination — Part 1: Basic requirements — Clearances and creepage distances*¹⁾.

EN 50153, *Railway applications — Rolling stock — Protective provisions relating to electrical hazards.*

EN 50163, *Railway applications — Supply voltages of traction systems.*

EN 50179, *Power installations exceeding 1 kV a.c.*¹⁾.

EN 60529, *Degrees of protection provided by enclosures* (IP Code) (IEC 529).

HD 384.4.41, *Electrical installations of buildings — Part 4: Protection for safety — Chapter 41: Protection against electric shock* (IEC 364-4-41, modified).

HD 384.4.47, *Electrical installations of buildings — Part 4: Protection for safety — Chapter 47: Application of protective measures for safety — Section 471: Measures of Protection against electric shock* (IEC 364-4-47, modified).

HD 366, *Classification of electrical and electronic equipment with regard to protection against electric shock.*

IEC 479-1, *Effects of current passing through the human body — Part 1: General aspects.*

ISO 3864, *Safety colours and safety signs.*

3 Definitions

For the purposes of this standard, the following definitions apply:

3.1

electric traction system

a railway electrical distribution network used to provide energy for rolling stock

NOTE The system may comprise:

- contact line systems;
- return circuit of electric traction systems;
- running rails of nonelectric traction systems, which are in the vicinity of, and conductively connected to the running rails of an electric traction system;
- electrical installations, which are supplied from contact lines either directly or via a transformer;
- electrical installations in power plants and substations, which are utilized solely for generation and distribution of power directly to the contact line;
- electrical installations of switching stations.

3.2 Traction power supply systems

3.2.1

(traction) substation

an installation the main function of which is to supply a contact line system and at which the voltage of a primary supply system, and in certain cases the frequency, is transformed to the voltage and the frequency of the contact line

¹⁾ In preparation

3.2.2**(traction) switching station**

an installation from which electrical energy can be distributed to different feeding sections or from which different feeding sections can be switched on and off or can be interconnected

3.2.3**feeding section**

a section of the traction power supply system which may be isolated from other sections or feeders of the system by means of switching devices

3.2.4**feeder**

an electrical connection between the contact line and a substation or a switching station

3.2.5**feeding point**

a point at which the feeders or line feeders are connected to the contact line

3.3 Contact line systems**3.3.1****contact line** [IEC 50 811-33-01]

a conductor system for supplying electric energy to vehicles through current-collecting equipment

3.3.2**overhead contact line system**

a system consisting of the following:

- all overhead wiring, including the catenary, the contact wire and the return wire, the earth wire, the lightning protection wire, the line feeder and the reinforcing feeder mounted on the supports;
- overhead conductor rails;
- the foundations, supporting structures and any components supporting, registering, terminating or insulating the conductors;
- equipment mounted on the supports for switching, detecting or protecting.

3.3.3**overhead contact line**

a contact line placed above or beside the roof of the vehicles and supplying vehicles with electric energy through roof-mounted current collection equipment

NOTE The reinforcing feeder is not included.

3.3.4**overhead line** [IEC 50 466-01-01]

an electric line whose conductors are supported above ground, generally by means of insulators and appropriate supports

NOTE Certain overhead lines may also be constructed with insulated conductors.

3.3.5**catenary**

a longitudinal wire supporting the contact wire or wires either directly or indirectly

3.3.6**contact wire** [IEC 50 811-33-15]

the electric conductor of an overhead contact line with which the current collectors make contact

3.3.7**lightning protection wire**

an earthed wire fitted above the overhead contact line to protect it against lightning

3.3.8**overhead contact line zone and pantograph zone**

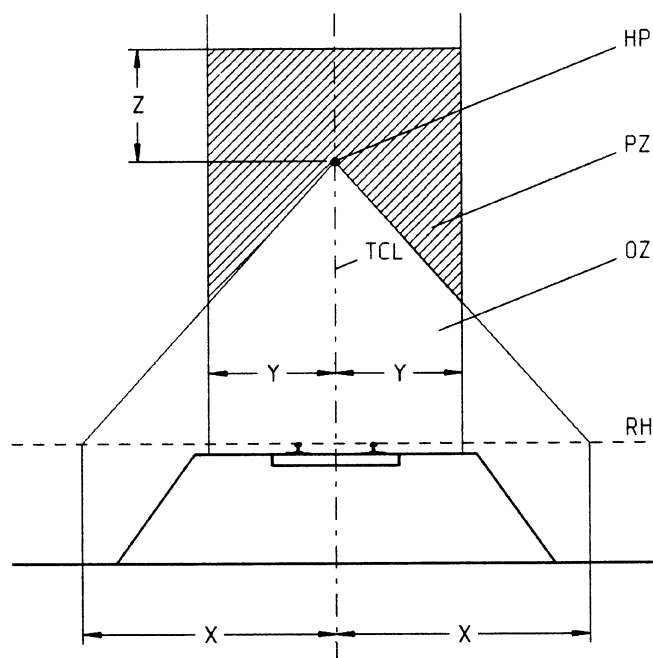
the zone whose limits are not exceeded in general, by a broken overhead contact line or a pantograph which is energized, in the event of dewirement or by broken fragments

structures and equipment may come accidentally in contact with a live broken overhead contact line or with live parts of a broken or dewired pantograph. Figure 1 defines the zone inside which such contact is considered to be probable. The parameters x , y , z shall be defined by national safety regulations. The point HP is the position of the highest conductor of the overhead contact line under all operational conditions considered in the centre of the track. The limits of the overhead contact line zone below the rail head are extended vertically downwards until the earth surface is reached. These limits, however, need not be extended beyond the upper surface of the deck when the railway runs over a bridge

in the case of out of running overhead contact lines the overhead contact line zone shall be extended accordingly

for conductor rail systems not mounted in the overhead position, no contact line zone is defined, while the limits of the pantograph shall be stipulated according to each specific case

NOTE The pantograph may be live because of dewirement or breakage as a result of being connected with other live pantographs or because electrical braking is in effect. The breaking of feeders or line feeders, which are not mechanically strained by current collectors, is not taken into consideration because the probability of a break is too small.



RH	Rail head
HP	Highest point of the overhead contact line
OZ	Overhead contact line zone
PZ	Pantograph zone
TCL	Track centre line

NOTE The stagger has been taken into consideration within the dimension of "y"

Figure 1 — Overhead contact line zone and pantograph zone

3.3.9 earth wire

a wire connecting supports collectively to earth or to running rails to protect people and installations in case of insulation fault and which may also be used as a return conductor

3.3.10 line feeder

an overhead conductor supported on the same structure as the overhead contact line to supply successive feeding points

3.3.11 reinforcing feeder

an overhead conductor mounted on the same structure as the overhead contact line, and directly connected to it at frequent intervals, in order to increase the effective cross-sectional area

3.3.12 supports (in electric traction) [IEC 50 811-33-19]

those parts which support the conductors and the associated insulators of an overhead contact line

3.3.13 mast (in electric traction)

a mainly vertical structure to provide for support, tensioning and registration of the overhead contact line

3.3.14 stagger

the displacement of the contact wire to opposite sides of the track centre at successive supports to avoid localized wear of the pantograph wearing strips

3.3.15 section insulator [IEC 50 811-36-15]

a sectioning point formed by insulators inserted in a continuous run of a contact line, with skids or similar devices to maintain continuous current collection

3.3.16 conductor rail

a rigid metallic section or rail mounted on insulators located near the running rails, with which the shoe gear makes contact

3.3.17**overhead conductor rail**

a rigid metallic section mounted with insulators on a structure forming an overhead contact line

3.3.18**out of running overhead contact line**

an overhead contact line, when it abandons the track centre line in order to reach a termination point on a mast or a structure and when it is not intended to be directly used for current collection

3.4 Return current circuits**3.4.1****return circuit**

all conductors which form the intended path for the traction return current and the current under fault conditions

NOTE The conductors may be for example:

- running rails;
- return conductor rails;
- return conductors;
- return cables.

3.4.2**track return system**

a system in which the running rails form a part of the return circuit

3.4.3**return conductor**

conductor which parallels the track and which is connected to the running rails at periodic intervals

3.4.4**return conductor rail**

a conductor rail used instead of the running rails for the traction return current

3.4.5**return cable**

an insulated conductor forming part of the return circuit and connecting the rest of the return circuit to the substation

3.4.6**rail joint bond**

a conductor ensuring the electrical continuity of a rail at a joint

3.4.7**insulated rail joint**

a mechanical rail joint which longitudinally separates the rail electrically

3.4.8**rail-to-rail cross bond**

an electrical bond that interconnects the running rails of the same track

3.4.9**track-to-track cross bond**

an electrical bond that interconnects tracks

3.4.10**track circuit** [IEC 50 821-03-01]

an electrical circuit of which the rails of a track section form a part, with usually a source of current connected at one end and a detection device at the other end for detecting whether this track section is clear or occupied by a vehicle

NOTE In a continuous signalling system, the track circuit may be used to transmit information between the ground and the train.

3.4.11**impedance bond**

an apparatus used on electric traction systems, usually at the ends of a double-rail track circuit, and designed to ensure the passage of the traction return current notwithstanding the presence of insulated rail joints

3.5 Earthing and bonding**3.5.1****earth** [IEC 50 826-04-01]

the conductive mass of the earth, whose electric potential at any point is conventionally taken as equal to zero

3.5.2**earth electrode** [IEC 50 826-04-02]

a conductive part or a group of conductive parts in intimate contact with and providing an electrical connection with earth

3.5.3**traction system earth**

the running rail, when it is utilized as a return circuit and is connected to earth intentionally. It includes all conductive parts connected thereto

3.5.4**tunnel earth**

the electrical interconnection of the reinforcing rods of reinforced concrete tunnels and in the case of other modes of construction the conductive interconnection of the metallic parts of the tunnel

NOTE In the case of single-phase a.c. traction systems, the tunnel earth is connected to the running rails and thus forms part of the traction system earth which may be supplemented by external earth connections.

3.5.5**traction system earthing**

connection between conductive parts and the traction system earth

3.5.6**direct traction system earthing**

the direct connection between conductive parts and the traction system earth

NOTE Earthing via impedance bonds, required by reason of track circuits considerations, is considered to be direct earthing.

3.5.7**open traction system earthing**

the connection of conductive parts to the traction system earth by a voltage-limiting device or by circuit-breakers, which make a conductive connection either temporarily or permanently if the limited value of the voltage is exceeded

3.5.8**protective conductor (symbol PE)**

[IEC 50 826-04-05]

a conductor required by some measures for protection against electric shock for electrically connecting any of the following parts:

- exposed conductive parts;
- extraneous conductive parts;
- main earthing terminal;
- earth electrode,
- earthed point of the source or artificial neutral.

3.5.9**PEN conductor [IEC 50 826-04-06]**

an earthed conductor combining the functions of both protective conductor and neutral conductor

NOTE The acronym PEN results from the combination of symbols PE for the protective conductor and N for the neutral conductor.

3.5.10**neutral conductor (symbol N)**

[IEC 50 826-01-03]

a conductor connected to the neutral point of a system and capable of contributing to the transmission of electrical energy

3.5.11**equipotential bonding [IEC 50 826-04-09]**

electrical connection putting various exposed conductive parts and extraneous conductive parts at a substantially equal potential

3.5.12**equipotential bonding conductor**

[IEC 50 826-04-10]

a protective conductor for ensuring equipotential bonding

3.6 Voltages and electrical circuits**3.6.1****nominal voltage**

voltage by which an installation or part of an installation is designated

NOTE The voltage of the contact line may differ from the nominal voltage by a quantity within permitted tolerances given in EN 50163.

3.6.2**rail potential**

the voltage occurring under operating conditions when the running rails are utilized for carrying the traction return current or under fault conditions between running rails and earth

3.6.3**accessible voltage**

that part of the rail potential under operating conditions which can be bridged by persons, the conductive path being conventionally from hand to both feet through the body or from hand to hand (horizontal distance of 1 m to a touchable part)

3.6.4**(effective) touch voltage**

voltage under fault conditions between parts when touched simultaneously

NOTE The value of the effective touch voltage may be appreciably influenced by the impedance of the person in contact with these parts.

3.6.5**traction return current**

the sum of the currents returning to the supply source (substation or regenerative braking vehicles)

3.6.6**stray current**

a current which follows paths other than the intended paths

3.6.7**short circuit [IEV 195-04-21]²⁾**

accidental or intentional conductive path between two or more points in a circuit forcing the voltages between these points to be relatively low

NOTE Any such conductive path whether between conductors or between a conductor and earth is regarded as a short circuit.

3.6.8**short-circuit current [IEV 195-05-18]²⁾**

the electric current flowing through the short-circuit

²⁾ In preparation

3.6.9**fault condition**

a non intended condition of the apparatus or equipment

3.6.10**rail to earth resistance**

the electrical resistance between the running rails and the earth

NOTE For d.c. traction systems in tunnels the measurement is made between the running rails and tunnel earth.

3.6.11**conductance per unit length**

the reciprocal value of the rail to earth resistance per unit length (S/km)

3.7 Current collection [IEC 50 811-32]**3.7.1****current collector [IEC 50 811-32-01]**

equipment fitted to the vehicle and intended to collect current from a contact wire or conductor rail

3.7.2**pantograph [IEC 50 811-32-02]**

apparatus for collecting current from one or more contact wires or overhead conductor rails, formed of a hinged device designed to allow vertical movement of the pantograph head

3.7.3**trolley [IEC 50 811-32-08]**

apparatus for collecting current from a contact wire by means of a grooved wheel or contact slipper mounted on a pole which is movable in any direction

3.7.4**shoegear [IEC 50 811-32-19]**

an assembly of parts for collecting current from a conductor rail

3.8 Electric shock [IEC 50 826-03]**3.8.1****live part**

a conductor or conductive part intended to be energized in normal use, by convention, this does not include the running rails and parts connected to them

3.8.2**exposed conductive part [IEC 50 826-03-02]**

a conductive part of electrical equipment, which can be touched and which is not normally live, but which may become live under fault conditions

NOTE A conductive part of electrical equipment which can only become live under fault conditions through an exposed conductive part is not considered to be an exposed conductive part.

3.8.3**extraneous conductive part [IEC 50 826-03-03]**

a conductive part not forming part of the electrical installation and liable to introduce a potential, generally the earth potential

3.8.4**electric shock [IEC 50 826-03-04]**

pathophysiological effect resulting from an electric current passing through a human or animal body

3.8.5**direct contact [IEC 50 826-03-05]**

contact of persons or livestock with live parts

3.8.6**indirect contact [IEC 50 826-03-06]**

contact of persons or livestock with exposed conductive parts which have become live under fault conditions

3.8.7**leakage current [IEC 50 826-03-08]**

a current which, in the absence of a fault, flows to earth or to extraneous conductive parts in a circuit

3.8.8**barrier [IEC 50 826-03-13]**

a part providing protection against direct contact from any usual direction of access

3.8.9**obstacle [IEC 50 826-03-14]**

a part preventing unintentional direct contact but not preventing direct contact by deliberate action

3.8.10**protective boarding**

a non-conducting physical barrier to protect persons from coming into contact with the live conductor rail

3.8.11**standing surface**

any point on a surface where persons may stand or walk about

3.9 Protective devices**3.9.1****voltage-limiting device**

a protective device against permanent existence of an inadmissible high touch/accessible voltage

3.9.2**surge divertor [IEC 50 811-31-09]**

a device intended to protect the electrical apparatus from high transient overvoltages and to limit the duration and frequently the amplitude of the follow-on current

3.9.3**residual current-operated protective device**

a protective device for detecting residual current and then isolating the circuit

3.10 Other definitions**3.10.1****closed electrical operating area**

any room or location which serves exclusively for the operation of electrical equipment and is kept secured by a means appropriate to the voltage and location. Access to such areas is permitted only to authorized persons

3.10.2**public area**

an area to which the public has unrestricted access

3.10.3**restricted area**

an area for which the railway authority has the responsibility and to which access is only permitted for authorized persons

3.10.4**anti-trespassing guard**

equipment provided to prevent entry to a restricted area, structure or building by an unauthorized person

3.10.5**public level crossing**

crossing where the public has a right of way over the railway on the level subject to compliance with the safety arrangements provided at the location

3.10.6**private level crossing**

crossing usually provided for field or occupation access. There would normally be limited or no public use

3.10.7**railway controlled crossing**

crossing to which there would be little, if any, public access and where the railway authority controls the area around the crossing and issues instructions (such as notices) for its use; for example depots and goods yards

3.10.8**authorized trackside walking route**

a safe path alongside the track. The use of this path is allowed only for authorized persons

4 Protective provisions against electric shock in installations for nominal voltages up to and including 1 kV a.c./1,5 kV d.c.

The requirements as defined in HD 384.4.41 and HD 384.4.47 shall apply with the following limitations and extensions.

4.1 Protection against direct contact**4.1.1 General**

No protection against direct contact is required for circuits with nominal voltages up to and including 25 V a.c./60 V d.c.. This shall not apply, however, if the circuit concerned is connected to the return circuit which may rise above earth potential.

4.1.1.1 If it is intended that work is to be carried out on energized overhead contact lines the system shall be constructed in such a manner that no parts of different potential are situated directly within the immediate working area in order to provide protection against direct contact.

This does not apply to locations for which this requirement is technically impossible, e.g. for section insulators, crossing points of overhead contact lines of tramways and trolleybuses and for trolleybus overhead contact lines.

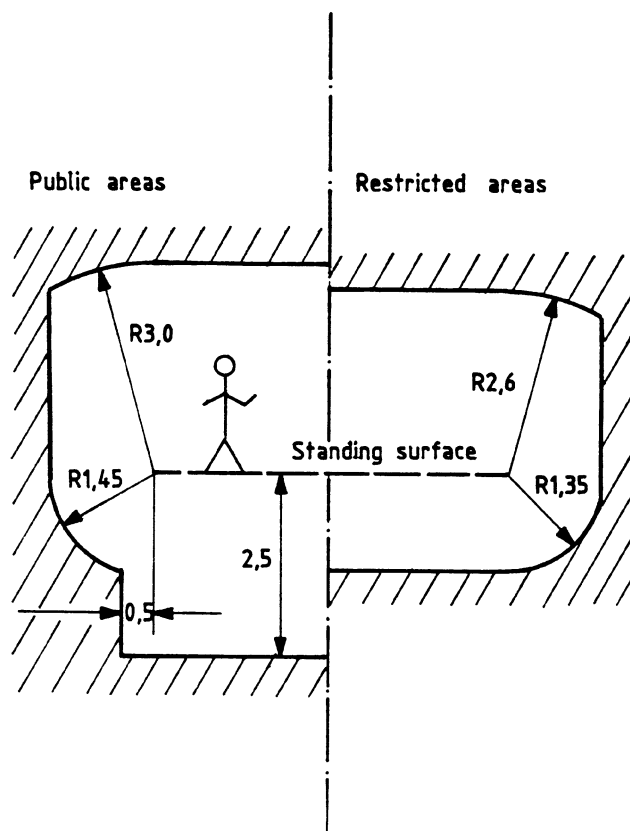
NOTE This requirement can be met, for example, by installation of additional insulation in the overhead contact line system to achieve neutral sections each at least 0,5 m long, which ensure that simultaneous contact with different potentials in the immediate working area is obviated completely or to a considerable extent.

4.1.1.2 Insulators whose design is such that supporting structures may become live under fault conditions, e.g. saddle insulators or buckle insulators, shall not be utilized for the overhead contact line system where live working shall be undertaken.

4.1.1.3 With the exception of synthetic rope in a length of more than 2 m (1 m under cover) all insulators which are directly connected to a live part shall be regarded as a live part when considering clearance dimensions in this standard.

4.1.2 Protection by clearance**4.1.2.1 Standing surface**

For standing surfaces, accessible to persons, clearance for touching in a straight line shown in Figure 2 shall be provided against direct contact with live parts of an overhead contact line system as well as any live parts on the outside of a vehicle (e.g. current collectors, roof conductors, resistors). This does not apply to conductor rail systems near the running rails (see **4.1.3.3.3**).



All dimensions are minimum dimensions in m.

Figure 2 — Clearances to accessible live parts on the outside of vehicles as well as to live parts of overhead contact line systems from standing surfaces accessible to persons for nominal voltages up to and including 1 kV a.c./1,5 kV d.c.

The clearances given in the following clauses are minimum values which shall be maintained at all temperatures and with additional and exceptional line loading. Due to national or regional existing practices, greater clearances or smaller mesh sizes may be prescribed by the relevant railway authority.

NOTE 1 Touching in a straight line implies that live parts are accessible by a person from a standing surface without use of objects.

NOTE 2 This protection is not mandatory if, although contact with live parts is possible, other provisions have been taken to ensure isolation from the supply.

NOTE 3 Figure 2 assumes that the standing surface does not afford protection against contact with live parts situated below or to the side thereof. Depending on its construction the surface may in practice meet the requirement respecting provision of obstacles. In this case the lower clearances applicable to obstacles may be utilized. The clearances were determined on the basis of the arms reach as defined in HD 384.4.41 to which a margin of safety was added. The margin of safety has been determined depending on the voltage of the contact line system and on whether the standing surface is in a restricted area or in a public area.

4.1.2.2 Standing surfaces for working persons

The clearances to be observed for persons working nearby energized overhead contact line systems shall be defined in the operational specifications. If operational specifications do not exist, clearances shown in Figure 2 or the clearances according to 4.1.3 shall be used.

NOTE Station platform roofs, working platforms and working planks at signal bridges, working platforms at signals, maintenance-ladders, work cages of hydraulic working platforms, working platforms of tower cars, which are utilized solely for execution of work at or in the vicinity of overhead contact line systems are not included.

4.1.2.3 Minimum height of overhead contact lines and associated feeders

Where a road carting normal vehicular traffic crosses (or coincides) with railway (or tramway) electrified by means of an overhead contact line and no road traffic restrictions are specified, unless otherwise fixed by national regulation, a minimum vertical clearance of 4,70 m shall be provided between the road surface and the lowest point of the overhead contact line and associated feeders.

If the required minimum clearance cannot be provided, and unless otherwise specified by national regulation, the maximum height of road vehicles permitted to pass under the overhead contact line shall be limited so as to guarantee the following minimum vertical clearances between the highest point of road vehicle (load included) and the live parts, of:

- 0,50 m, where only road traffic signs indicating the maximum permissible vehicle height are utilized;

— 0,30 m, where additional fixed barriers (e.g. a rigid obstacle or a firmly fixed metallic wire made visible by means of a suspended warning sign) are erected on both sides of the crossing, physically limiting such vehicle height.

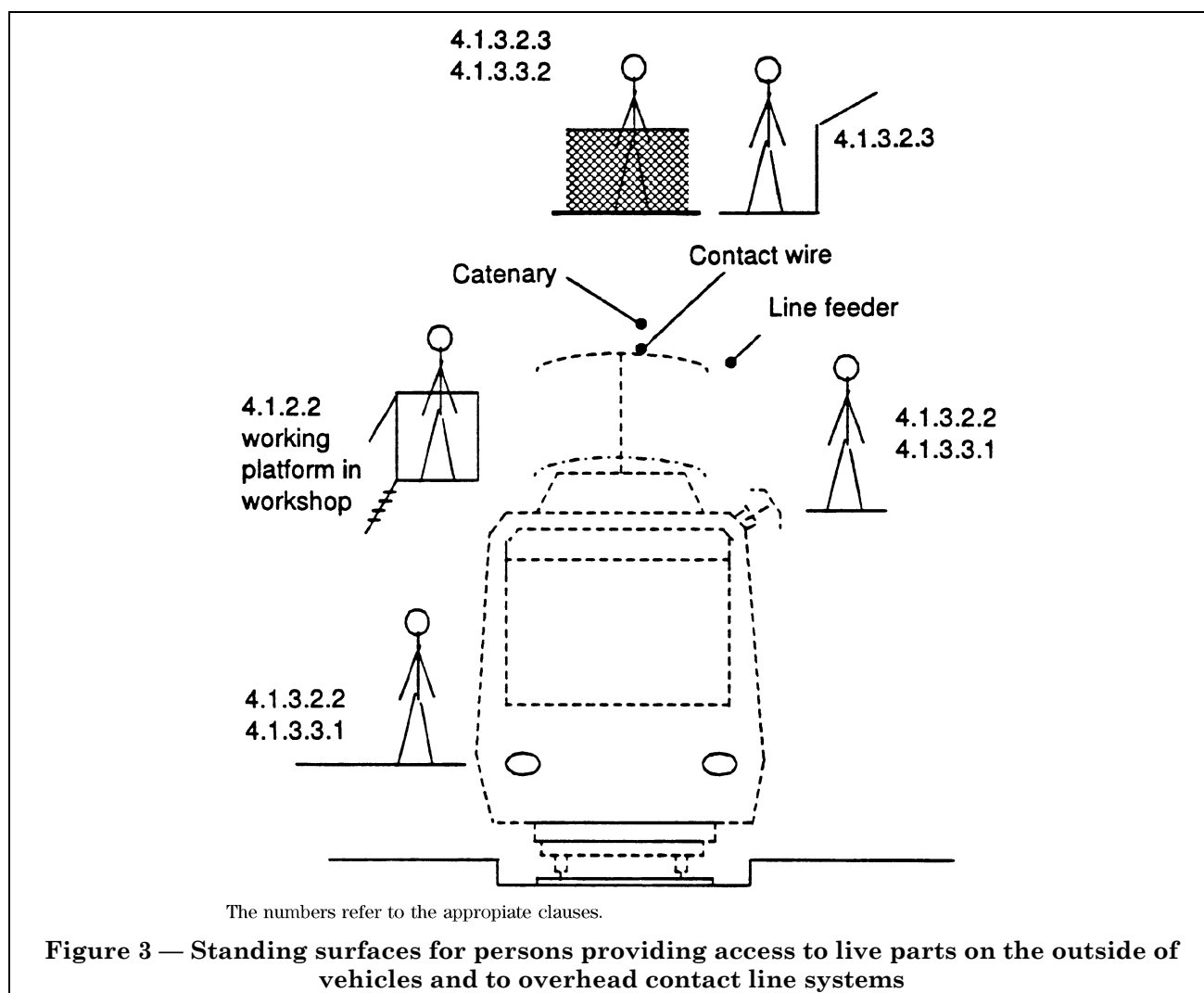
4.1.3 Protection by obstacles

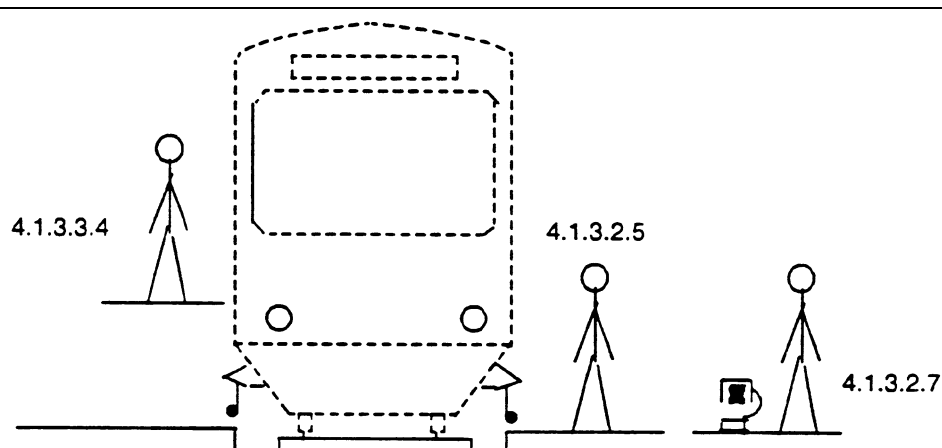
4.1.3.1 General

If the clearances given in 4.1.2 cannot be maintained, obstacles shall be provided as protection against direct contact with live parts. The design of obstacles is dependent: on the location of the standing surfaces shown in Figure 3 and Figure 4 relative to the live parts, on the clearance between the obstacle and the live parts, and on whether the standing surface is a restricted area or a public area.

The dimensions of the obstacles shall be such that live parts cannot be touched in a straight line by persons on a standing surface.

Where the obstacle is made of conductive material the requirements of 4.2.2 shall apply.





NOTE The numbers refer to the appropriate clauses.

Figure 4 — Standing surfaces for persons providing access to live parts on the outside of vehicles and to conductor rail systems

4.1.3.2 Obstacles for standing surfaces in restricted areas

4.1.3.2.1 The obstacles shall meet the requirements for protective obstacles Class IP2X as defined in EN 60529 where the provisions given in 4.1.3.2.2 are not satisfied.

4.1.3.2.2 With standing surfaces adjacent to live parts on the outside of vehicles or adjacent to live parts of a contact line system, obstacles of mesh screens construction with a maximum 1 200 mm² mesh size and having a height “h” of at least 1,8 m may be utilized where the live parts are situated above the standing surface (see Figure 5, example a).

If the live parts are situated at the same height or lower than the standing surface, the height “h” of the obstacle shall be such that a clearance of 1,35 m from the top of the obstacle is maintained as shown in Figure 2 (see Figure 5, example b).

The clearance between the obstacle and the live parts shall be at least 0,3 m. If this clearance is not maintained, the obstacles shall be designed in accordance with clause 4.1.3.2.1 or be of solid-wall design and leave no gap between the obstacle and the standing surface (see Figure 5, examples c and d). The dimension “d” between the obstacle and live parts, shown in Figure 5, shall be determined as described in EN 50124-1³⁾.

4.1.3.2.3 For standing surfaces above live parts on the outside of vehicles or above live parts of a contact line system, the obstacles shall be designed according to 4.1.3.2.1 or shall be solid-walled (see Figure 6, example a). If the clearance between the obstacles and the live parts is at least 0,6 m, walkways which include a mesh construction with a maximum 1 200 mm² mesh size, may be provided as obstacles (see Figure 6, example b).

The length of the solid-walled or mesh construction standing surface shall correspond to the pantograph zone and extend beyond the live parts of a contact line system by at least 0,5 m on each side. In the case of conductors not being used for current collection (e.g. line feeders, reinforcing feeders, out of running overhead contact lines) there shall be a width of at least 0,5 m on each side of the conductor provided that movement due to dynamic and thermal effects has been taken into account.

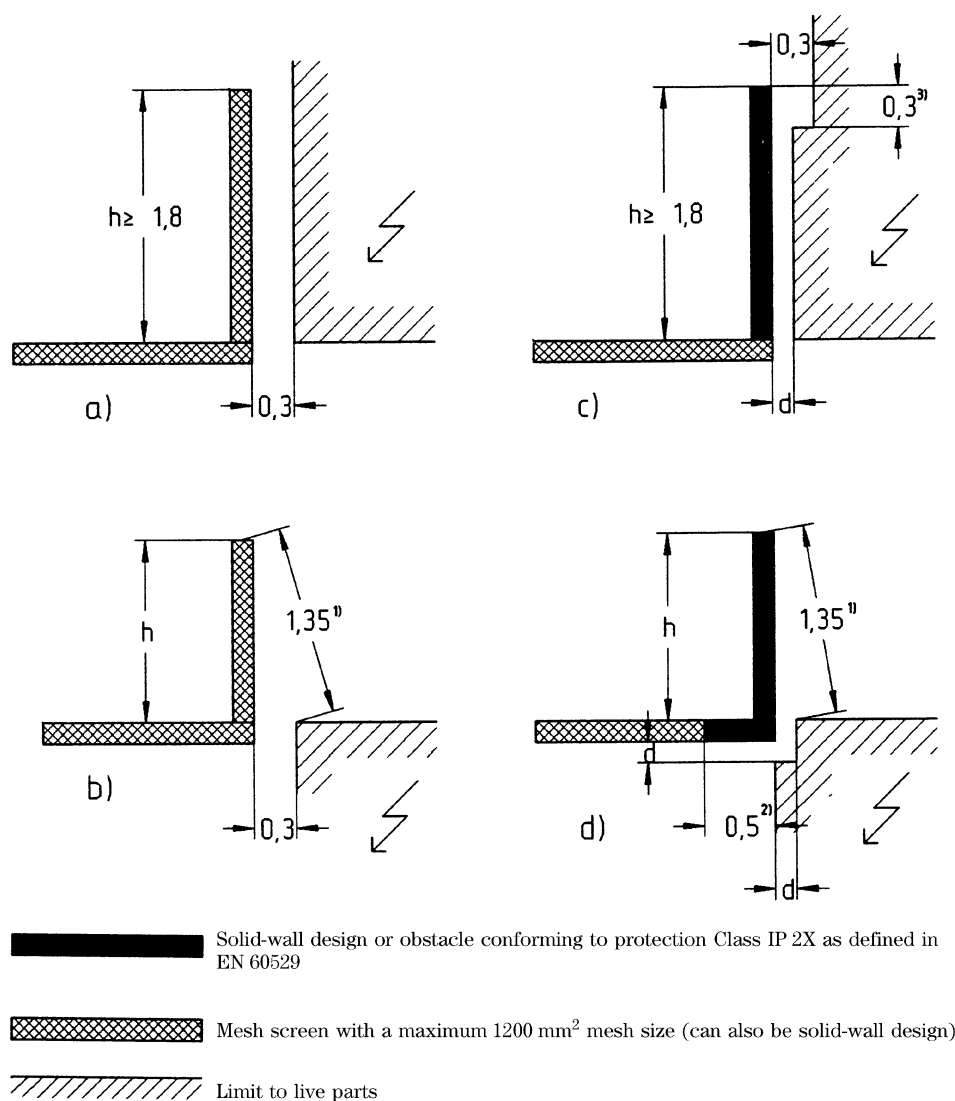
Obstacles of mesh screen construction with a maximum 1 200 mm² mesh size shall be at least provided at the sides of such standing surfaces. The height “h” of the obstacles shall be such that a clearance of 1,35 m from the top of the obstacles is maintained as depicted in Figure 2 (see Figure 6, examples a and b). These obstacles shall be at least as long as the length of the standing surface above live parts.

NOTE The height of the side obstacles, if necessary, will generally correspond to the height of the necessary safety railing but should have a height of at least 1,0 m.

The dimension “d” between the obstacle and live parts, shown in Figure 6, shall be determined as described in EN 50124-1⁴⁾.

³⁾ In preparation

⁴⁾ In preparation



'd' is the clearance in air between the obstacle and live parts.

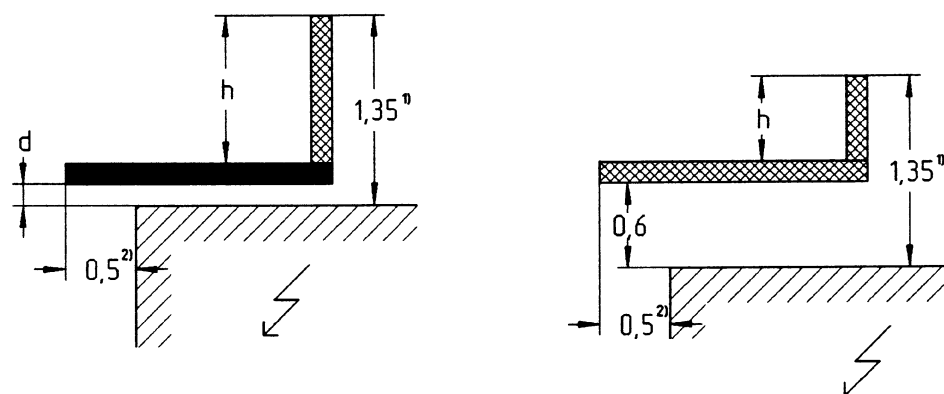
1) The dimension 1,35 is derived from figure 2.

2) The dimension 0,5 is based on the requirements given in 4.1.3.2.3.

3) The dimension 0,3 may be reduced by the same extent as the height 'h' exceeds the value of 1,8 m.

All dimensions are minimum dimensions in m.

Figure 5 — Examples of obstacles for standing surfaces in restricted areas for protection against direct contact with adjacent live parts on the outside of vehicles or adjacent live parts of a contact line system for nominal voltages up to and including 1 kV a.c./1,5 kV d.c. (see 4.1.3.2.2)



a)

b)

■ Solid-wall design or obstacle conforming to protection Class IP 2X as defined in EN 60529

▨ Mesh screen with a maximum 1200 mm² mesh size (can also be solid-wall design).

▨ Limit to live parts.

'd' is the clearance in air between the obstacle and live parts.

1) The dimension 1,35 is derived from figure 2.

2) The dimension 0,5 is based on the requirements given in 4.1.3.2.3

All dimensions are minimum dimensions in m.

Figure 6 — Examples of obstacles for standing surfaces in restricted areas for protection against direct contact when above live parts on the outside of vehicles or live parts of a contact line system for nominal voltages up to and including 1 kV a.c./1,5 kV d.c. (see 4.1.3.2.3)

4.1.3.2.4 In the case of overhead contact lines suspended beneath structures (tunnels, underpasses) where work is to be executed from an insulated platform whilst the line is energized, insulated obstacles or obstacles insulated with respect to the structure of minimum width 1,0 m shall be provided on both sides of the track centre if the clearance to structural elements connected to earth is less than 1,0 m (see Figure 7). These obstacles shall be 0,5 m longer than the edge of the structure.

4.1.3.2.5 The requirements given in 4.1.3.2.1 can be waived in the case of vehicles with shoe gear for conductor rails if the shoe gear does not extend considerably outside the vehicle gauge. In depots the internal operational ways shall be clearly distinguishable.

4.1.3.2.6 For workshops where it may be impracticable to place a protective boarding around the conductor rail, a conductor rail system shall not be used. A trolley system or other systems shall be used for supplying the vehicle. In this case, national or locally defined working practices shall ensure safety around any live parts of a vehicle energized in a workshop, such as shoe gear.

4.1.3.2.7 In the case of bottom-contact conductor rail systems, the surface of the conductor rail not touched by the current collector shall be protected by means of protective boarding.

In the case of side contact conductor rail systems, the top and side surfaces not touched by the current collector shall be protected by means of protective boarding.

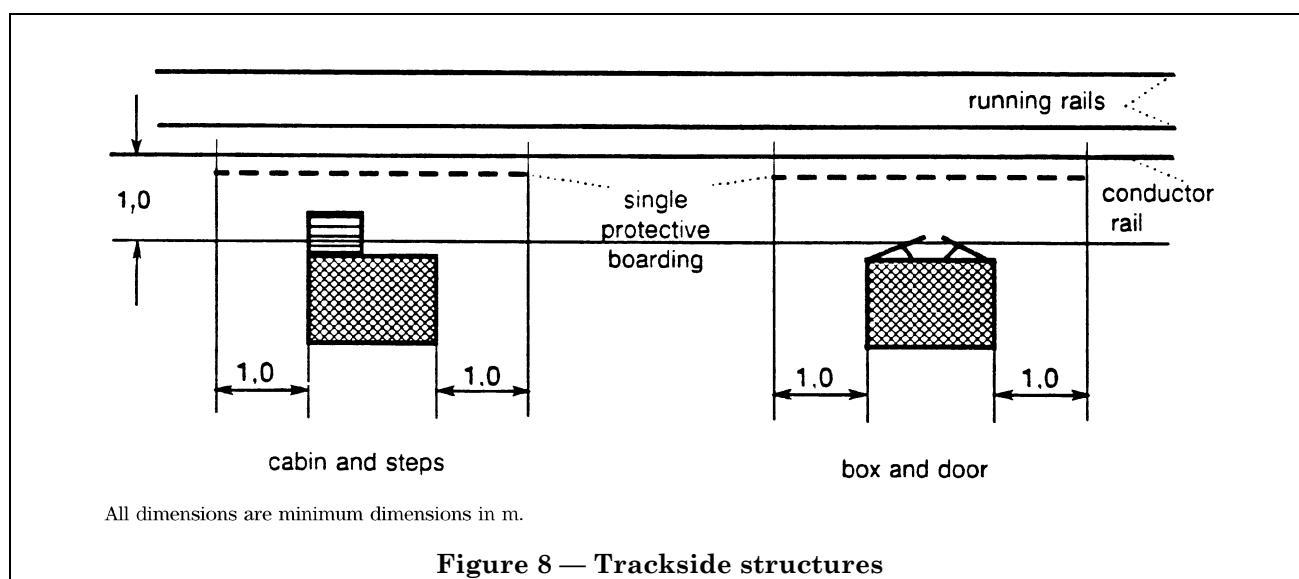
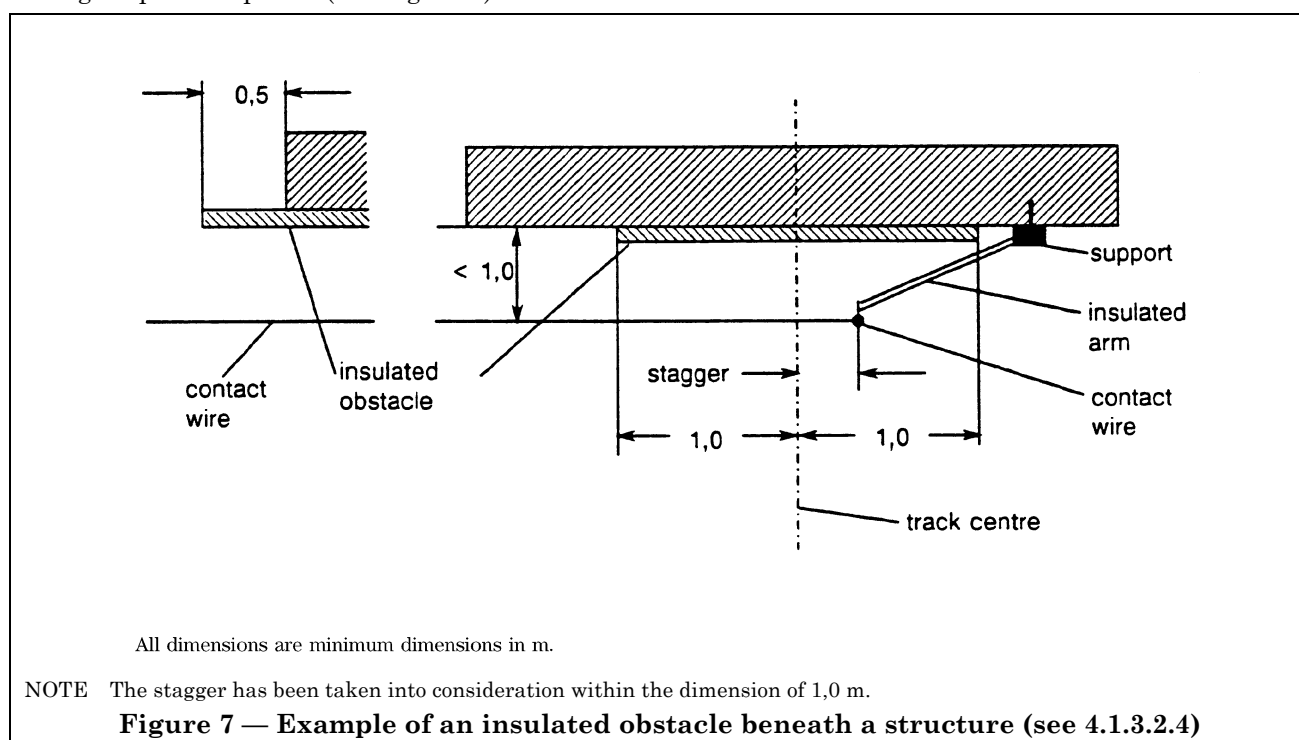
4.1.3.2.8 For top contact conductor rail systems the following requirements apply:

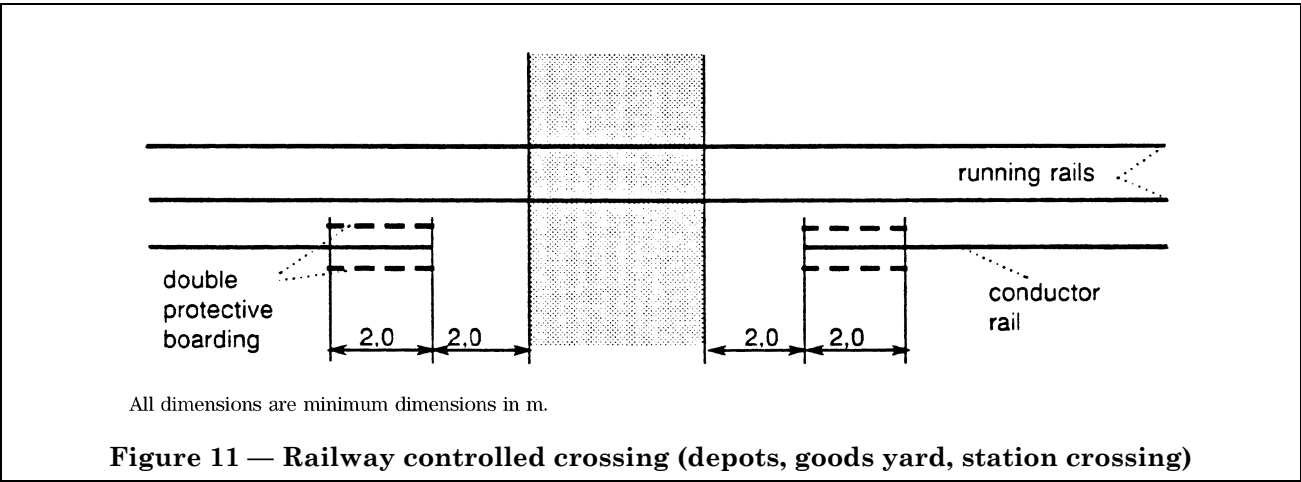
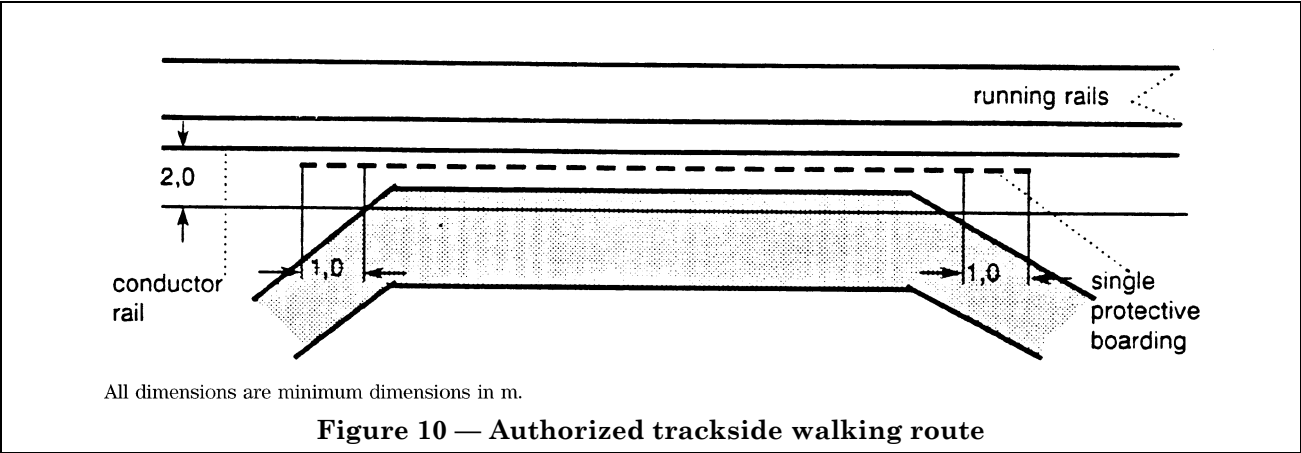
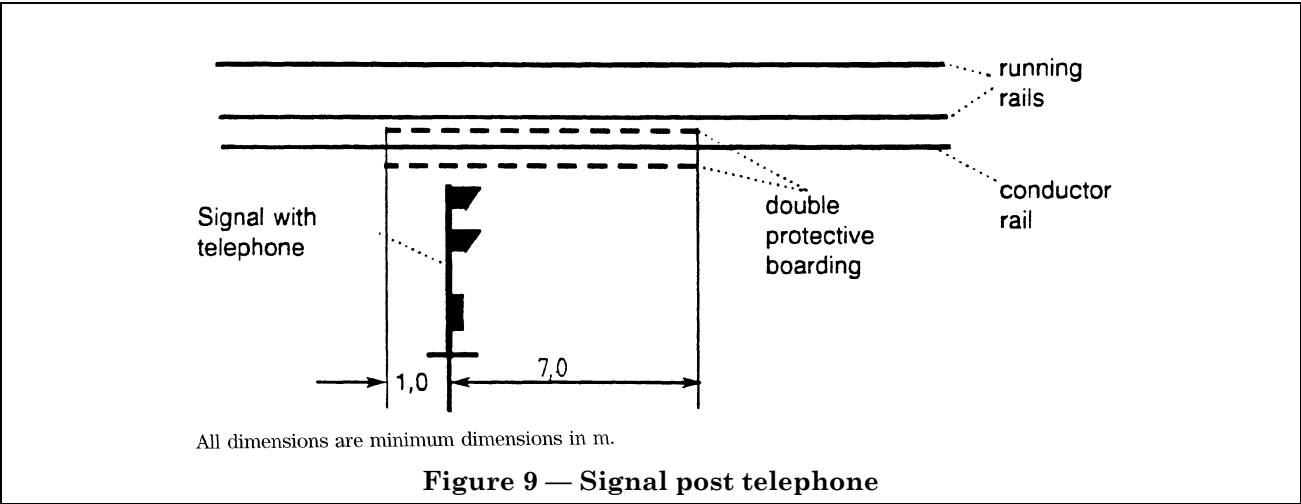
4.1.3.2.8.1 Where the egress from a building or an equipment location facing towards the track is within 1,0 m of the conductor rail and no physical barrier is provided, then single protective boarding shall be provided on the outside of the conductor rail, extending 1,0 m either side of the location (see Figure 8).

4.1.3.2.8.2 At signal post/telephone locations double protective boarding shall be provided for a distance of 8,0 m, commencing 7,0 m on the approach side of the signal post/telephone (see Figure 9).

4.1.3.2.8.3 Where an authorized trackside walking route is located within 2,0 m of a conductor rail and no physical barrier is provided, then single protective boarding shall be provided on the outside of the conductor rail, extending 1,0 m either side of the end of the route (see Figure 10).

4.1.3.2.8.4 For railway controlled crossings the conductor rail shall be terminated 2,0 m from the crossing with double protective boarding provided for a minimum distance of 2,0 m (see Figure 11).



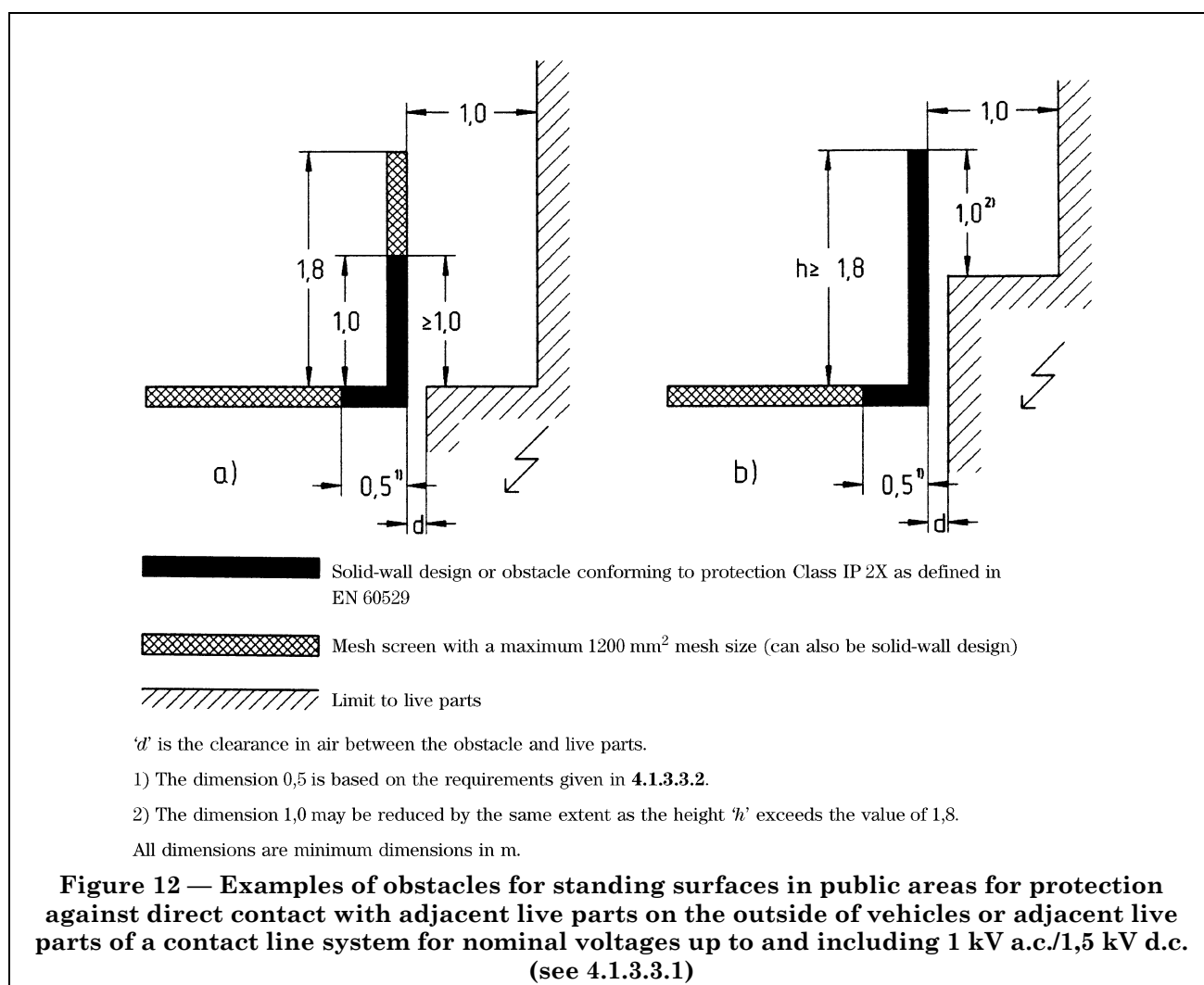


4.1.3.3 Obstacles for standing surfaces in public areas

4.1.3.3.1 With standing surfaces adjacent to live parts on the outside of vehicles or adjacent to live parts of a contact line system, if the clearances shown in Figure 2 cannot be achieved, obstacles shall be provided and shall leave no gap with respect to the standing surface.

If the clearance between the obstacle and live parts is at least 1,0 m (see Figure 12, example a) the obstacle shall be of solid-wall design or Class IP2X (as defined in EN 60529) 1,0 m high, surmounted by a mesh construction with a maximum 1 200 mm² mesh size to an overall height of at least 1,8 m. Such mesh shall be arranged so as to ensure that the obstacle remains unclimbable. If this clearance is not achieved, the obstacle shall be of solid-wall design or Class IP2X 1,8 m high (see Figure 12, example b). The dimension “*d*” between the obstacle and live parts, shown in Figure 12, shall be determined as described in EN 501241-1⁵⁾.

The tops of such obstacles shall be designed to prevent access for standing or walking on.



⁵⁾ In preparation

4.1.3.3.2 For standing surfaces above live parts on the outside of vehicles or above live parts of a contact line system shall be of solid-wall design.

The length of the solid-walled standing surface shall correspond to the pantograph zone and extend beyond the live parts of a contact line system by at least 0,5 m on each side. In the case of conductors not being used for current collection (e.g. line feeders, reinforcing feeders, out of running overhead contact lines) there shall be a width of at least 0,5 m on each side of the conductor provided that movement due to dynamic and thermal effects has been taken into account.

Along the sides of these standing surfaces, obstacles shall be provided in order to prevent any direct contact with live parts on the outsides of vehicles and to live parts of the contact line system even where, for example, a rod or liquid jet is utilized. These obstacles shall be provided over a length that corresponds at least to the length of the standing surface in solid-wall design.

Where horizontal obstacles extend beyond the vertical obstacle by at least 0,5 m, it becomes acceptable to refer the 1,45 m lateral clearance required by Figure 2 to the top of the vertical obstacle, instead of referring it to the edge of the standing surface (see Annex A, Figure A.1, example a). The height of the vertical obstacle shall be consequentially increased whenever necessary, in order to achieve this clearance. The horizontal obstacle shall be impossible to stand on.

NOTE To achieve this, the horizontal obstacle should be designed so that it is obviously not a standing surface, or it should be inclined upwards or downwards (see Annex A, Figure A.1, example a).

Whenever this horizontal obstacle is not present, the vertical obstacle shall conform to the requirements of **4.1.3.3.1** (see Annex A, Figure A.1, example b). The alternative solution (shown in annex A, Figure A.1, example c) is also acceptable.

The tops of such obstacles shall be designed to prevent access for standing or walking on.

Any vertical obstacle shall be of solid-wall design or Class IP2X as defined in EN 60529 at least to a height of 1,0 m (see Annex A, Figure A.1) except in the case of the above said horizontal obstacle, where a railing is sufficient, if the clearances required by Figure 2 are present.

NOTE In addition to the requirements for electrical safety there may be additional requirements for obstacles to provide for security according to national regulations.

4.1.3.3.3 In the case of trackside platforms for vehicles with shoe gear for conductor rails, the requirements given in **4.1.3.3.1** are complied with if there is only a small clearance between the platform and the vehicle. If the clearance is not small, additional protection shall be provided to prevent access in a straight line.

NOTE This can be achieved for example by use of shrouded shoe gear.

4.1.3.3.4 Conductor rails shall be on the side of the track away from the platform wherever practical. This applies to all cases except where one track lies between two platforms.

4.1.3.3.5 *Top contact conductor rail systems*

4.1.3.3.5.1 Anti-trespassing guards constructed of non-conductive material shall be installed parallel to the track for a distance of 2,0 m when measured along the line of the track, at both public and private level crossings. The conductor rail shall be terminated 0,3 m from the edge of the anti-trespassing guard (see Figure 13).

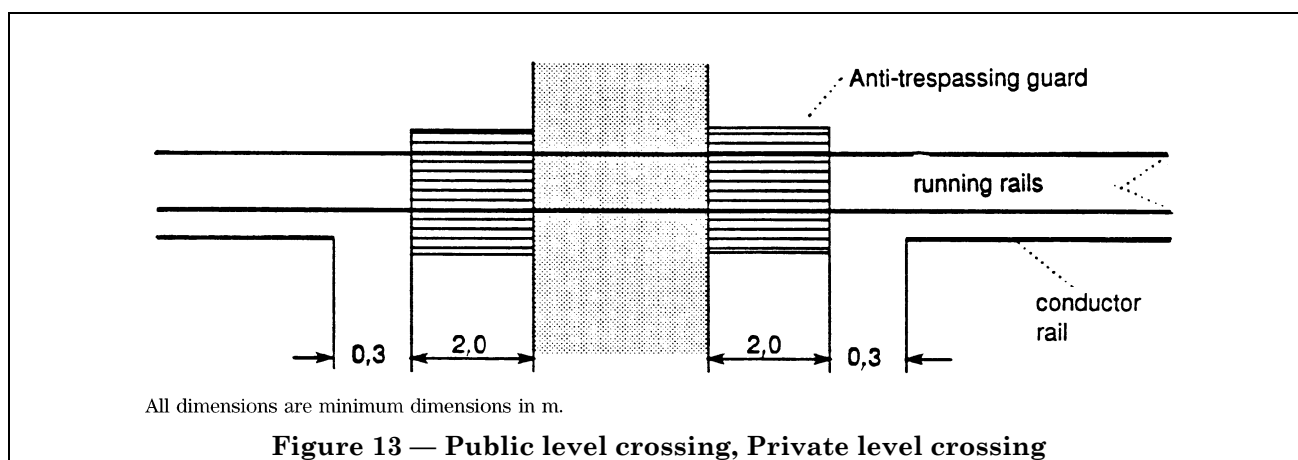
4.1.3.3.5.2 Anti-trespassing guards shall comprise a surface difficult for persons and animals to walk upon.

4.1.3.3.5.3 At public level crossings provided with gates that are protected in accordance with **4.1.3.3.1** except for the necessary clearance between the gate and the standing surface and which prevent access to the railway when the road is open, the conductor rail shall be terminated 0,3 m from the edge of the crossing. No anti-trespassing guards or protective boarding are required.

4.1.3.3.5.4 At private level crossings provided with gates in accordance with **4.1.3.3.5.3** and which prevent public access to the railway, the conductor rail shall be terminated 0,3 m from the edge of the crossing. Double protective boarding shall be provided for a minimum distance of 2,0 m.

4.1.3.4 *Anti-climbing provisions*

Anti-climbing provisions are not normally necessary. In justified cases however, anti-climbing provisions can become necessary. The railway authority shall specify where it is necessary to equip masts and structure anchor wires with anti-climbing provisions.



NOTE Provisions are required only for masts and anchor wires which can be climbed without any aids and if there is a risk of coming into unsafe proximity to live parts of the overhead contact line equipment when climbing.

4.1.4 Warning signs

Warning signs shall be used in areas where there is a serious risk of coming within the limits of live parts of a contact line system given in 4.1.2.1. Such warning signs shall be placed in a prominent position and be readily visible adjacent to the point of access. The sign shall be as defined in ISO 3864 (see Annex B) and if required an appropriate supplementary sign may be used.

4.2 Protection against indirect contact

4.2.1 General

Protection against indirect contact for nominal voltages up to 50 V a.c./120 V d.c. shall be provided unless the relevant requirements as defined in HD 384.4.41 clause 411.1 are met. For higher nominal voltages the protective provisions given in 4.2.2 through 4.2.4 shall be provided.

4.2.2 Traction system earthing

4.2.2.1 In a.c. traction systems, direct traction system earthing is the preferred method for system faults and safety of persons. This method utilizes the return circuits for conduction of the current under fault conditions.

All exposed conductive parts liable to become live from the voltage of the contact line under fault conditions shall be directly connected to the traction system earth.

NOTE Connection to the running rails is the preferred method to achieve this.

If it is determined that the running rails or exposed conductive parts cannot be directly connected to the traction system earth (e.g. track circuits), then a voltage-limiting device shall be used to provide a path for the current in the event of the running rails or exposed conductive parts becoming live.

4.2.2.2 To minimize stray currents in a d.c. traction system, direct earthing of the running rails is not desirable. All exposed conductive parts which are not insulated from earth shall, therefore, be connected to earth and shall not be bonded to the return circuit.

In this case, voltage-limiting devices shall be used to make an open connection from the exposed conductive parts to the return circuit to allow the interruption of the current in a short time to limit the voltage given in Table 4 or Table 5 or in 7.3.3.

NOTE An alternative method is to insulate from earth any masts or structures and then connect them directly to the return circuit.

4.2.2.3 Parts of the support system for overhead contact line systems (neutral parts), which by reason of protection-at-work provisions are insulated both with respect to the traction system earth and to the overhead contact line, shall not be connected to the traction system earth.

NOTE 1 Supporting structures for overhead contact lines need not be connected to the traction system earth if the insulation of the overhead contact line has been doubled or reinforced as defined in HD 366.

NOTE 2 Supports of conductor rails need not be connected to the traction system earth if the insulation between the conductor rails and the supports has been doubled or reinforced as defined in HD 366 and where the supports are insulated mounted.

NOTE 3 Insulated mounting of the supports can be achieved, for example, by attachment to wooden sleepers or by interposition of insulated parts between the support and the supporting structure.

4.2.2.4 An insulator shall be installed in the anchor of wooden masts external to the zone as shown in Figure 2 where the supporting structures of the overhead contact line system are not connected to the traction system earth.

4.2.3 Protection by use of Class II equipment

Protection by use of Class II equipment or by equivalent insulation shall meet the requirements as defined in HD 384.4.41, clause 413.2.

4.2.4 Exceptions

No special provisions are required for protection against electric shock with the exposed conductive parts equipment which is either mounted or fixed on insulators in closed electrical operating areas. This also applies to electrical equipment external to closed electrical operating areas, which is normally not accessible when it is energized. Such exposed conductive parts shall be provided with a warning sign "Broken arrow" as defined in ISO 3864 (see Annex B). Such marking can be dispensed with where provision is made for switching-off the equipment in normal operation before access to the equipment can be gained.

4.3 Protective provisions for wholly or partially conductive structures and for metallic structures located in the overhead contact line zone or the pantograph zone

4.3.1 For wholly or partially conductive structures (e.g. steel structures, reinforced concrete structures) and metallic structures (e.g. masts for overhead contact lines, reinforced concrete masts, metallic fences, drainpipes, running rails of non-electric traction systems) which will become live by a broken overhead contact line or a broken or dewired pantograph protective provisions against the remaining of dangerous touch voltage given in 4.2 shall be taken if necessary.

These protective provisions shall be harmonized with the provisions specified for d.c. traction systems as defined in EN 50122-2⁶⁾ for reduction of corrosion due to stray currents.

4.3.2 For conductive structures of small dimensions which do not support or contain electrical equipment it is not necessary to take any protective provisions. Such structures include for example: gully covers, signal posts, level crossing posts, individual masts, warning labels, trash cans, fences and mesh constructions up to 15 m in length measured parallel to the overhead contact line zone and also not more than 2,0 m measured in a direction at right angle to and away from the overhead contact line zone.

NOTE This excludes items utilized for protective provisions given in 4.2 and 6.2.

4.3.3 These protective provisions shall not normally be required for conductive materials temporarily stored in the vicinity of the track, e.g. running rails unless required by the operational specifications.

4.3.4 Instead of the protective provisions specified for conductive structures or conductive structural components located in the pantograph zone given in 4.2, an obstacle may be installed in the pantograph zone. The obstacle shall be situated between the overhead contact line and the structures or structural components and shall have a width at least equivalent to that of the pantograph zone and should extend by at least 0,5 m beyond the end of the structure or structural component. The conductive supporting elements of the overhead contact line connected to the structure shall be protected by the obstacle so that in the event of a fault no voltage will be transmitted to the structure.

In cases when the current collector is guided by the overhead contact line (e.g. trolleybus), the pantograph zone width can be reduced by extending the obstacle downwards on both sides of the overhead contact line by at least 0,05 m below the overhead contact line.

4.4 Traction power supply systems with nominal voltages up to and including 1,5 kV d.c. in which the running rails or other support systems are not utilized for carrying traction return current (e.g. trolleybus).

4.4.1 Contact wires shall be laid in such a way that they are insulated for the nominal voltage relative to one another and to earth.

4.4.2 Unearthed overhead contact lines shall have an equipment for monitoring the state of the insulation between each contact wire and earth.

4.4.3 If the negative contact wire is earthed, it shall only be earthed at one point per feeding section of the contact line where stray currents protection is required with the exception of 4.4.5.

4.4.4 When the negative contact wire is earthed, no provision for disconnection of the negative is allowed. When the negative contact wire is not earthed, disconnection of the negative contact wire is required and shall only be permitted in conjunction with disconnection of the positive contact wire. Return cables shall conform to the details given in 9.1.

4.4.5 Where there is a common traction power supply system for a trolleybus system and a tramway system, one contact wire of the trolleybus system shall be connected to the traction system earth at several points if the running rail system is continuously connected.

⁶⁾ In preparation

5 Protective provisions against electric shock in installations for nominal voltages in excess of 1 kV a.c./1,5 kV d.c. up to 25 kV a.c. or d.c. to earth

5.1 Protection against direct contact

5.1.1 General

In overhead contact line systems one of the following protective provisions shall be utilized for protection against direct contact:

- protection by clearance;
- protection by obstacles.

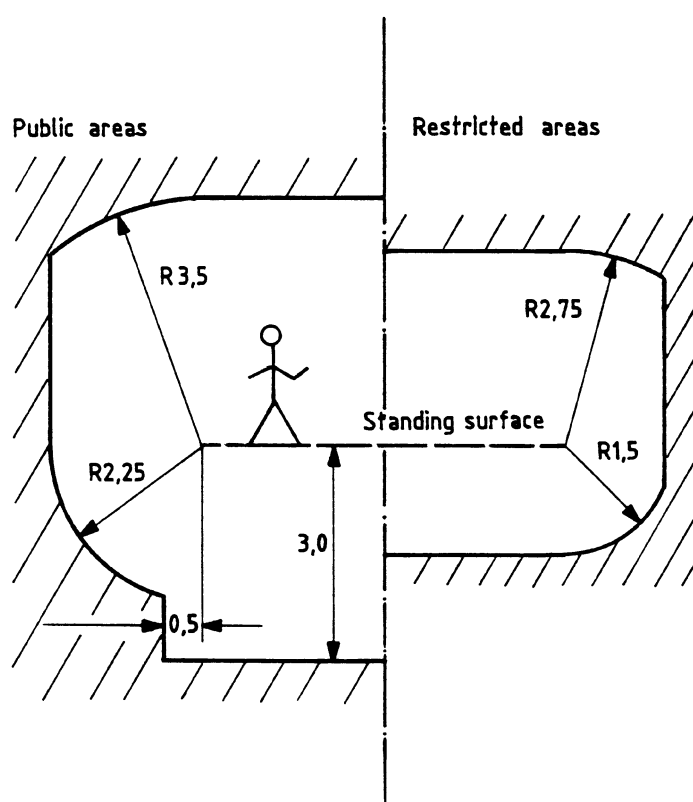
All insulators which are connected to a live part shall be regarded as a live part when considering clearance dimensions in this standard.

5.1.2 Protection by clearance

5.1.2.1 Standing surface

For standing surfaces, accessible to persons, clearance for touching in a straight line shown in Figure 14, shall be provided against direct contact with live parts of an overhead contact line system as well as any live parts on the outside of a vehicle (e.g. current collectors, roof conductors, resistors).

The clearances given in the following clauses are minimum values which shall be maintained at all temperatures and with additional and exceptional line loading. Due to national or regional existing practices, greater clearances or smaller mesh sizes may be prescribed by the relevant railway authority.



All dimensions are minimum dimensions in m

Figure 14 — Clearances to accessible live parts on the outside of vehicles as well as to live parts of overhead contact line systems from standing surfaces accessible to persons for nominal voltages in excess of 1 kV a.c./1,5 kV d.c. up to 25 kV a.c. or d.c. to earth

NOTE 1 Touching in a straight line implies that live parts are accessible by a person from a standing surface without use of objects.

NOTE 2 This protection is not mandatory if, although contact with live parts is possible, other provisions have been taken to ensure isolation from the supply.

NOTE 3 Figure 14 assumes that the standing surface does not afford protection against contact with live parts situated below or to the side thereof. Depending on its construction the surface may in practice meet the requirement respecting provision of obstacles. In this case the lower clearances applicable to obstacles may be utilized.

The clearances were determined on the basis of the arms reach as defined in HD 384.4.41 to which a margin of safety was added. The margin of safety has been determined depending on the voltage of the contact line system and on whether the standing surface is in a restricted area or in a public area.

5.1.2.2 Standing surfaces for working persons

The clearances to be observed for persons working nearby energized overhead contact line systems shall be defined in the operational specifications.

If operational specifications do not exist, clearances shown in Figure 14 or the clearances according to 5.1.3 shall be used.

NOTE Station platform roofs, working platforms and working planks at signal bridges, working platforms at signals, maintenance-ladders, work cages of hydraulic work platforms, working platforms of tower cars which are utilized solely for execution of work at or in the vicinity of overhead contact line systems are not included.

5.1.2.3 Minimum height of overhead contact lines and associated feeders

Where a road carrying normal vehicular traffic crosses with railway electrified by means of an overhead contact line and no road traffic restrictions are specified, unless otherwise fixed by national regulation, a minimum vertical clearance of 5,50 m shall be provided between the road surface and the lowest point of the overhead contact line and associated feeders.

If the said minimum clearance cannot be provided and unless otherwise specified by national regulation, the maximum height of road vehicles permitted to pass under the overhead contact line shall be limited so as to guarantee the following minimum vertical clearances between the highest point of road vehicle (load included) and the live parts, of:

- 1,0 m, where only road traffic signs indicating the maximum permissible vehicle height are utilized;
- 0,50 m, where additional fixed barriers (e.g. a rigid obstacle or a firmly fixed metallic wire made visible by means of a suspended warning sign) are erected on both sides of the crossing, physically limiting such vehicle height.

5.1.2.4 Feeders above loading roads

Feeders or line feeders may only in urgent cases be situated above loading roads or loading sidings. In such cases, there shall be a distance of at least 12 m between surfaces of the loading platform (ramps) and feeders.

5.1.2.5 Clearance between overhead contact line systems and trees

A clearance of 2,5 m shall be maintained under any conditions between the overhead contact line system and branches of trees or bushes.

5.1.3 Protection by obstacles

5.1.3.1 General

If the clearances given in 5.1.2 cannot be maintained, obstacles shall be provided as protection against direct contact with live parts.

The design of obstacles is dependent: on the location of the standing surfaces shown in Figure 15 relative to the live parts, on the clearance between the obstacle and the live parts, and on whether the standing surface is a restricted area or a public area.

The dimensions of the obstacles shall be such that live parts cannot be touched in a straight line by persons on a standing surface.

Where the obstacle is made of conductive material the requirements of 5.2.2 shall apply.

5.1.3.1.1 Only obstacles of the following type shall be provided:

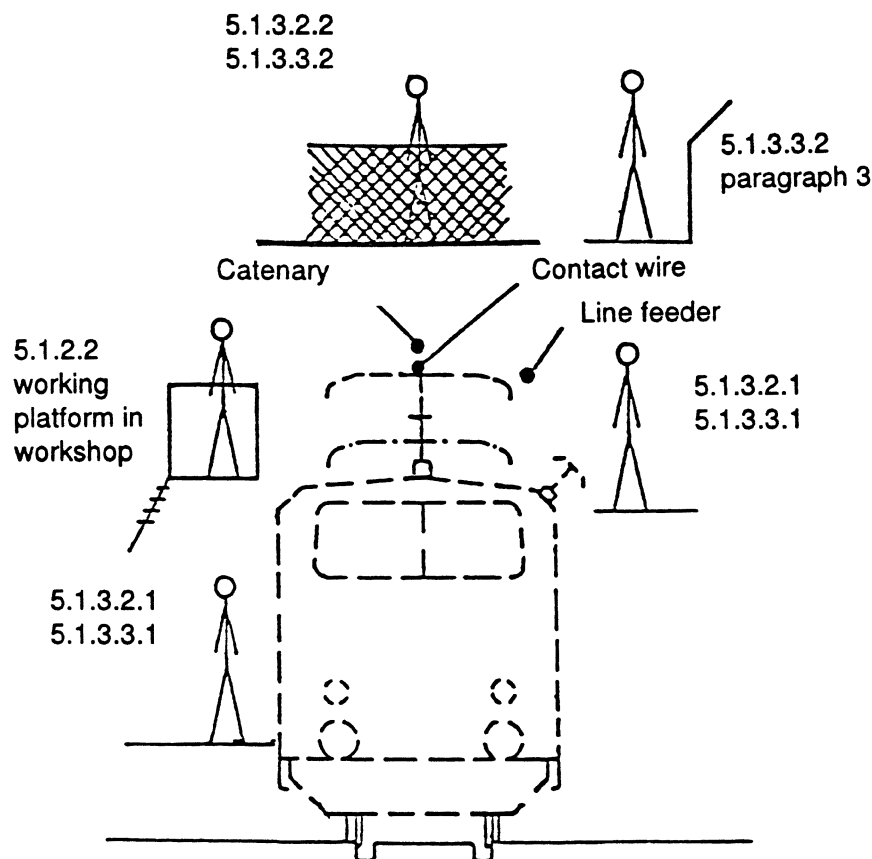
- solid-wall design or solid-wall doors;
- mesh constructions, when they are of conductive material and when they are earthed.

Mesh constructions of non-conductive material or plastic coated metal shall not be utilized.

Non-conductive obstacles shall be of solid-wall design. They are to be surrounded by a bare conductor that shall be connected to the traction system earth.

The obstacle shall be constructed to prevent non-intentional (accidental) contact with parts of the body. Obstacles shall be fixed mechanically and reliably. They shall only be removable with tools. In locations or rooms which are accessible to the public, these obstacles shall not even be removable with tools without destruction.

Conductive obstacles shall be secured to ensure that the distance to live parts is maintained.



The numbers refer to the appropriate clauses

Figure 15 — Standing surfaces for persons providing access to live parts on the outside of vehicles and to overhead contact line systems in excess of 1 kV a.c./1,5 kV d.c. up to 25 kV a.c. or d.c. to earth

5.1.3.1.2 Minimum clearances shall be maintained between obstacles and live parts. The clearance in air shall correspond to the relevant minimum clearance as defined in EN 50124-1⁷⁾ to which the following addition shall be added:

- 30 mm for solid-walls or solid-wall doors, if buckling or warping cannot be excluded.
- 100 mm for mesh constructions, where no other minimum clearance is given in **5.1.3.2.1** and **5.1.3.3.1**.

5.1.3.2 *Obstacles for standing surfaces in restricted areas*

5.1.3.2.1 With standing surfaces adjacent to live parts on the outside of vehicles or adjacent to live parts of an overhead contact line system, obstacles of mesh screens construction with a maximum 1 200 mm² mesh size and having a height “h” of at least 1,8 m may be utilized, where the live parts are situated above the standing surface (see Figure 16, example a).

If the live parts are situated at the same height or lower than the standing surface, the height “h” of the obstacle shall be such that a clearance of 1,5 m from the top of the obstacle is maintained as shown in Figure 14 (see Figure 16, example b).

⁷⁾ In preparation

The clearance between the obstacle and live parts shall be at least 0,6 m. If this clearance is not maintained, the obstacles shall be of solid-wall design and leave no gap between the obstacle and the standing surface (see Figure 16, examples c and d). The dimension “*d*” between the obstacle and live parts, shown in Figure 16, shall be determined as described in 5.1.3.1.2.

5.1.3.2.2 For standing surfaces above live parts on the outside of vehicles or above live parts of an overhead contact line system shall be of solid-walled design.

The length of the solid-walled standing surface shall correspond to the pantograph zone and extend beyond the live parts of an overhead contact line system by at least 0,5 m on each side. In the case of conductors not being used for current collection (e.g. line feeders, reinforcing feeders, out of running overhead contact lines) there shall be a width of at least 0,5 m on each side of the conductor provided that movement due to dynamic and thermal effects has been taken into account.

Obstacles of mesh screen construction with a maximum 1 200 mm² mesh size shall be provided at the sides of such standing surfaces. The height “*h*” of the obstacles shall be such that a clearance of 1,5 m from the top of the obstacles is maintained as shown in Figure 14 (see Figure 17). These obstacles shall be at least as long as the length of the standing surface above live parts.

NOTE The height of the side obstacles, if necessary, will generally correspond to the height of the necessary safety railing but should have a height of at least 1,0 m.

The dimension “*d*” between the obstacle and live parts, shown in Figure 17, shall be determined as described in 5.1.3.1.2.

5.1.3.3 *Obstacles for standing surfaces in public areas*

5.1.3.3.1 With standing surfaces adjacent to live parts on the outside of vehicles or adjacent to live parts of an overhead contact line system, if the clearances shown in Figure 14 cannot be achieved, obstacles shall be provided and shall leave no gap with respect to the standing surface.

If the clearance between the obstacles and live parts is at least 1,0 m (see Figure 18, example a) the obstacle shall be of solid-wall design 1,0 m high, surmounted by a mesh construction with a maximum 1 200 mm² mesh size to an overall height of at least 1,8 m. Such mesh shall be arranged so as to ensure that the obstacle remains unclimbable.

If this clearance is not achieved, the obstacles shall be of solid-wall design 1,8 m high (see Figure 18, example b). The dimension “*d*” between the obstacle and live parts, shown in Figure 18, shall be determined as described in 5.1.3.1.2.

The tops of such obstacles shall be designed to prevent access for standing or walking on.

5.1.3.3.2 For standing surfaces above live parts on the outside of vehicles or above live parts of an overhead contact line system shall be of solid-wall design.

The length of the solid-walled standing surface shall correspond to the pantograph zone and extend beyond the live parts of an overhead contact line by at least 0,5 m on each side. In the case of conductors not being used for current collection (e.g. line feeders, reinforcing feeders, out of running overhead contact lines) there shall be a width of at least 0,5 m on each side of the conductor provided that movement due to dynamic and thermal effects has been taken into account.

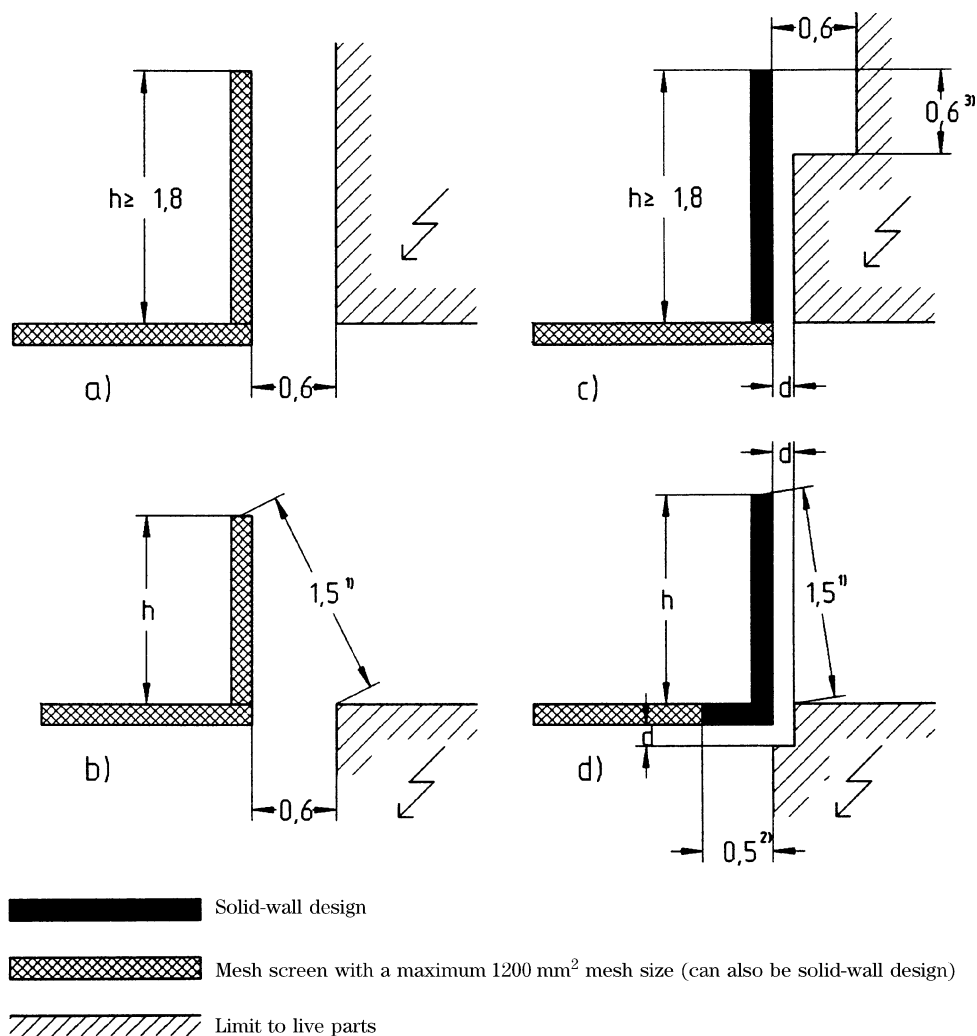
Along the sides of these standing surfaces, obstacles shall be provided in order to prevent any direct contact with live parts on the outsides of vehicles and to live parts of the overhead contact line system even where, for example, a rod or liquid jet is utilized. These obstacles shall be provided over a length that corresponds at least to the length of the standing surface in solid-wall design.

Where horizontal obstacles extend beyond the vertical obstacle by at least 1,5 m, it becomes acceptable to refer the 2,25 m lateral clearance required by Figure 14 to the top of the vertical obstacle, instead of referring it to the edge of the standing surface (see Annex A, Figure A.2, example a). The height of the vertical obstacle shall be consequentially increased whenever necessary, in order to achieve this clearance. The horizontal obstacle shall be impossible to stand on.

NOTE To achieve this, the horizontal obstacle should be designed so that it is obviously not a standing surface, or it should be inclined upwards or downwards (see Annex A, Figure A.2 example a).

Whenever this horizontal obstacle is not present, the vertical obstacle shall conform to the requirements of 5.1.3.3.1 (see Annex A, Figure A.2, example b). The alternative solution (shown in Annex A, Figure A.2, example c) is also acceptable. The tops of such obstacles shall be designed to prevent access for standing or walking on.

Any vertical obstacle shall be of solid-wall design at least to a height of 1,0 m (see Annex A, Figure A.2), except in the case of the above said horizontal obstacle, where a railing is sufficient, if the clearances required by Figure 14 are present.



' d ' is the clearance in air between the obstacle and live parts.

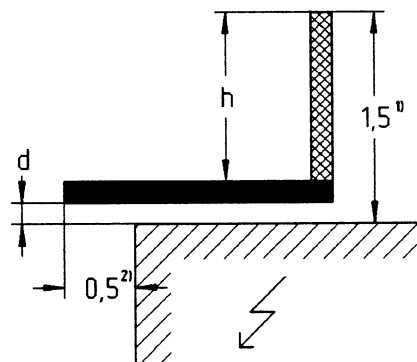
1) The dimension 1,5 is derived from figure 14.

2) The dimension 0,5 is based on the requirements given in 5.1.3.2.2.

3) The dimension 0,6 may be reduced by the same extent as the height ' h ' exceeds the value of 1,8 m.

All dimensions are minimum dimensions in m.

Figure 16 — Examples of obstacles for standing surfaces in restricted areas for protection against direct contact with adjacent live parts on the outside of vehicles or adjacent live parts of an overhead contact line system for nominal voltages in excess of 1 kV a.c./1,5 kV d.c. up to 25 kV a.c. or d.c. to earth (see 5.1.3.2.1)



 Solid-wall design

 Mesh screen with a maximum 1200 mm² mesh size (can also be solid-wall design)

 Limit to live parts

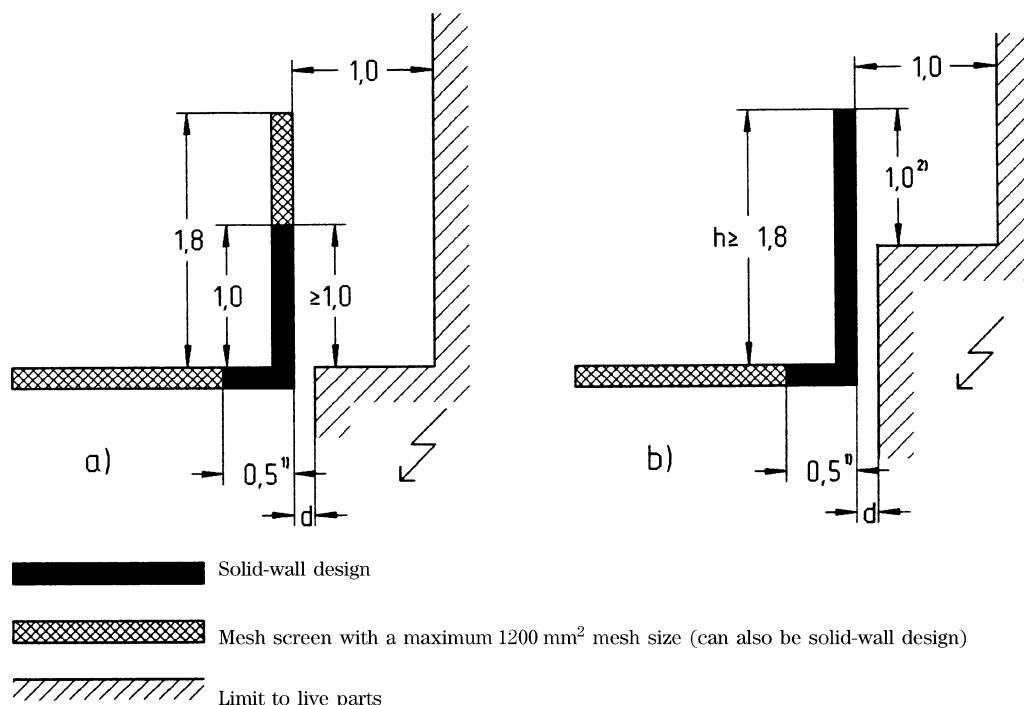
'd' is the clearance in air between the obstacle and live parts.

1) The dimension 1,5 is derived from figure 14

2) The dimension 0,5 is based on the requirements given in 5.1.3.2.2

All dimensions are minimum dimensions in m.

Figure 17 — Examples of obstacles for standing surfaces in restricted areas for protection against direct contact when above live parts on the outside of vehicles or live parts of an overhead contact line system for nominal voltages in excess of 1 kV a.c./1,5 kV d.c. up to 25 kV a.c. or d.c. to earth (see 5.1.3.2.2)



'd' is the clearance in air between the obstacle and live parts.

1) The dimension 0,5 is based on the requirements given in 5.1.3.3.2.

2) The dimension 1,0 may be reduced by the same extent as the height 'h' exceeds the value of 1,8.

All dimensions are minimum dimensions in m.

Figure 18 — Examples of obstacles for standing surfaces in public areas for protection against direct contact with adjacent live parts on the outside of vehicles or adjacent live parts of an overhead contact line system for nominal voltages in excess of 1 kV a.c./1,5 kV d.c. up to 25 kV a.c. or d.c. to earth (see 5.1.3.3.1)

NOTE In addition to the requirements of electrical safety there may be additional requirements for obstacles to provide for security according to national regulations.

5.1.3.4 Anti-climbing provisions

Anti-climbing provisions are not normally necessary. In justified cases however, anti-climbing provisions can become necessary. The railway authority shall specify where it is necessary to equip masts and structure anchor wires with anti-climbing provisions.

NOTE Provisions are required only for masts and anchor wires which can be climbed without any aids and if there is a risk of coming into unsafe proximity to live parts of the overhead contact line equipment when climbing.

5.2.2 Traction system earthing

5.2.2.1 In a.c. traction systems, direct traction system earthing is the preferred method for system faults and safety of persons. This method utilizes the return circuits for conduction of the current

5.1.4 Warning signs

Warning signs shall be used in areas where there is a serious risk of coming within the limits of live parts of an overhead contact line system given in 5.1.2.1. Such warning signs shall be placed in a prominent position and readily visible adjacent to the point of access. The sign shall be as defined in ISO 3864 (see Annex B) and if required an appropriate supplementary sign may be used.

5.2 Protection against indirect contact

5.2.1 General

Provisions for protection against indirect contact shall be provided for exposed conductive parts and components of overhead contact line systems.

under fault conditions.

All exposed conductive parts liable to become live from the contact line voltage under fault conditions shall be directly connected to the traction system earth.

NOTE Connection to the running rails is the preferred method to achieve this.

If it is determined that the running rails or exposed conductive parts cannot be directly connected to the traction system earth (e.g. track circuits), then a voltage-limiting device shall be used to provide a path for the current in the event of the running rails or exposed conductive parts becoming live.

5.2.2.2 To minimize stray currents in a d.c. traction system, direct earthing of the running rails is not desirable. All exposed conductive parts which are not insulated from earth shall, therefore, be connected to earth and shall not be bonded to the return circuit.

In this case, voltage-limiting devices shall be used to make an open connection from the exposed conductive parts to the return circuit to allow the interruption of the current in a short time to limit the voltage given in Table 4 or Table 5 or in **7.3.3**.

NOTE An alternative method is to insulate from earth any masts or structures and then connect them directly to the return circuit.

If in industrial d.c. railway systems (e.g. surface coal mining) protection against inadmissible touch voltage can only be achieved by direct earthing of the running rails, special protective provisions to minimize stray currents shall be considered (see also EN 50122-2)⁸⁾.

5.2.2.3 For overhead contact line systems up to and including 3 kV d.c., and where permitted by national regulations, an exception to the requirements given in **5.2.2.2** may be allowed for the metallic supports of the overhead contact line suspensions in tunnels outside the zone shown in Figure 14. This exception requires the use of insulators, restoring their dielectric property even after a flashover.

Due to the characteristics of such insulators and in order to reduce the troubles to the maintenance service, often difficult in tunnels, it is permitted that such metallic parts, normally inaccessible, shall not be earthed. In this case the clamping of the metallic supports to the tunnel vault shall be achieved by using a grout having a good insulating property.

Where national regulations allow to perform maintenance work on structures adjacent to live parts, provisions in accordance with **5.1.2.2** shall apply considering the support is also a live part.

5.3 Protective provisions for wholly or partially conductive structures and for

metallic structures located in the overhead contact line zone or the pantograph zone

5.3.1 For wholly or partially conductive structures (e.g. steel structures, reinforced concrete structures) and metallic structures (e.g. masts for overhead contact lines, reinforced concrete masts, metallic fences, drainpipes, running rails of non-electric traction systems) which will become live by a broken overhead contact line or a broken or dewired pantograph protective provisions against the remaining of dangerous touch voltage given in **5.2** shall be taken if necessary.

These protective provisions shall be harmonized with the provisions specified for d.c. traction systems as defined in EN 50122-2⁸⁾ for reduction of corrosion due to stray currents.

5.3.2 For a.c. traction systems conductive structures not greater than 2 m in horizontal direction which do not support or contain electrical equipment it is not necessary to take any protective provisions. All d.c. traction systems shall conform to the requirements in **4.3.2**.

5.3.3 These protective provisions are not required for conductive materials temporarily stored in the vicinity of the track, e.g. running rails unless required by the operational specifications.

5.3.4 Instead of the traction system earthing specified for conductive structural components located in the pantograph zone, an obstacle may be installed in the pantograph zone. The obstacle shall be situated between the overhead contact line and the structural components and shall be connected with the traction system earth. It shall have a width at least equivalent to that of the pantograph zone and should extend by at least 0,5 m beyond the end of the structural component.

6 Additional protective provisions

6.1 Protective provisions where track systems, which are utilized for carrying traction return current, or overhead contact line systems pass through plant areas in which flammable liquids or gases are utilized in hazardous zones.

NOTE Hazardous zones are defined in national safety regulations.

6.1.1 The requirements given in **6.1.2** to **6.1.8** shall apply to installations where protective provisions against explosion due to sparking are prescribed by other standards.

Explosion due to sparking may, for example, result from:

- contact with an overhead contact line;
- rupture of a contact wire;

⁸⁾ In preparation

- voltage between running rails utilized for carrying traction return current and earth (rail potential);
- discharge of static electricity to the traction system earth.

NOTE Examples of such installations are loading installations or parts of refineries or chemical firms or tank depots.

6.1.2 All pipework carrying flammable liquids or gases shall be continuously conductively interconnected. Consequently, protective provisions against sparking as a result of partial voltages from a potential gradient area in the soil on rupture of a contact wire are not required.

6.1.3 Rail joints of tracks, where these are utilized for carrying traction return current, shall be provided by rail joint bonds.

- Cross bonds shall be provided at intervals not exceeding 70 m between:
- the rails of a track;
- neighbouring tracks (see Figure 21).

In special cases where additional return conductors are utilized, these shall be installed level with the overhead contact line.

6.1.4 Tracks and adjacent conductive installations shall be conductively interconnected (equipotential bonding). Loading sidings shall be conductively connected to the charging nozzles of the loading installation.

6.1.5 In the case of loading sidings without overhead contact lines which lie outside the overhead contact line zone of a neighbouring track but through which could flow the traction return current, insulated rail joints shall be installed in the running rails of this siding outside the hazardous zone and outside the overhead contact line zone of a neighbouring track (see Figure 19).

The running rails of the track section separated by insulated rail joints shall be connected nearby their two ends by rail-to-rail crossbonds. Furthermore, they shall be interconnected conductively with the charging nozzles of the installation by equipotential bonding conductors.

NOTE The insulated rail joints should not be installed in places where it can be expected that vehicles, which may short-circuit the insulated rail joints with their wheel sets, bodies, drawbars and buffers, will be parked.

6.1.6 If a loading siding, above which there is no overhead contact line, is isolated by means of insulated rail joints and lies within the overhead contact line zone of the neighbouring track, the isolated track section shall be connected to the traction system earth by a voltage-limiting device (see Figure 20). The voltage-limiting device shall be located outside the hazardous zone.

Provision shall be made for switching off and earthing by switches (traction system earth) of the overhead contact line, also if the overhead contact line zone of a neighbouring track overlaps the hazardous zone of the loading siding.

6.1.7 In the case of loading sidings with an overhead contact line, the isolation of tracks by means of insulated rail joints is not permissible because the tracks are utilized for carrying traction return current. The overhead contact line of the loading siding shall be interruptable and shall be connectable to the traction system earth by a means which ensures that the loading procedure is not initiated unless the overhead contact line is earthed (see Figure 21).

6.1.8 The charging nozzle connected to the loading siding given in **6.1.4** shall either be also conductively connected to the remainder of the loading installation, or shall be isolated by means of isolation pieces.

6.1.8.1 Conductive connections (e.g. a conducting nozzle) to the loading installation shall only be used when insulated rail joints are provided unless precautions are taken to prevent damage to the electrical equipment of the loading installation from the traction return current.

6.1.8.2 If the charging nozzle is insulated with respect to the loading installation, the isolation pieces shall be located so that the loading installation lies outside the overhead contact line zone, as a result of which earthing (traction system earth) as given in **4.3** and **5.3** can be dispensed with (see Figure 22). Where this is not the case, the loading installation shall be connected to traction system earth by means of a voltage-limiting-device (see Figure 23).

6.1.8.3 If isolation pieces are installed in the refilling pipe outside the overhead contact line zone and flashover or voltage breakdown of the isolation pieces is likely due to lightning strike, the isolation pieces shall be bridged by means of surge divertors (see Figure 22). Surge divertors utilized in hazardous zones shall be of flameproof design. Care shall be taken to ensure that accidental bridging of isolation pieces or surge divertors (e.g. by tools) is prevented.

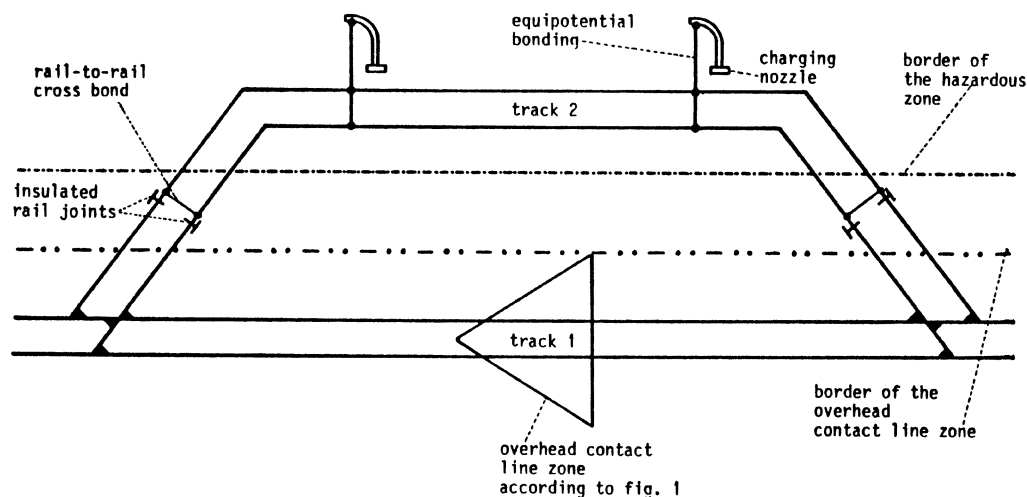


Figure 19 — Disposition of insulated rail joints (double-track illustration) in the case of the loading siding having no overhead contact line and lying outside the overhead contact line zone of the neighbouring track (see 6.1.5)

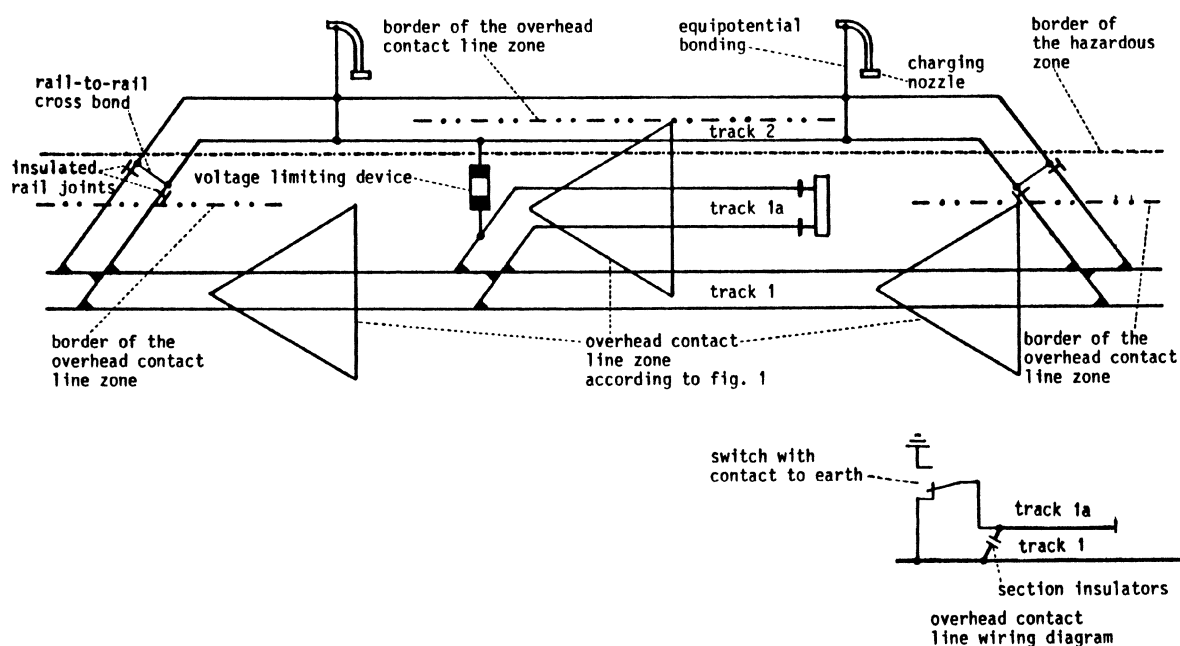


Figure 20 — Disposition of insulated rail joints (double-track illustration), open traction system earthing (voltage-limiting device) and connection of the overhead contact line in the case of the loading siding being within the overhead contact line zone of the neighbouring track (see 6.1.6)

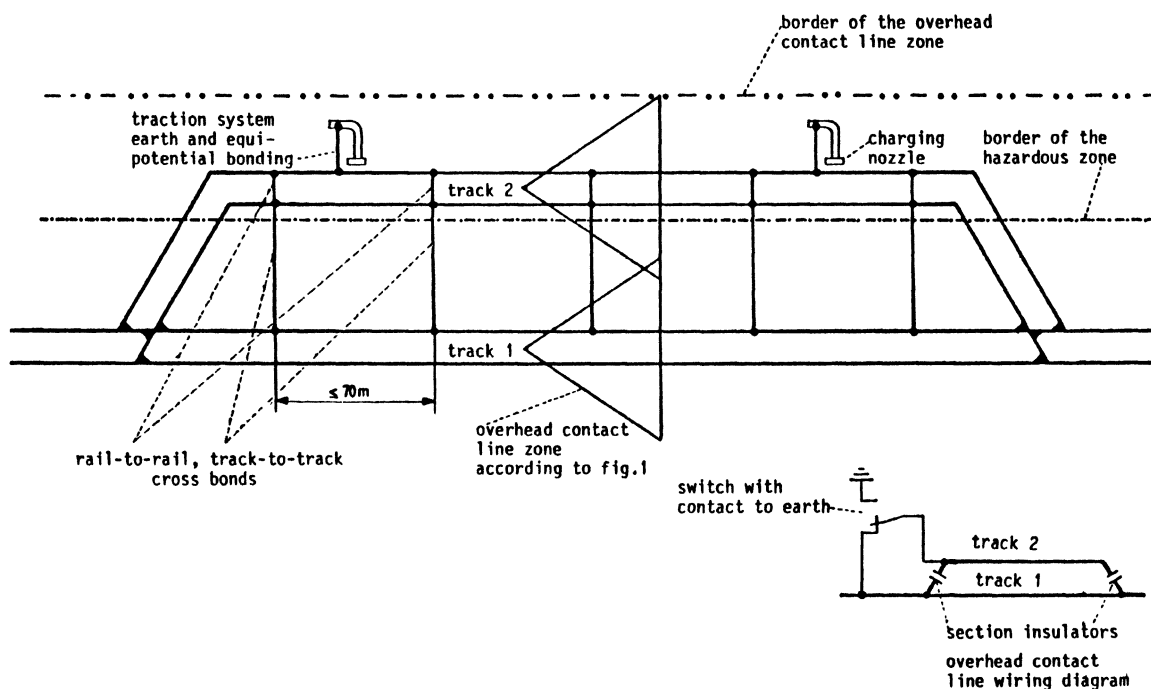


Figure 21 — Disposition of rail-to-rail crossbonds and track-to-track crossbonds (double-track illustration) and connection of the overhead contact line in case of the loading siding having an overhead contact line (see 6.1.3 and 6.1.7)

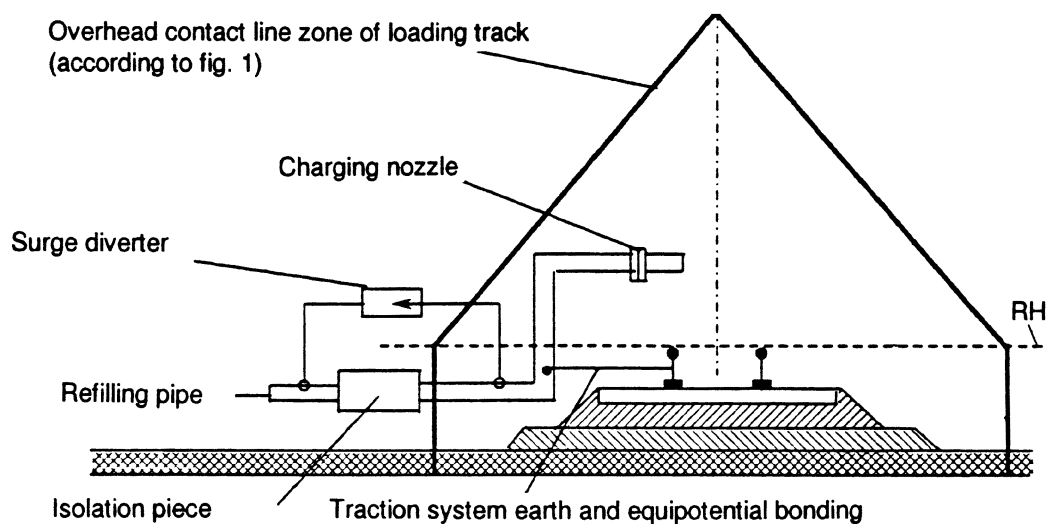


Figure 22 — Location of a surge diverter if isolation pieces are between charging nozzle and refilling pipe outside the overhead contact line zone of a loading siding with over-head contact line and if there is a possibility of flashovers of the isolation pieces through lightning strikes (see 6.1.8.2 and 6.1.8.3)

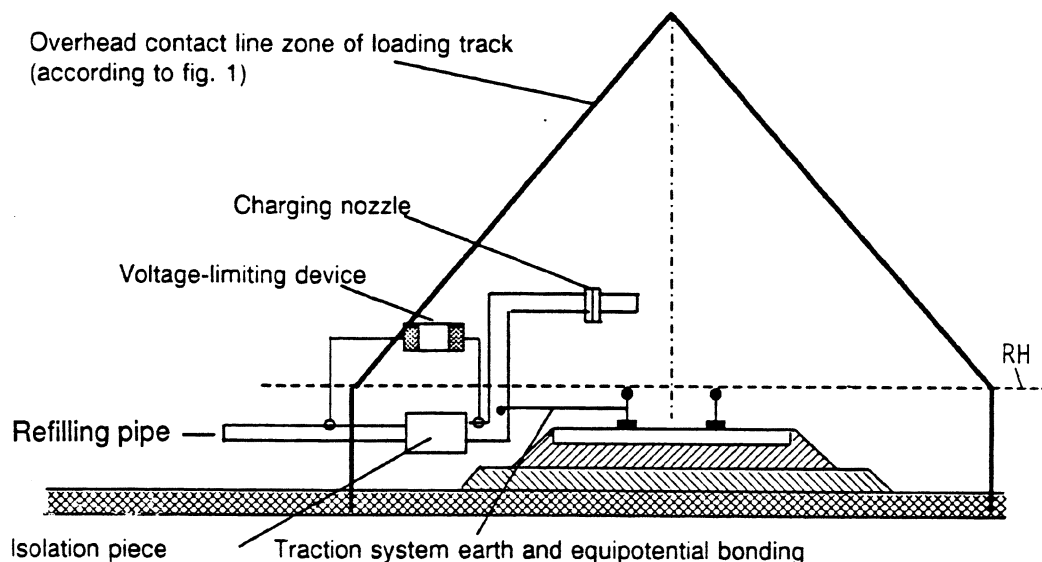


Figure 23 — Location of a voltage-limiting device if isolation pieces are between charging nozzle and refilling pipe inside the overhead contact line zone of a loading siding with overhead contact line (see 6.1.8.2)

6.2 Protective provisions for electric power, telecommunications and other electrical installations against the danger of the traction power supply system

6.2.1 Endangered installations:

The following installations are endangered, for example:

Consumer installations, railway- or road signalling installations, overhead lines, telecontrol equipment for remote devices. All these installations are characterized by being able, under particular circumstances, to transfer a dangerous voltage over long distances; therefore no part of such systems shall be considered as objects of "Small dimension" as given in 4.3.2 and 5.3.2.

NOTE The above installations could be endangered by the following:

- a broken overhead contact line or a broken or dewired pantograph which may transfer the overhead contact line voltage in these installations when they are located in the overhead contact line zone or the pantograph zone, or when they have conductive connections to installations which are located in these zones;

— traction return currents, which may overload protective conductors, neutral conductors, PEN-conductors, or which may impair protective provisions where exposed conductive parts are intentionally and functionally connected to the traction system earth or to the return circuit.

6.2.2 Related provisions

The following protective provisions as given in 6.2.3 and 6.2.4 shall apply for stationary electrical installations as given in 6.2.1 in addition to provisions required by other relevant standards. These protective provisions shall be harmonized with the provisions specified for d.c. traction systems as defined in EN 50122-2 for reduction of effects due to stray currents.

6.2.3 Protective provisions for electrical installations in the overhead contact line zone or the pantograph zone

6.2.3.1 Exposed conductive parts

Any exposed conductive part within the overhead contact line zone or the pantograph zone shall be connected directly or via a voltage-limiting device to the return circuit, or otherwise it shall be protected by a robust obstacle. If the obstacle is not made of insulating material, it shall be connected to the return circuit. In d.c. traction systems such obstacles shall conform to the requirements as defined in EN 50122-2⁹⁾.

⁹⁾ In preparation

6.2.3.2 Equipment of protection Class II

If electrical equipment of protection Class II (see HD 366) is utilized for traction systems of nominal voltages up to and including 1 kV a.c./1,5 kV d.c., then its insulation shall correspond to the nominal voltage of the contact line.

6.2.4 Protective provisions for installations which are endangered by the traction return current

6.2.4.1 General

The danger of damage or functional impairment of protective provisions against electric shock by traction return currents shall be obviated by adoption of the protective provisions given in 6.2.4.2 to 6.2.4.4.

NOTE Because there are known different protective provisions, some typical examples for such special protective provisions are shown in the Figure 24 to Figure 33.

6.2.4.2 A.C. traction systems

If the star point of the transformer is connected to the traction system earth (see Figure 28), the requirements as defined in HD 384.4.41 shall apply. The requirements given in 6.2.4.2.1 and 6.2.4.2.2 shall apply only where the earthed star point of the transformer is connected to an earth other than the traction system earth.

6.2.4.2.1 Power supply from TT system

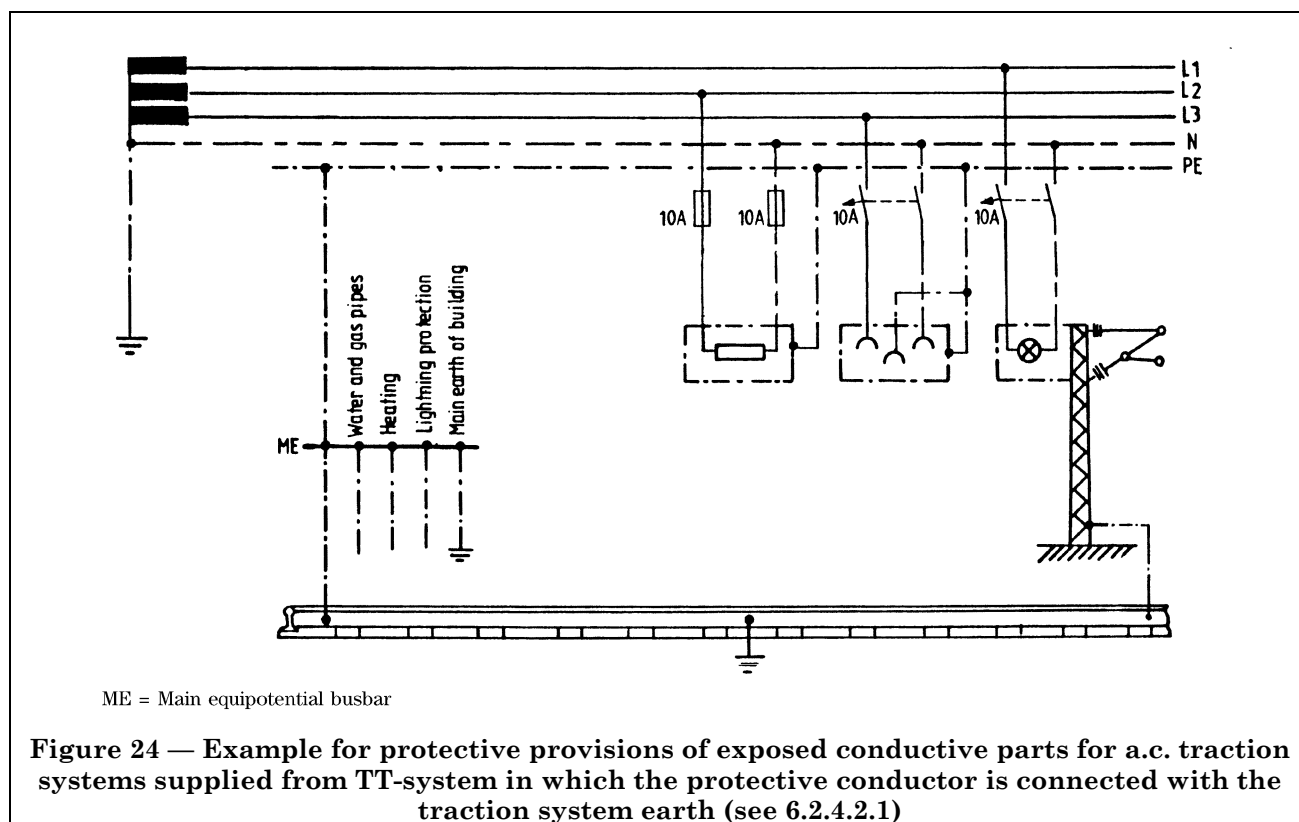
The traction system earth shall be used as an earth electrode. Since the neutral conductor could be endangered by traction return currents in the case of a fault connection between a live part and an exposed conductive part, the overcurrent protective device shall also protect the connection to the neutral conductor as shown in Figure 24.

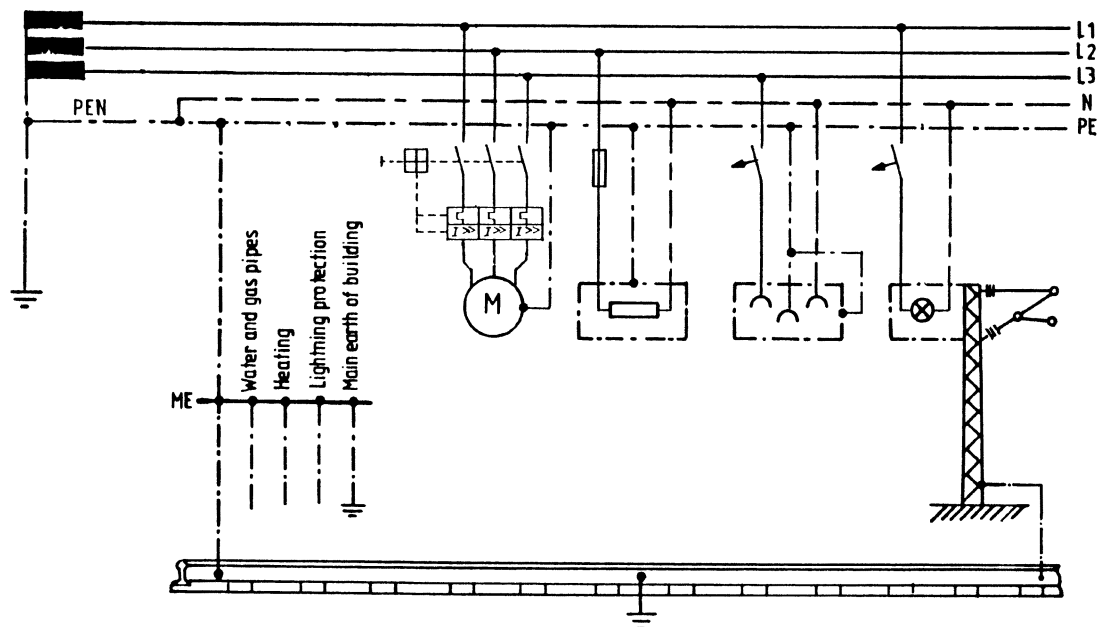
NOTE Overcurrent protective devices may generally be used as protective devices for automatic tripping, up to a nominal current of 10 A, because the low earthing resistance that is necessary for higher nominal currents cannot be guaranteed (see Figure 24).

6.2.4.2.2 Power supply from TN system

On utilization of overcurrent protective devices for automatic tripping (see Figure 25), the PEN conductors or the protective conductors shall be connected to the traction system earth only where an accessible voltage not exceeding 50 V occurs as a result of the rail potential, and the PEN conductor or the protective conductor is not endangered by the traction return current.

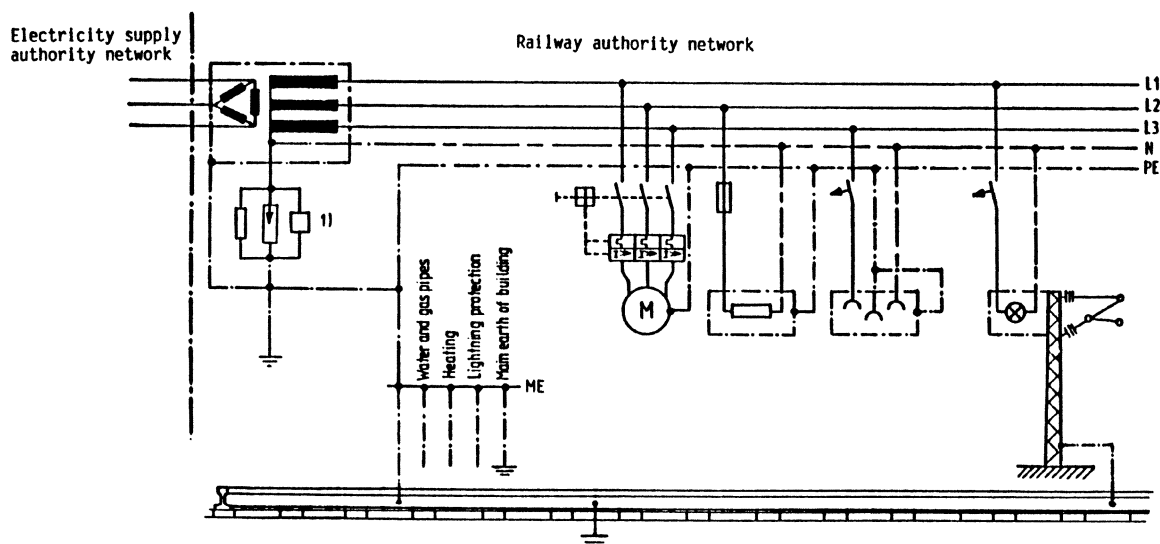
If the equipment has a traction system earth via one running rail and if this running rail has an equipotential bonding conductor to the main equipotential busbar, then the PE conductor incorporated in the power supply cable need not be connected to the exposed conductive part of the equipment.





ME = Main equipotential busbar

Figure 25 — Example for protective provisions of exposed conductive parts for a.c. traction systems supplied from TN system in which the PEN conductor and the protective conductor are connected with the traction system earth (see 6.2.4.2.2)



1) isolating monitoring

ME = Main Equipotential busbar

Figure 26 — Example for protective provisions of exposed conductive parts for a.c. traction systems supplied from an IT-System, in which the PE is connected with the starpoint of the transformer (see 6.2.4.2.3)

6.2.4.2.3 Power supply from IT system

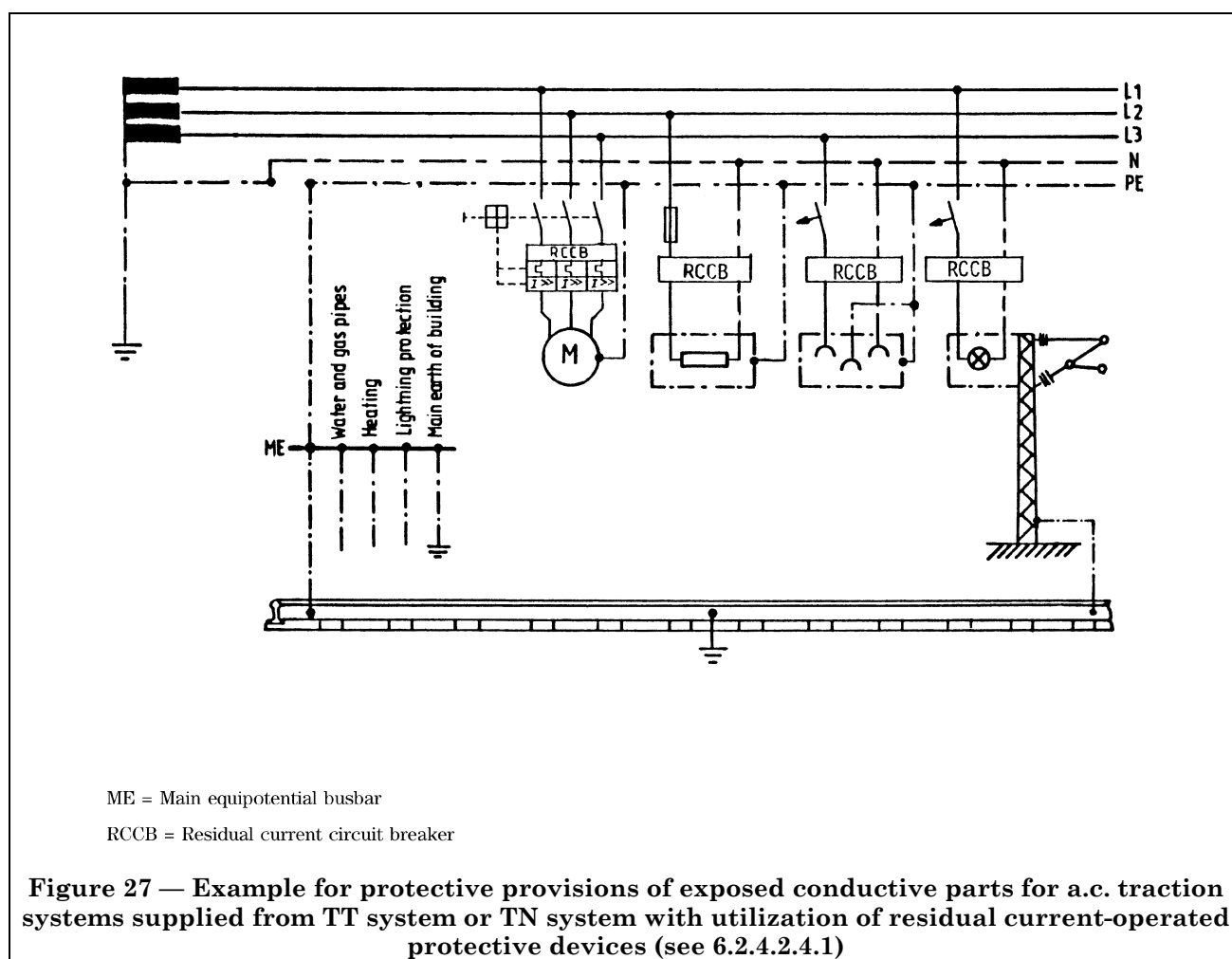
For certain cases where continuity of supply is essential for safety critical reasons, then an IT system shall be provided. An example is shown in Figure 26.

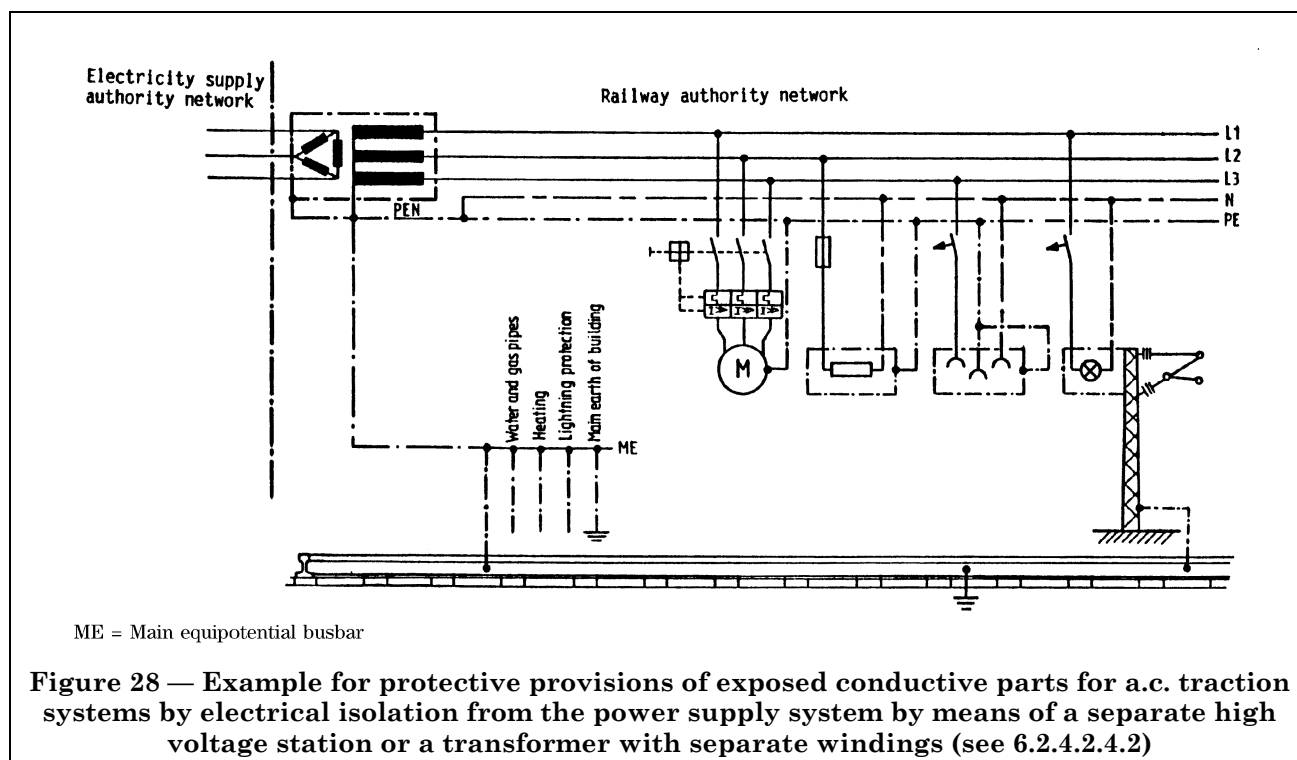
6.2.4.2.4 Alternative protective provisions

6.2.4.2.4.1 Residual current-operated protective devices for automatic tripping may be utilized (see Figure 27), in which case the exposed conductive parts of the electrical equipment may be connected to the traction system earth.

6.2.4.2.4.2 When electrical isolation of the electrical equipment from the power supply system is by means of a separate high voltage station or a transformer with separate windings (see Figure 28), then the exposed conductive parts of the electrical equipment of the electrically isolated installation, the PEN conductors or neutral conductors and the protective conductors shall be connected to the traction system earth.

If the equipment has a traction system earth via one running rail and if this running rail has an equipotential bonding conductor to the transformer star point, then the PE conductor incorporated in the power supply cable need not be connected to the exposed conductive part of the equipment.





6.2.4.3 D.C. traction systems

For d.c. traction systems the following requirements shall be observed:

6.2.4.3.1 Electrical equipment outside the overhead contact line zone or the pantograph zone

The railway network shall be separated, by a transformer with separate windings, from the public power supply system.

6.2.4.3.1.1 Running rails insulated from earth

If work is carried out on vehicles, on running rails, or on conductive parts which are connected to the return circuit, using Class I electrical equipment which is plugged into wall sockets supplied from the railway authority network, these wall sockets shall be connected to the railway authority network by residual-current protective devices or by transformers with separate windings. If these residual-current protective devices are used, the protective conductor shall be connected to PE conductor via an appropriate device such as a capacitor. A discharge resistor shall be connected in parallel with a capacitor.

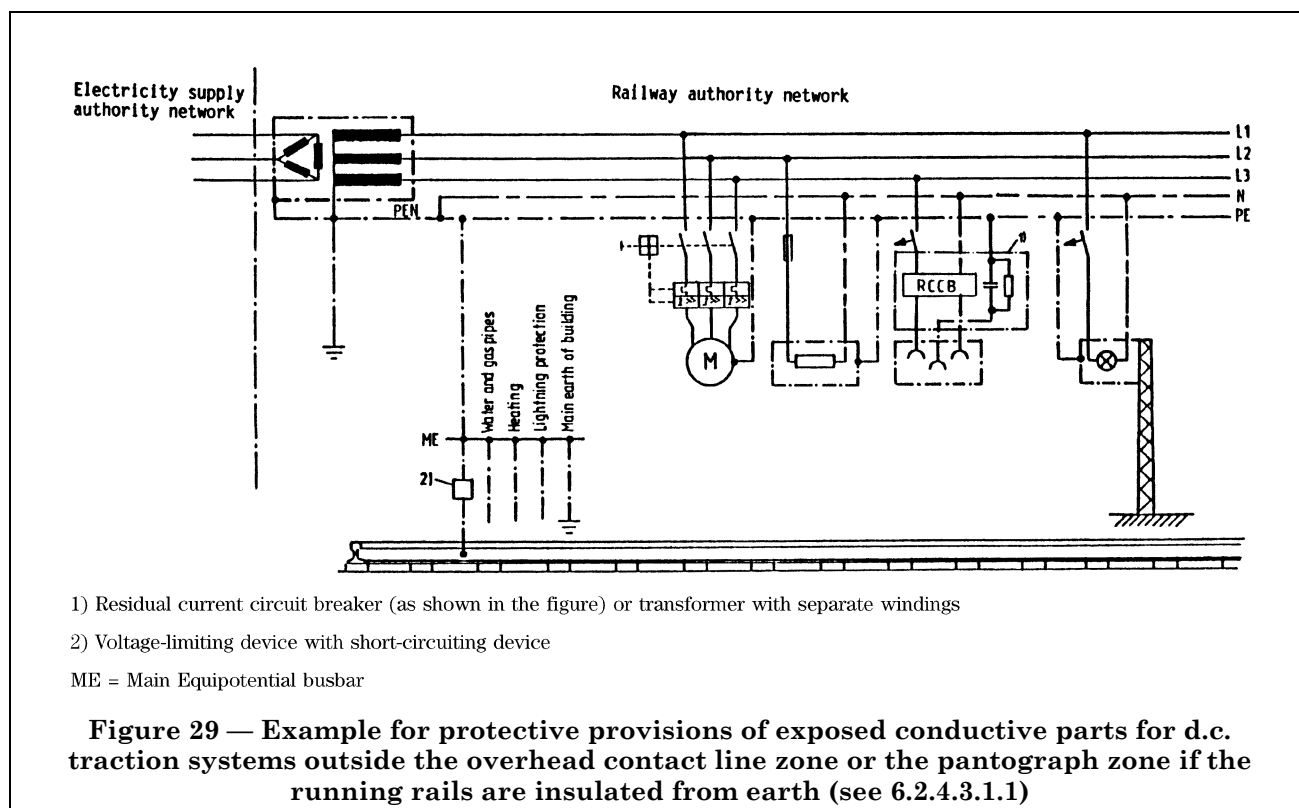
NOTE The appropriate device prevents traction return currents flowing via the protective conductor to the main earth of a building.

If inadmissible touch/accessible voltages may occur between protective conductor/neutral conductor and running rails, a voltage-limiting device with short-circuiting device shall be installed between protective conductor/neutral conductor and running rails (see Figure 29).

If the short-circuiting device has operated:

- either it shall automatically reset after a maximum of 10 s;
- or, if it is not reset, a procedure shall be established to note and rectify the cause of such event rapidly.

The short-circuiting device shall be designed to operate under the highest envisaged value of the current under fault conditions. Once closed, it shall not open until this current has reduced to a safe value.



6.2.4.3.1.2 Running rails in connection with earth or foundations

If the running rails are directly connected to earth or foundations of buildings as for example in workshops or depots the protective conductor or the PEN conductor may be connected to the running rails.

In order to reduce the danger of stray currents the running rails shall be separated from the main line by insulated rail joints and the traction power shall be supplied by separate rectifiers or other means (see Figure 30).

NOTE The insulated rail joints should not be installed in places where it can be expected that vehicles, which may short-circuit the insulated rail joints with their wheel sets, bodies, drawbars and buffers, will be parked.

6.2.4.3.2 Electrical equipment inside the overhead contact line zone or the pantograph zone

6.2.4.3.2.1 Running rails insulated from earth

Individual electrical equipment in Class I which is supplied from a TN system shall be protected by a residual-current protective device. The exposed conductive parts shall be connected to the PE conductor via an appropriate device such as a capacitor and shall be connected moreover with the running rails via a voltage-limiting device by an additional conductor of a sufficient cross-section. A discharge resistor shall be connected in parallel to the appropriate device (see Figure 31).

NOTE The appropriate device prevents traction return currents flowing via the protective conductor to the main earth of building, if the voltage-limiting device has operated.

If numerous items of electrical equipment have to be supplied, a railway network which is additionally separated from the public network via a transformer with separated windings shall be provided.

6.2.4.3.2.2 Running rails in connection with earth or foundations

If the running rails are directly connected to earth or foundations of buildings as for example in workshops or depots the protective conductor or the PEN conductor may be connected to the running rails.

In order to reduce the danger of stray currents the running rails shall be separated from the main line by insulated rail joints and the traction power shall be supplied by separate rectifiers or other means (see Figure 32).

NOTE The insulated rail joints should not be installed in places where it can be expected that vehicles, which may short-circuit the insulated rail joints with their wheel sets, bodies, drawbars and buffers, will be parked.

Exposed conductive parts of Class I shall be connected to the running rails by an additional protective conductor with a sufficient cross-section (see Figure 32).

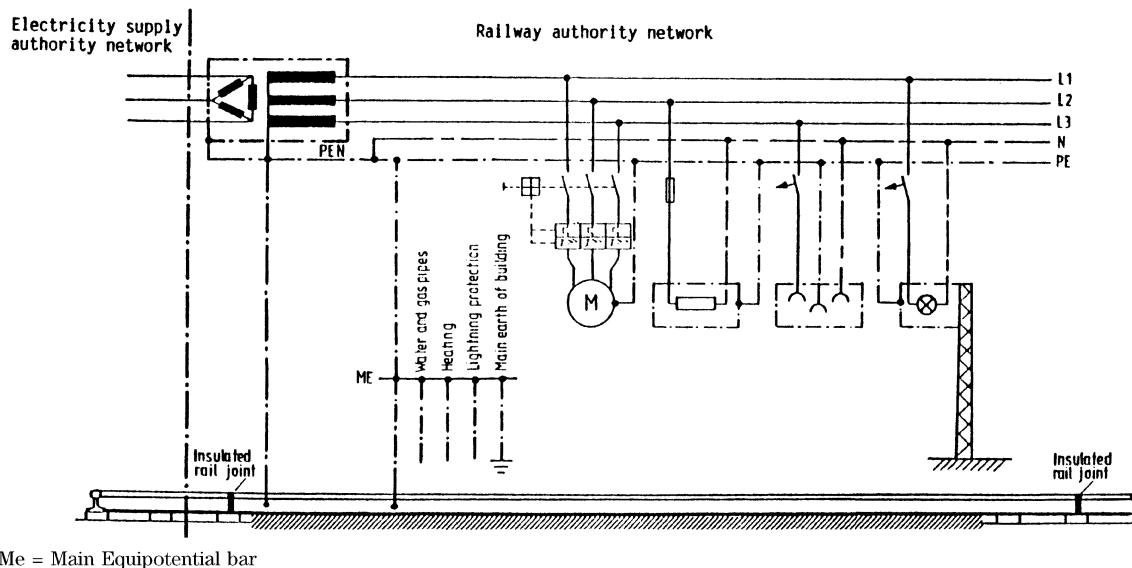


Figure 30 — Example for protective provisions of exposed conductive parts for d.c. traction systems outside the overhead contact line zone or the pantograph zone if the running rails are directly connected to earth or to foundations of buildings (see 6.2.4.3.1.2)

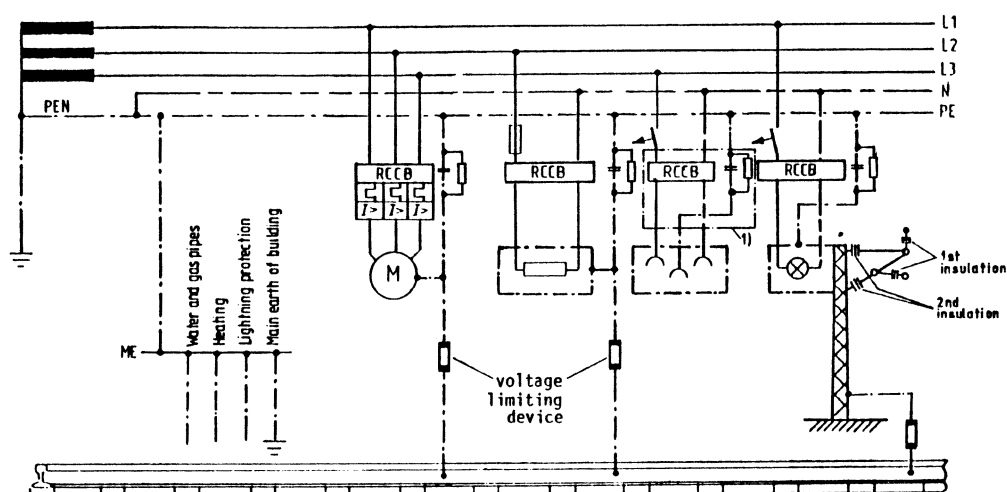
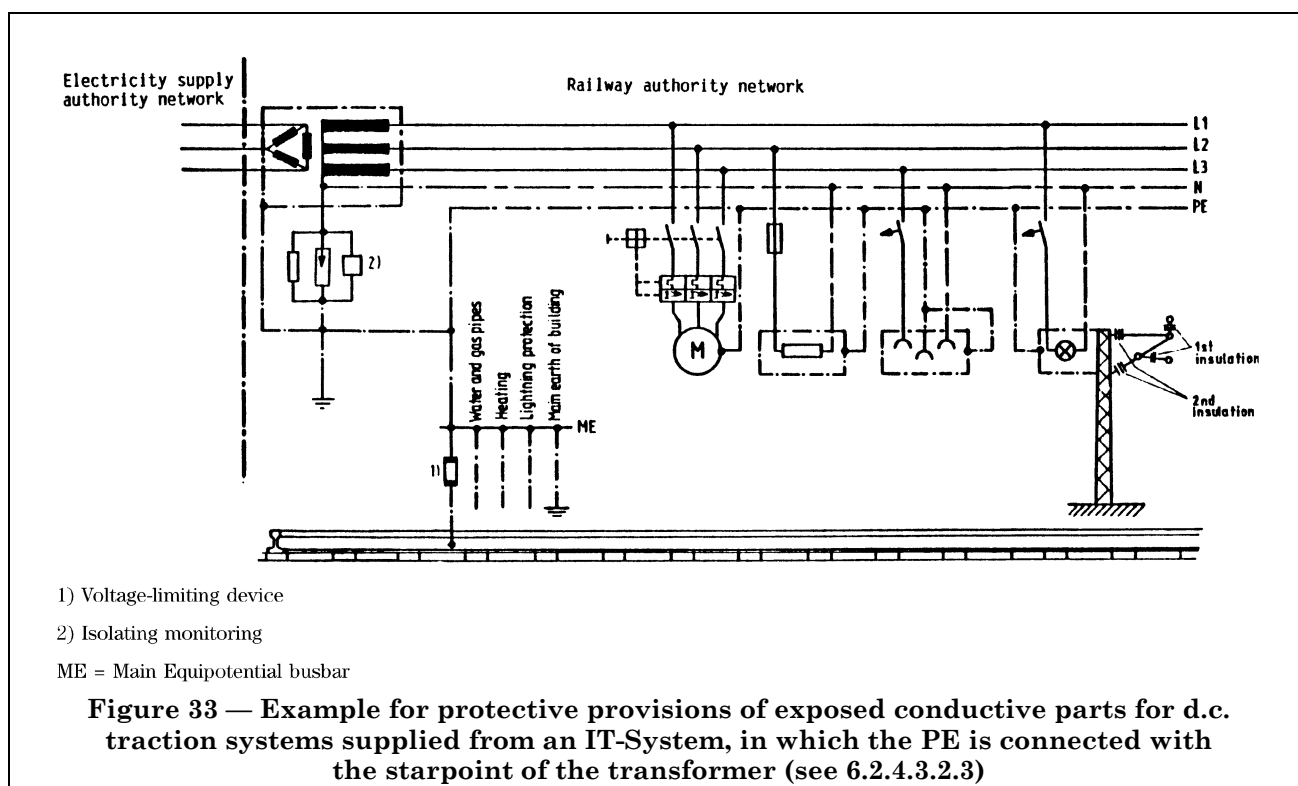
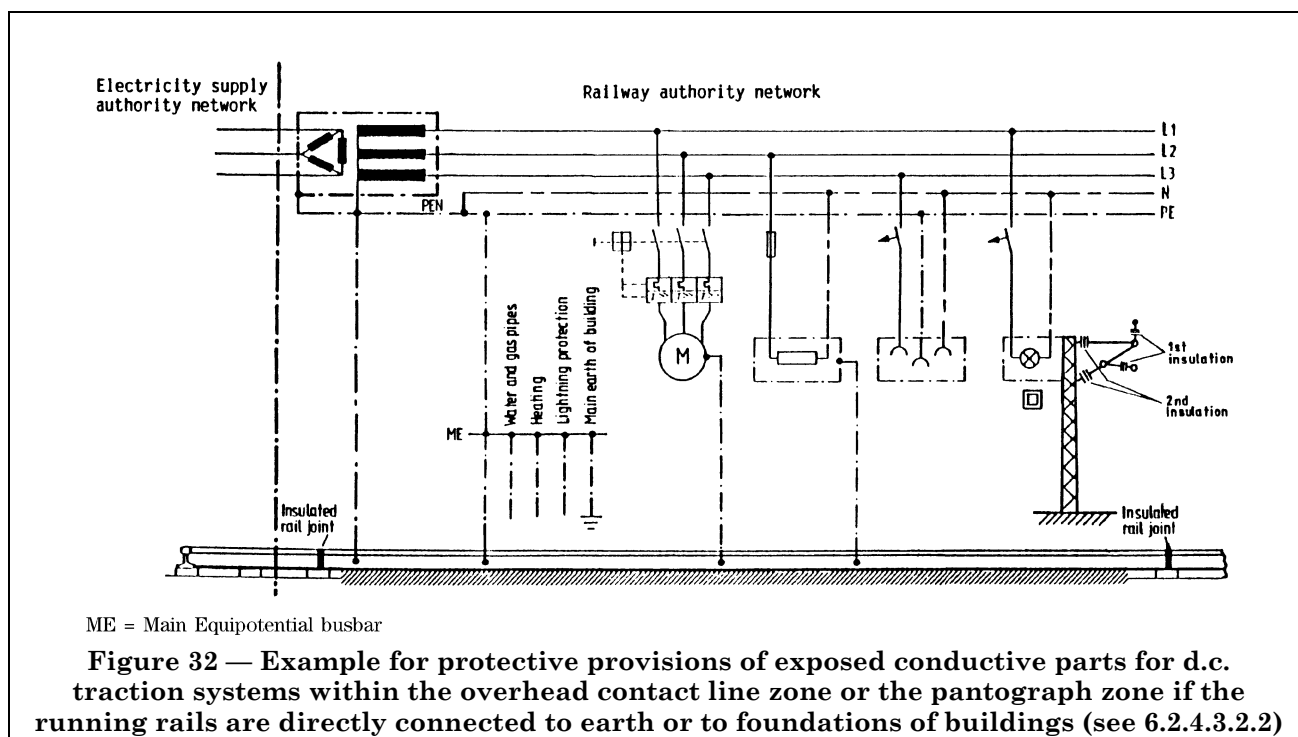


Figure 31 — Example for protective provisions of exposed conductive parts for d.c. traction systems supplied from TN system within the overhead contact line zone or the pantograph zone if the running rails are insulated from earth (see 6.2.4.3.2.1)



6.2.4.3.2.3 Power supply from IT system

For certain cases where continuity of supply is essential for safety critical reasons, then an IT-system shall be provided. An example is shown in Figure 33. The main equipotential busbar (ME) shall be connected with the running rails by a voltage-limiting device.

6.3 Protective provisions for cables against the danger of the traction power supply system

6.3.1 Multiple connection of metallic sheaths, armour or shields of power cables to the traction system earth is permissible only where an inadmissible temperature rise cannot occur as a result of such earthing, if multiple connection of metallic sheaths, armour or shields of power cable to the traction system earth is necessary, it shall be ensured that the metallic sheaths, armour or shields of power cables do not carry inadmissible high traction return currents.

6.3.2 If metallic sheaths, armour or shields of cables are connected to the traction system earth of a.c. traction systems, the connection shall only be permissible where the metallic sheaths, armour or shields are not connected to another earthing system and:

- a) either these are connected only once to the traction system earth or;
- b) where multiple connection to the traction system earth is utilized and at least two tracks exist, the connection shall only be made at the location where all running rails are cross-bonded.

6.3.3 The metallic sheaths, armour or shields of single-phase a.c. feeder cables shall be connected to the traction system earth. Where necessary, special provisions shall be adopted to ensure that an inadmissible temperature rise does not occur in the metallic sheaths, armour or shields due to induction with the supply current as a result of earthing at both ends or an inadmissible high voltage as a result of earthing at one end only.

6.3.4 The metallic sheaths, armour or shields of d.c. feeder cables may be connected to the return circuit only where they are insulated with respect to earth by means of an electrically insulating sheath. The cables shall be provided with protection against an inadmissible high touch voltage. They shall be tripped on occurrence of an inadmissible high touch voltage on the metallic sheaths, armour or shields in the event of a fault or on occurrence of an excessively high current under fault conditions.

7 Protection against the danger of rail potential

7.1 General

7.1.1 The local and time-dependent rail potential leads to touch voltages in fault conditions or accessible voltages in operating conditions.

The values given in 7 shall be the maximum values considered acceptable. All voltages (including d.c. voltages) given in the following tables are rms- values, over the considered interval.

For voltages which vary significantly, particular care shall be given to the choice of the time interval so as to ensure that the most unfavourable condition has been considered.

NOTE The values given in the tables in 7 are based on the assumption, that the current path through the human body is from hand to both feet. For maintenance purposes it may be necessary to consider more dangerous currents paths through the human body. Additional protective provisions may then be necessary.

7.1.2 Attention is directed to the problem that may occur if a vehicle is touched. In this case the voltage, which is applied to a person touching the vehicle can be greater than the corresponding accessible voltage measured on the running rails, due to the impedance between the vehicle body and the running rails (conductor). Therefore all values for accessible voltages given in 7 shall apply inclusive of the voltage due to the impedance referred to in EN 50153 subclause 6.4.3.

7.1.3 For a relevant fault duration, the correct operation of the protection devices and switches are taken into account.

NOTE For all tables in 7 the values for intermediate durations can be calculated by linear interpolation.

7.1.4 The permissible touch/accessible voltages are distinguished according to the time duration shown in Table 1. The short time conditions shall apply only for touch voltages.

Table 1 — Time duration conditions

Conditions	<i>t</i>
Short time	≤ 0,5
Temporary	0,5 < <i>t</i> ≤ 300
Permanent	> 300

where:

t is the time duration of current flow in s.

NOTE 1 The calculation of the values in the following tables are shown in Annex D.

NOTE 2 Measurement methods for touch/accessible voltages are given in Annex E.

7.2 A.C. traction systems

7.2.1 Short time conditions

The touch voltages shall not exceed the values shown in Table 2.

Table 2 — Maximum permissible touch voltages U_t in a.c. traction systems as a function of short time conditions

t	U_t
0,02	940
0,05	935
0,1	842
0,2	670
0,3	497
0,4	305
0,5	225

where:

t is the time duration of current flow in s;

U_t is the touch voltage in V (rms).

7.2.2 Temporary conditions

The accessible voltages shall not exceed the values shown in Table 3.

Table 3 — Maximum permissible accessible voltages U_a in a.c. traction systems as a function of temporary conditions

t	U_a
0,6	160
0,7	130
0,8	110
0,9	90
1,0	80
≤ 300	65

where:

t is the time duration of current flow in s;

U_a is the accessible voltage in V(rms).

7.2.3 Permanent conditions

The accessible voltages shall not exceed 60 V(rms) except in workshops and similar locations where the limit shall be 25 V(rms).

7.2.4 In order to determine whether an inadmissible high touch/accessible voltage can occur, the rail potential at the relevant point shall be evaluated both for normal operation and for fault conditions.

7.2.5 If the rail potential is determined by calculation, such calculation shall employ the maximum operating current and short-circuit current flowing in the running rail and also the initial short-circuit current shall be taken for the short-circuit calculation (Guide values for the specific rail potential can be found in Annex C).

If the guide values shown in Figure C.1 in conjunction with the potential decrease, shown in Figure C.2, lead to a higher touch/accessible voltage than given in Table 2 or Table 3 or in **7.2.3**, or if the actual conditions are different to those to which the guide values refer, then further detailed investigation is necessary. This investigation shall be confirmed by measurements.

7.2.6 In tunnels, if an additional return conductor is used, it shall be of adequate cross-section and shall be longitudinally interconnected and connected to the running rails at appropriate intervals. If the reinforcing rods are within the overhead contact line zone or the pantograph zone, national safety regulations shall apply as necessary.

If reinforcing rods are connected in sections (with gaps along the length), each section shall be connected to the running rails or each section shall be connected together and then connected to the running rails.

In dimensioning the reinforcing rods or additional return conductors utilized for carrying traction return current, the following factors shall be taken into account:

- the expected short-circuit current;
- the expected traction return current component taking the skin effect into account, if relevant;
- the geometrical arrangement in the tunnel (electromagnetic coupling).

7.2.7 If the values given in Table 2 or Table 3 or in **7.2.3** are exceeded, the following provisions shall be considered to reduce the touch/accessible voltage directly or by means of a rail potential reduction:

- reduction of the rail to earth resistance, e.g. by means of additional earth electrodes;
- equipotential bonding;
- reinforcement of the return circuit taking electromagnetic coupling into account;
- voltage-limiting device;
- insulation of the standing surface;
- potential grading by means of appropriate surface earth electrodes;
- reduction of the tripping time needed to interrupt the short circuit-current.

7.3 D.C. traction systems

7.3.1 Short time conditions

The touch voltages shall not exceed the values shown in Table 4.

Table 4 — Maximum permissible touch voltages U_t in d.c. traction systems as a function of short time conditions

t	U_t
0,02	940
0,05	770
0,1	660
0,2	535
0,3	480
0,4	435
0,5	395

where:

- t is the time duration of current flow in s;
- U_t is the accessible voltage in V.

7.3.2 Temporary conditions

7.3.2.1 The accessible voltages shall not exceed the values shown in Table 5.

Table 5 — Maximum permissible accessible voltages U_a in d.c. traction systems as a function of temporary conditions

t	U_a
0,6	310
0,7	270
0,8	240
0,9	200
1,0	170
≤ 300	150

where:

- t is the time duration of current flow in s;
- U_a is the accessible voltage in V.

7.3.2.2 For workshops and similar locations **7.3.3** shall apply.

7.3.3 Permanent conditions

The accessible voltages shall not exceed 120 V except in workshops and similar locations where the limit shall be 60 V.

7.3.4 In order to determine whether an inadmissible high touch/accessible voltage can occur, the rail potential at the relevant point shall be evaluated both for normal operation and for fault conditions on the basis of the voltage drop in the return conductors.

7.3.5 If the rail potential is determined by calculation, such calculation shall employ the maximum operating current and short-circuit current flowing in the running rail and also the initial short-circuit shall be taken for short-circuit calculation (Guide values for the specific rail potential can be found in Annex C).

If the guide values shown in the Figure C.3 or Figure C.4 in conjunction with the potential decrease calculated according **C.1.2** lead to a higher touch/accessible voltage than given in Table 4 or Table 5 or in **7.3.3** or if the actual conditions are different to those to which the guide values refer, then further detailed investigation is necessary. This investigation shall be confirmed by measurements.

7.3.6 In tunnels the reinforcing rods shall not be connected directly to the running rails. A connection may be made to the tunnel earth.

NOTE Provisions may be needed to avoid any inadmissible voltage between the reinforcing rods and the running rails (see Table 4 and Table 5) by voltage-limiting devices.

7.3.7 If the values given in Table 4 or Table 5 or in **7.3.3** are exceeded, the following provisions shall be considered to reduce the touch/accessible voltage directly or by means of a rail potential reduction:

- improvement of the earthing provided in **4.2.2.2** and **5.2.2.2** of the conductive structures, in order to reduce the rail potential in fault conditions;
- voltage-limiting device;
- reinforcement of the return conductor;
- insulation of the standing surface;
- potential grading by means of appropriate surface earth electrodes;
- reduction of the tripping time needed to interrupt the short-circuit current.

8 Substations and switching stations

Substations and switching stations shall conform to the Standard EN 50179¹⁰⁾ for all matters concerning electrical safety and earthing. This shall apply also with respect to the presence of a.c. or d.c. electric systems relating to electrical traction, with the following additions:

8.1 The permissible touch/accessible voltages caused by the traction system under fault conditions or in operating conditions shall not exceed values shown in Table 2 to Table 5 or given in **7.2.3** or **7.3.3**.

¹⁰⁾ In preparation

8.2 In a.c. traction systems a direct connection shall be made between the return circuit and the earthing system of the substation or the switching station. In d.c. traction systems such a direct connection shall not be applied in order to minimize stray currents, except where safety considerations require direct earthing.

8.3 For insulation co-ordination the requirements given in EN 50124-1¹¹⁾ shall apply.

8.4 Circuit breakers in switchgear installations for contact lines shall be installed:

- only in one conductor of each supply circuit in switchgear installations;
- in the connection with the overhead contact line or the conductor rail;
- in the insulated conductor, if one conductor is not intentionally insulated.

9 Return current circuits and earthing conductors¹²⁾

9.1 Each substation shall be connected to the running rails, return conductors or return conductor rails by at least two return cables, either directly or via impedance bonds. The return cables shall be sufficient for the maximum load current, thereby allowing for the failure of one return cable. Fuses, non lockable switches and joint straps which can be released without a tool shall not be installed in the return circuit.

Whenever a switch is installed in the return circuit, another switch shall be installed in the supply circuit, and the return circuit switch shall be interlocked so that it cannot be opened before the supply switch is open.

9.2 Rail joints in the running rails used for carrying traction return current shall be provided with rail joint bonds. This also applies to the components of crossings and points. In the case of a.c. traction systems above 1 kV, the tight joint fish-plates generally provide an adequate conducting bridge.

In the case of track circuits for which rail joint bonds are impossible for technical reasons, the traction return current shall, where necessary, be returned by means of impedance bonds.

9.3 Track-to-track cross bonding shall be provided at suitable intervals in order to ensure the continuity of the traction return current path, thereby enabling the safety criteria for touch or accessible voltage to be achieved, under both normal operating and fault conditions.

The specific arrangements for connections of track-to-track cross bonding shall take account of the track-circuiting arrangements in use and the overall traction power supply system design.

NOTE 1 In areas without track circuits only, the running rails of each track may be connected by rail-to-rail cross bonds, as required.

NOTE 2 Steel sleepers or riveted or screwed rail tie bars between the running rails provide sufficient transverse connection if the running rails are secured to them without any intermediate insulating layers.

9.4 At interruptions in tracks, for example at movable bridges or track weighbridges, conductors shall be arranged so as to ensure that the traction return current is not interrupted.

9.5 All the above conductors shall be designed in such a way that they satisfy the thermal loads which can be produced by the traction return current in the case of normal operations or short-circuit. For mechanical reasons they shall have a minimum cross-section of 50 mm².

9.6 All earthing conductors and earthing electrodes shall be designed and constructed as required in EN 50179, clauses **9.2.2.1** to **9.2.2.3**¹¹⁾. This applies in particular to materials choice and to dimensioning, in consideration of mechanical strength, and thermal effects of corrosion.

In d.c. traction systems the thermal dimensioning of such conductors shall be based on the value of the expected prospective short-circuit current at the location.

10 Means of achieving safe isolation

10.1 For a system with a conductor rail safe emergency, isolation should be achieved by means of a portable short-circuiting device which causes the circuit breakers to trip and maintains the shorting connection. The device shall be capable of being applied at any point on the track. Alternative means may be considered.

10.2 For cases of emergency at overhead contact lines an effective procedure shall be established for tripping the circuit breakers subsequently ensuring that the overhead contact line cannot be accidentally re-energized.

10.3 If section insulators are used as isolating gaps in overhead contact lines to isolate feeding sections, the minimum clearance in air may be reduced. A smaller clearance than the normal minimum clearance in air is allowable, such that for safety reasons, the reduced clearances shall not be less than:

- 100 mm in 15 kV traction systems;

¹¹⁾ In preparation.

¹²⁾ See also EN 50122-2 (in preparation).

— 150 mm in 25 kV traction systems.

NOTE Section insulators are inserted between the contact wires and they have contact with the pantograph. For this reason, their mechanical design must meet the same requirements as specified for the contact wire dynamics. These requirements impose limits concerning the possible clearances between different live parts, between live parts and earthed parts and the size and the weight. Because of that, it is necessary to allow these smaller distances.

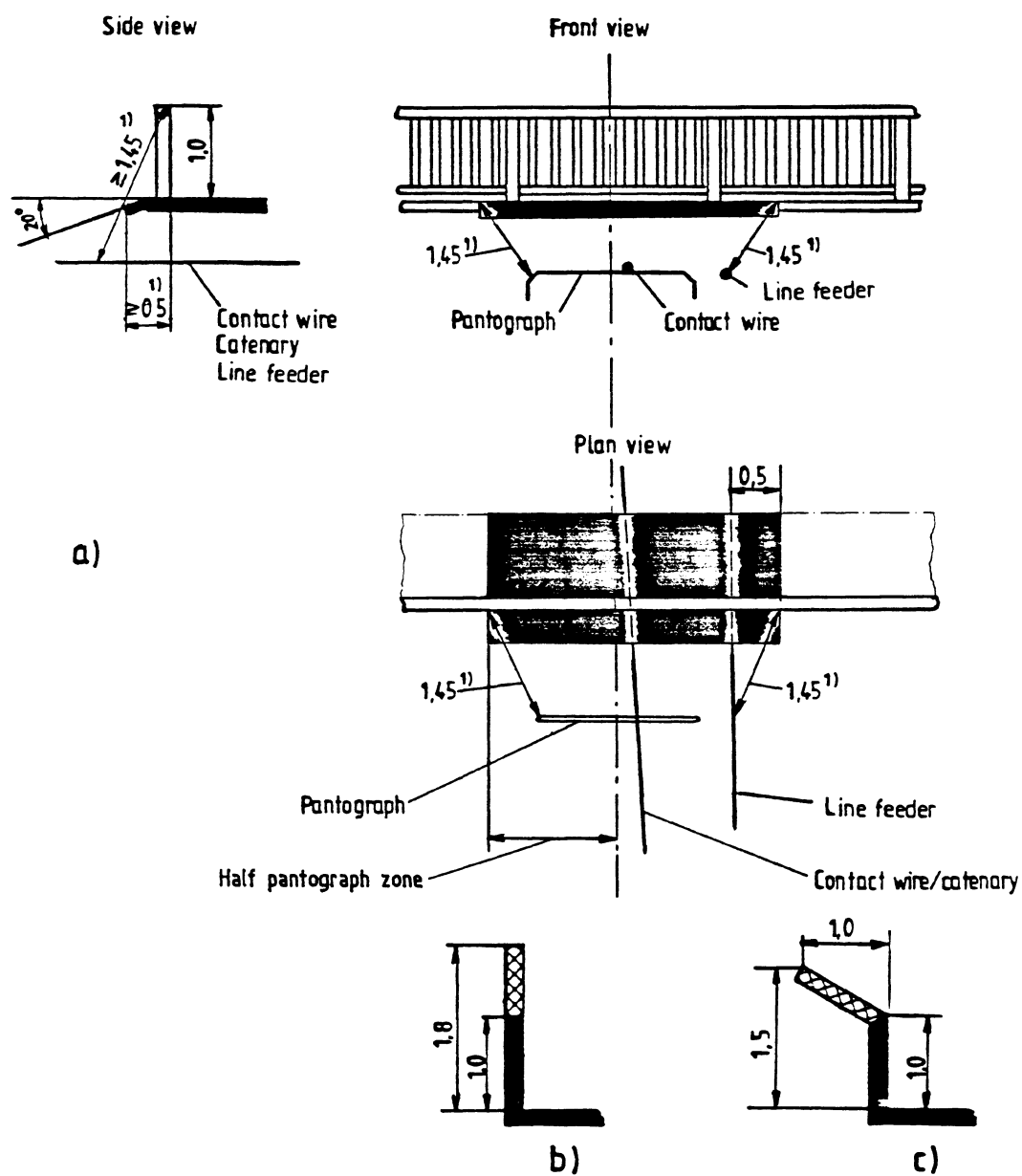
10.4 In overhead contact line systems, isolating gaps may be bridged by insulators without provisions against leakage currents from the terminal on one side to the terminal on the other side if regulations impose that all maintenance work on one side of the overhead contact line is performed only on a section of line, which is earthed or bonded to the running rails, as required.

11 Removing of decommissioned overhead contact lines

For a track which is permanently decommissioned the overhead contact line and associated feeders shall be removed.

Annex A (informative)

Typical obstacles



█ Solid-wall design or obstacle conforming to degree of protection IP2X as defined in EN 60529

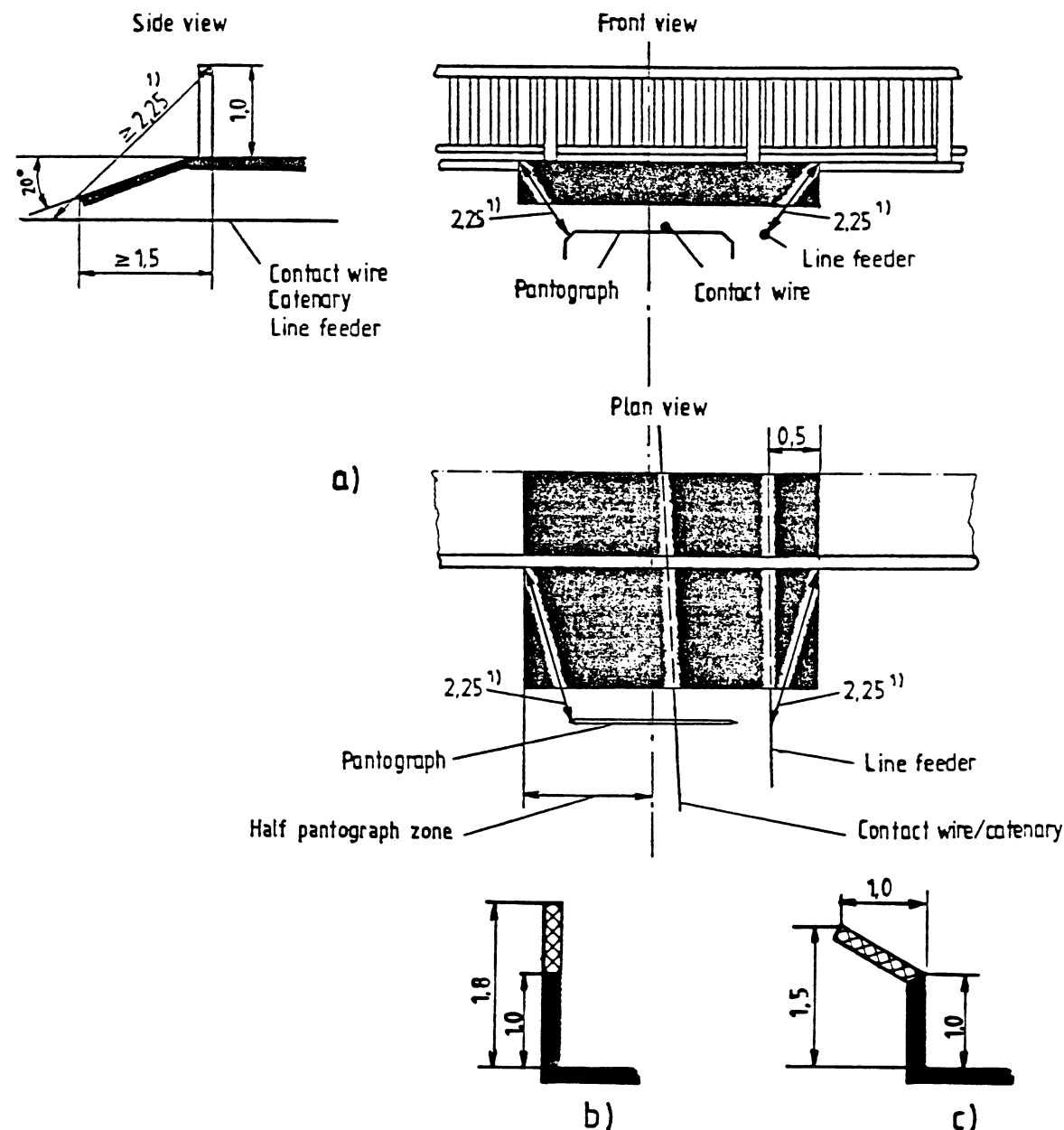
▤ Mesh screen with a maximum 1200 mm² mesh size (can also be solid-wall design)

▬ Railing, mesh construction (can also be solid-wall design)

1) The dimension is taken from figure 2 in 4.1.2.1.

All dimensions are minimum dimensions in m.

Figure A.1 — Examples of obstacles along the sides of standing surfaces in public areas for protection against direct contact when above live parts on the outside of vehicles or live parts of an overhead contact line system for nominal voltages up to and including 1 kV a.c./1,5 kV d.c. (see 4.1.3.3.2)



■ Solid-wall design

▨ Mesh screen with a maximum 1200 mm² mesh size (can also be solid-wall design)

□ Railing, mesh construction (can also be solid-wall design)

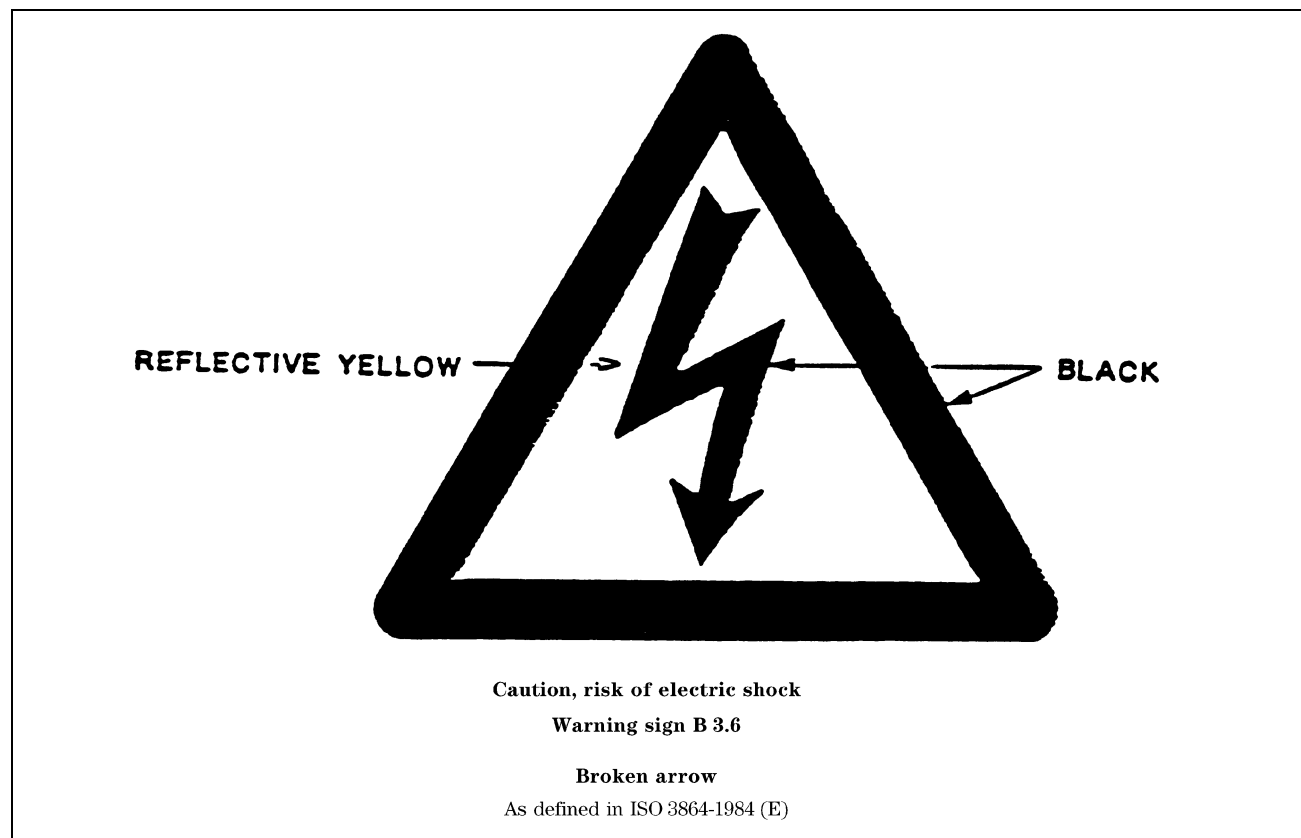
1) The dimension is taken from figure 14 in 5.1.2.1.

All dimensions are minimum dimensions in m.

Figure A.2 — Examples of obstacles along the sides of standing surfaces in public areas for protection against direct contact when above live parts on the outside of vehicles or live parts of an overhead contact line system for nominal voltages in excess of 1 kV a.c./1,5 kV d.c. up to 25 kV a.c. or d.c. (see 5.1.3.3.2)

Annex B (normative)

Warning sign



Annex C (informative)

Guide values for specific rail potential

C.1 A.C. traction systems

C.1.1 Guide values for the specific rail potential related to the traction return current in relation to the conductance per unit length are shown in Figure C.1; these are referred to the following conditions:

- double track system, four rails connected in parallel;
- 100 A current injected in one of the two tracks;
- average feeder length;
- average soil resistivity ($40 \Omega \cdot \text{m}$ to $200 \Omega \cdot \text{m}$).

NOTE 1 Empirical values for the conductance per unit length G' for double-track sections are:

- on-surface sections with concrete sleepers 0,4 S/km to 1,7 S/km;
- on-surface sections with wood/steel sleepers 1,7 S/km to 7,0 S/km;
- tunnel sections with connection between tracks and reinforcing rods 7,0 S/km to 15,0 S/km.

NOTE 2 These values could be increased to take into account the neighbouring parts connected to the traction system earth and to multi-track section in railway stations.

C.1.2 The value at which the rail potential for a.c. traction systems, given in 7.2 can act as an accessible voltage should be investigated. Guide values for the potential decrease measured at right angle away from the track of a.c. traction systems, where the running rails are directly earthed (traction system earth), are given in Figure C.2 for average soil earthing resistance.

C.2 D.C. traction systems

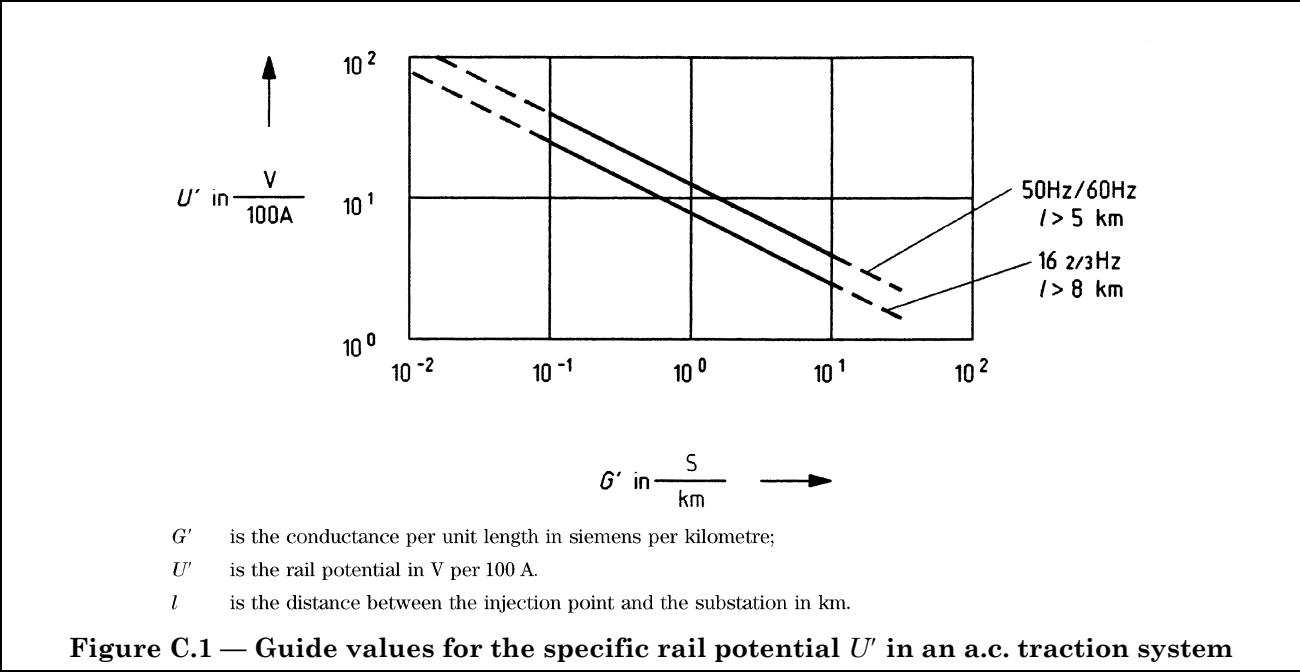
C.2.1 Guide values for the specific rail potential related to the traction return current in relation to the conductance per unit length and for different distances from the injection point to the substation, are shown in Figure C.3; these are referred to the following conditions:

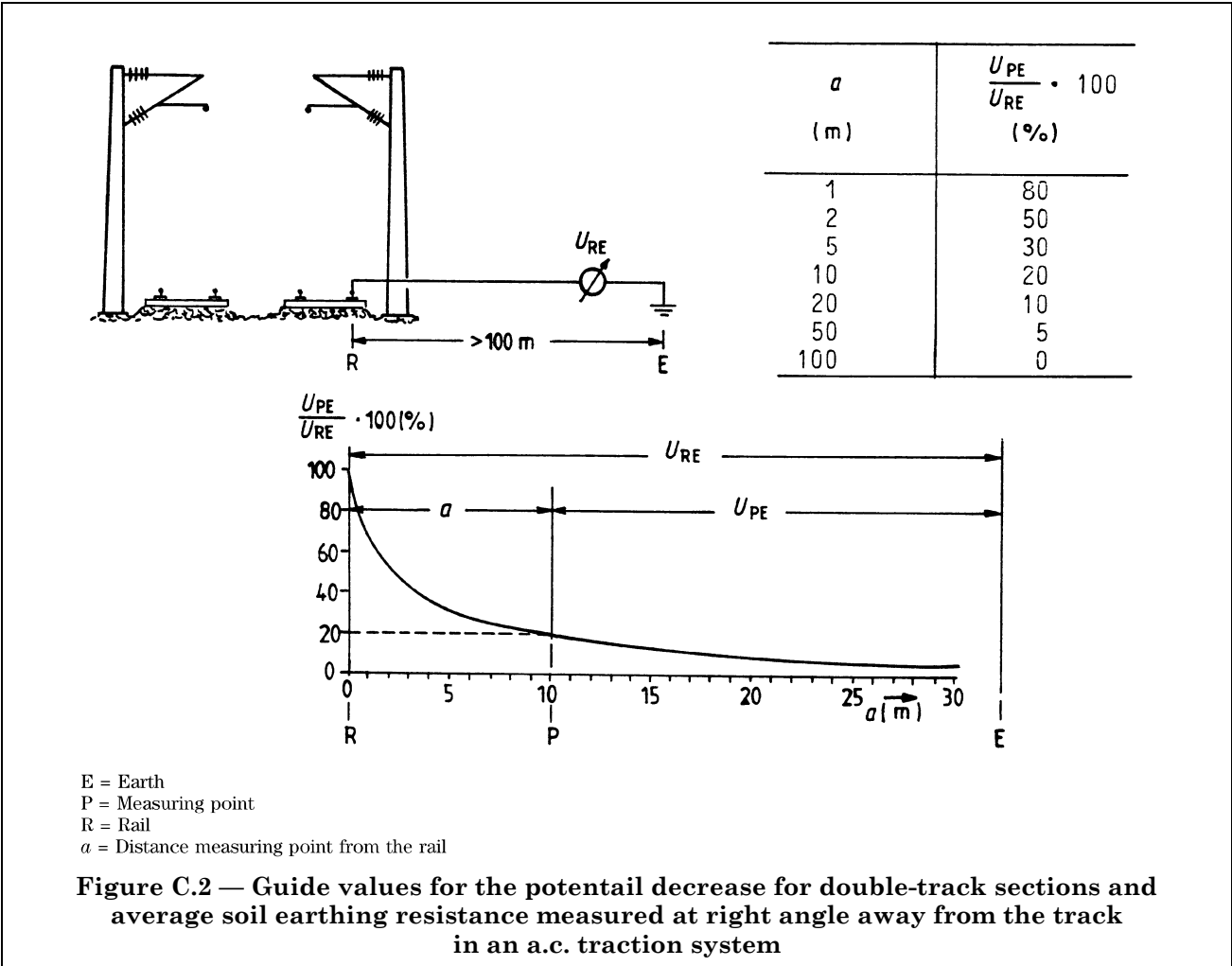
- double track system, four rails connected in parallel;
- 100 A current injected in one of the two tracks;

— 50 kg/m to 60 kg/m rails, welded or efficiently bonded.

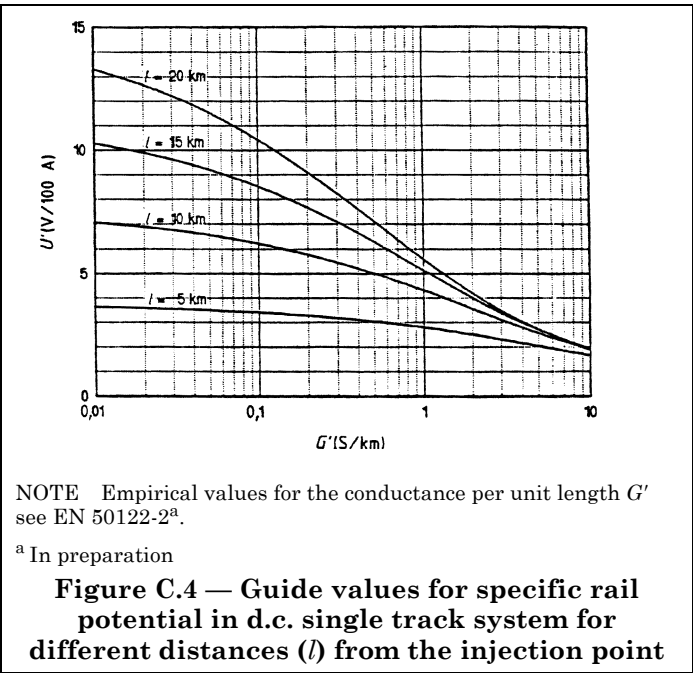
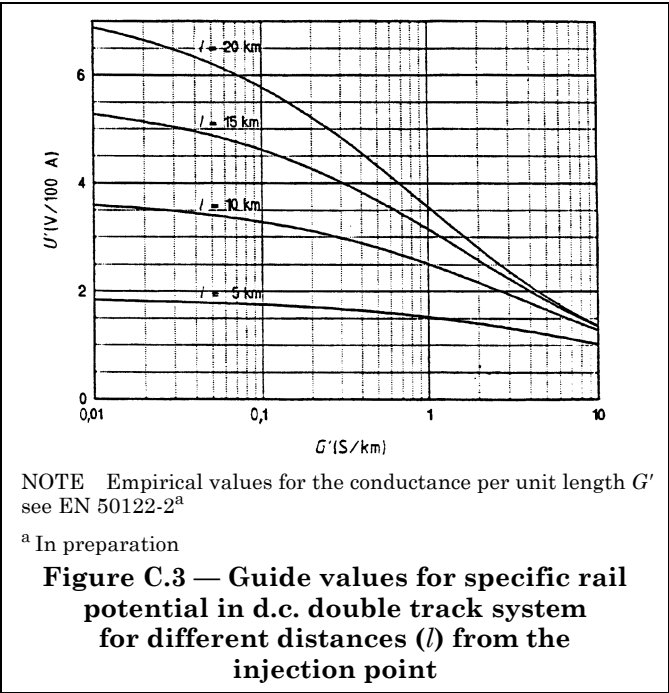
Where a track is fed from two substations operating in parallel, the rail potential at the injection point is the sum of the two potentials corresponding to the current supplied by each substation.

The graphs shown in Figure C.4 can be applied to a single track system.





C.2.2 The value at which the rail potential for d.c. traction systems, given in 7.3 can act as an accessible voltage should be investigated. Guide values for the potential decrease are the same as for a.c. traction case (see Figure C.2) in the same conditions. If as it is normally the case, the running rails are not directly earthed, the effect of the rail to earth resistance has to be taken into account.



Annex D (informative)
Touch/accessible voltage and body current

D.1 The calculation of the values for the permissible touch/accessible voltage in 7.2 and 7.3 are based on:

- IEC-Publication 479-1:1994 “Effects of current passing through the human body”;
- EN 50179 “Power installations exceeding 1 kV a.c.”¹³⁾

¹³⁾ In preparation

The following assumptions are made:

- current path: one hand to both feet;
- 50 % probability of body impedance higher than assumed value;
- 0 % probability of ventricular fibrillation (see curve c_1 IEC- Publication 479-1:1994;
- additional resistance $R_a = 1\,000\ \Omega$ for old wet shoes in short time conditions.

NOTE Proceeding on the assumption that in most cases there is an additional resistance to the total body resistance because of shoes and moreover that the probability of danger is very small within time intervals of less than 1 s, an additional resistance of $1\,000\ \Omega$ (this value of old wet shoes corresponds with EN 50179) was taken into account in the following way to calculate the values:

- for time intervals up to 0,5 s an additional resistance of $1\,000\ \Omega$ was added to the values of U_{c1} (curve c_1);
- for time intervals longer than 1,0 s and up to 300 s the curve c_1 of IEC 479-1 without any additional resistance was taken into account;
- for time intervals between 0,5 s and 1,0 s intermediate values by interpolation between the value U_t ($t = 0,5$ s) and U_{c1} ($t = 1,0$ s) were used.

The resistance of the standing surface can be taken into account, according to EN 50179¹³⁾ additionally at any time interval.

D.2 The total human body impedance at 50 % probability is indicated in IEC 479-1:1994 Table 1, relating hand to hand current path. By applying the reduction factor 0,75 for the hand to feet current path, the following table was obtained:

Table D.1 — Body impedance Z and body current I_B

U_t	a.c. system			d.c. system		
	$Z(100)$	$Z(75)$	$I_B(75)$	$R(100)$	$R(75)$	$I_B(75)$
25	3 250	2 438	10	3 875	2 906	9
50	2 625	1 969	25	2 990	2 243	22
75	2 200	1 650	45	2 470	1 853	40
100	1 875	1 406	71	2 070	1 553	64
125	1 625	1 219	103	1 750	1 313	95
220	1 350	1 013	217	1 350	1 013	217
700	1 100	825	848	1 100	825	848
1 000	1 050	788	1 270	1 050	788	1 270

where

$$I_{B(75)} = \frac{U_t}{Z(75)} \times 10^3 \text{ or } \frac{U_t}{R(75)} \times 10^3 \text{ is the body current in mA,}$$

U_t is touch voltage in V (rms);

$Z(100)$ is the total body impedance in Ω ;

$Z(75)$ is 75 % of the total body impedance in Ω ;

$R(100)$ is the total body resistance in Ω ;

$R(75)$ is 75 % of the total body resistance in Ω .

D.3 In the Figure D.1 and Figure D.2 there are developed the:

- current/time curve c_1 shown in IEC-Publication 479-1:1994;
- touch voltage $U_t = f(I_B)$ obtained from the values in Table D.1.

D.3.1 A.C. traction systems

On the basis of the curves in Figure D.1 the following Table D.2 was obtained:

Table D.2 — Body currents and touch/accessible voltages as a function of the time duration in a.c. traction systems

t	I_{c1}	U_{c1}	U_t	U_i
0,02	500	440	940	—
0,05	500	435	935	—
0,1	442	400	842	—
0,2	350	320	670	—
0,3	252	245	497	—
0,4	145	160	305	—
0,5	100	125	225	—
0,6	78	105	—	160
0,7	66	95	—	130
0,8	58	90	—	110
0,9	52	85	—	90
1,0	50	80	—	—

$$U_t = U_{c1} + R_A \times I_{c1} \times 10^{-3}$$

Where

I_{c1} is the current in mA which corresponds to c_1 in IEC-Publication 479-1;

R_A 1 000 Ω is the resistance for old wet shoes;

t is the time duration of current flow in s;

U_{c1} is the accessible voltage in V(rms) and corresponds to I_{c1} (see Figure D.1);

U_t is the touch voltage in V(rms) for short time conditions for $0,02 \text{ s} \leq t \leq 0,5 \text{ s}$.

U_i values for the accessible voltage in V(rms), which are interpolated between the values U_t ($t = 0,5 \text{ s}$) and U_{c1} ($t =$

D.3.2 D.C. traction systems

On the basis of the curves in Figure D.2 the following Table D.3 was obtained:

Table D.3 — Body currents and touch/accessible voltages as function of time duration in d.c. traction system

t	I_{c1}	U_{c1}	U_t	U_i
0,02	500	440	940	—
0,05	410	360	770	—
0,1	340	320	660	—
0,2	275	260	535	—
0,3	240	240	480	—
0,4	215	220	435	—
0,5	195	200	395	—
0,6	180	190	—	310
0,7	175	185	—	270
0,8	165	180	—	240
0,9	160	175	—	200
1,0	150	170	—	—

$$U_t = U_{c1} + R_A \times I_{c1} \times 10^{-3}$$

Where

I_{c1} is the current in mA which corresponds to curve c_1 in IEC-Publication 479-1;

R_A = 1 000 Ω is the resistance for old wet shoes;

t is the time duration of current flow in s;

- U_{c1} is the accessible voltage in V and corresponds to I_{c1} (see Figure D.2);
 U_t is the touch voltage in V for short time conditions for $0,02 \text{ s} \leq t \leq 0,5 \text{ s}$
 U_i values for the accessible voltage in V(rms), which are interpolated between the values U_t ($t = 0,5 \text{ s}$) and U_{c1} ($t = 1,0$).

Annex E (normative)

Measurement methods for touch/accessible voltages

The measurement of the touch/accessible voltage shall be carried out as follows:

The touch voltage/accessible voltage shall be measured with a voltmeter having an internal resistance of 1 k Ω .

Each measuring electrode, for the simulation of feet, shall have a total area of 400 cm² and shall be pressed on the earth with a minimum force of 500 N.

Alternatively, a sensor driven 20 cm into the earth, may be used instead of the measuring electrode.

To measure the touch voltages/accessible voltage for concrete or dried up soil a wet cloth or water film shall be placed between the foot electrodes and earth. The foot electrodes shall be placed at a distance not less than 1 m from the exposed conductive part.

A measuring electrode, e.g. a tip electrode, shall be used for simulation of a hand. In this case paint coatings (but not insulations) shall be pierced reliably.

One clamp of the voltmeter shall be connected to the hand-electrode, the other clamp shall be connected to the foot electrode. It is sufficient to carry out such measurements by random checks of an installation.

NOTE In order to get the upper limit of the touch voltages/accessible voltages, the measurement by a voltmeter with a high internal resistance and a 10 cm deep-driven sensor is often suitable.

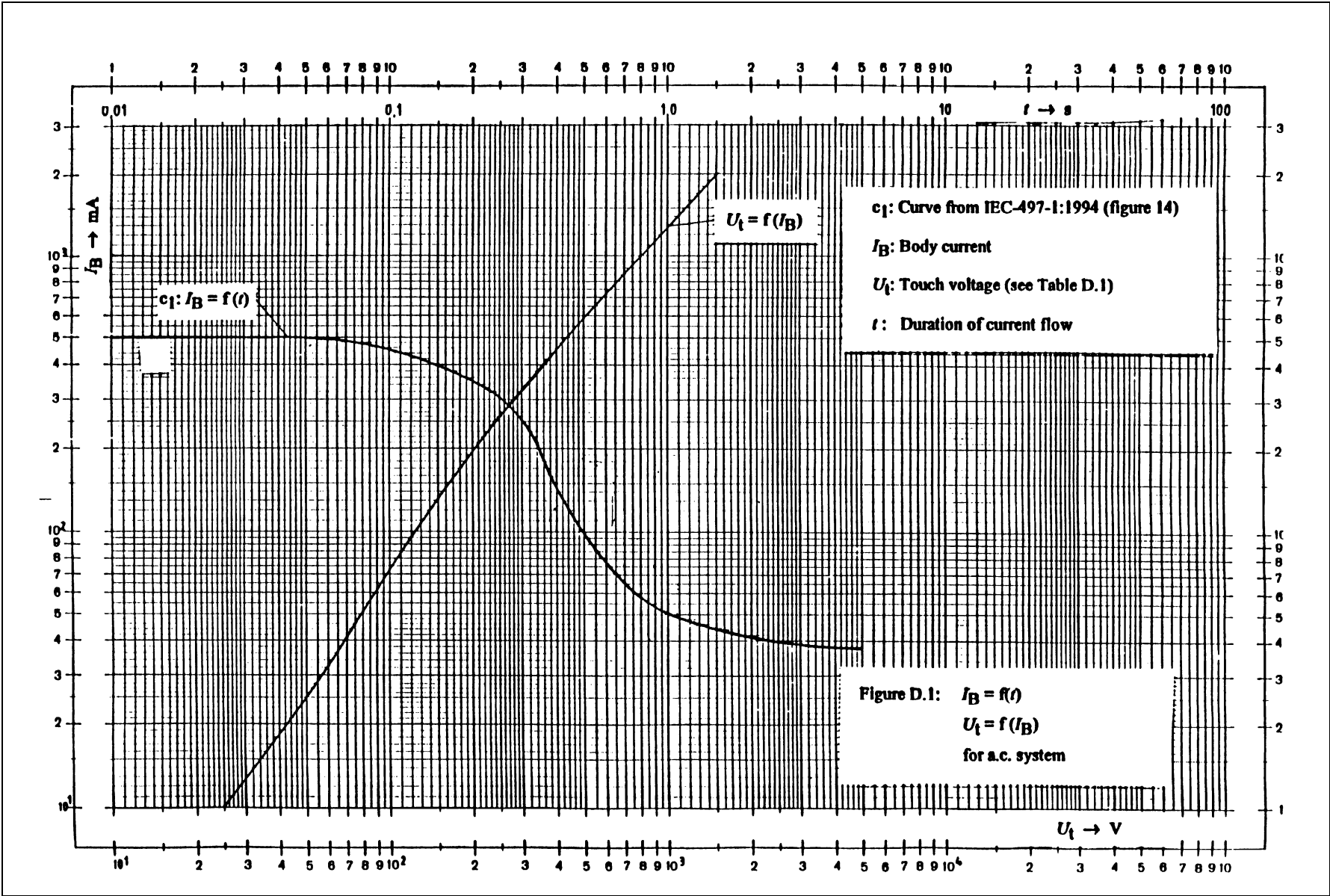
At measuring points at which the resistance to earth of the foot electrode does not exceed several hundred ohms, using a combination of a high and low-resistance measurement device has proved to be successful. If the voltage breaks down considerably during switch over from a high internal resistance to the internal resistance of 1 k Ω , it can be concluded that the earth contact resistances in the measuring circuit are too high.

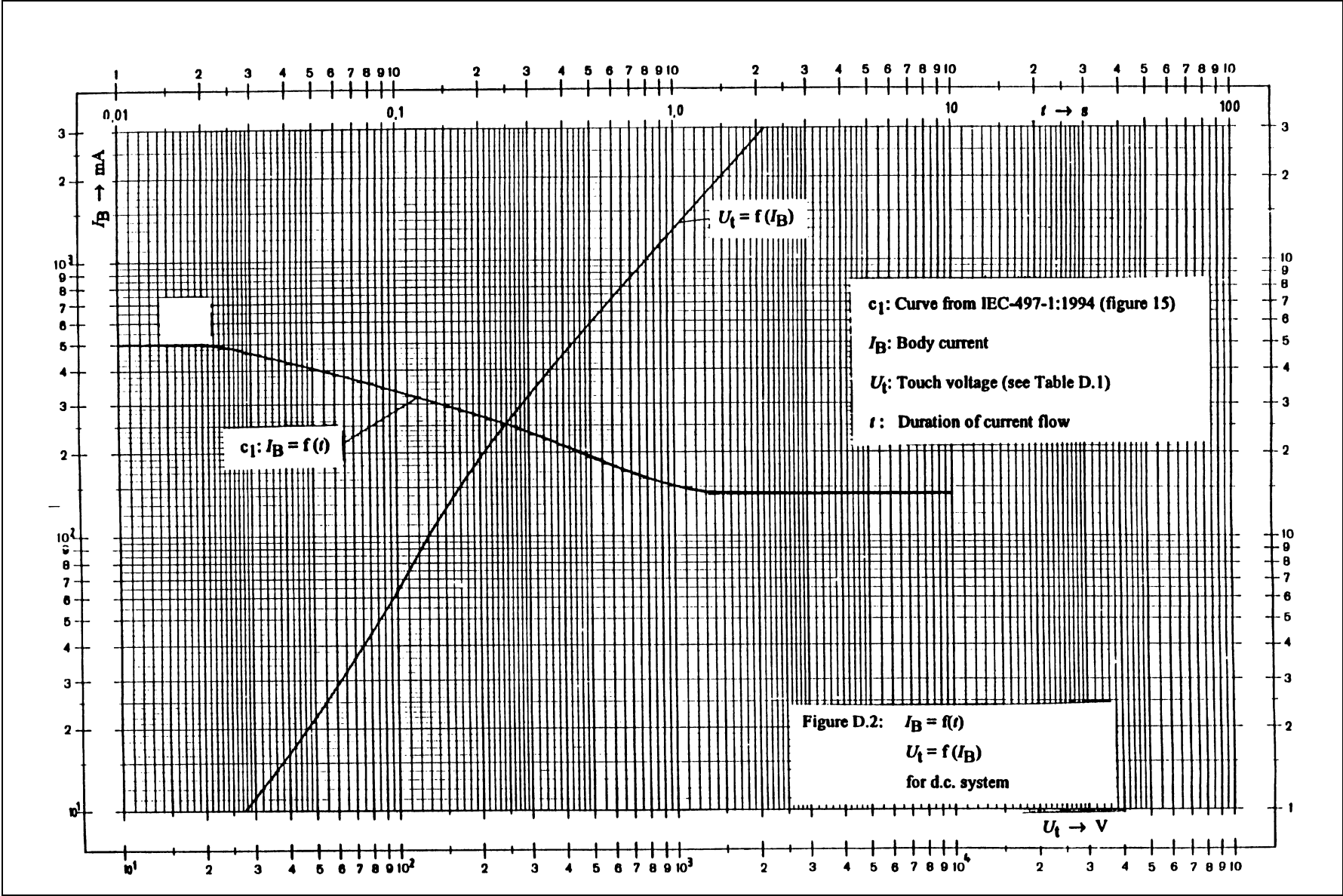
Annex F (informative)

Bibliography

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^a In preparation





Annex G (normative)
Special national conditions

Special national conditions: National characteristic or practice that cannot be changed even over a long period, e.g. climatic conditions, electrical earthing conditions. If it affects harmonization, it forms part of the European Standard.

For the countries in which the relevant special national conditions apply these provisions are normative, for other countries they are informative.

Clause Special national condition

3.3.8 Austria

In the vicinity of 15 kV overhead contact lines the following values will apply for the Figures X, Y and Z:

Figure	Dimensions (in m)
X	4,00
Y	2,00
Z	8,00 - $\frac{HP}{RH}$

4.1.2.3 Austria

Besides the details contained in 4.1.2.3, the terms of ÖVE-T 1 and ÖVE-T 5 will also apply.

5.1.2.1 United Kingdom

The dimension of R3,5 mm clearance in public areas shown in Figure 14 shall be amended to R2,75 minimum for use in the case of future electrification of existing railway lines with restricted infrastructure clearances

5.1.2.3 Austria

Besides the details contained in 4.1.2.3, the terms of ÖVE-T 1 and ÖVE-T 5 will also apply.

6.1.7 Austria

Isolating pieces must withstand for one minute a test voltage equal to two thirds of the nominal voltage of the overhead contact systems without being damaged (ÖVE-T 20).

Annex H (informative)

A-deviation

A-deviation: National deviation due to regulations, the alteration of which is for the time being outside the competence of the CEN/CENELEC member.

This European Standard does not fall under any Directive of the EC.

In the relevant CENELEC countries these A-deviations are valid instead of the provisions of the European Standard until they have been removed.

Clause Deviation

Austria

4.1.2.2 (Electrical Engineering Regulation 1993, BGBl. Nr. 47/1994)

The clearances to be observed for persons working nearby energized overhead contact line systems must be defined in national specifications. If national specifications do not exist, clearances shown in Figure 2 or at least the clearances according to 4.1.3 shall be used.

Figure 2 (Electrical Engineering Regulation 1993, BGBl. Nr. 47/1994) (Employee Protection Law, BGBl. Nr. 450/1994)

Dimensions "Public areas":

above the standing surface:	R 3,00
beside the standing surface:	R 1,75
below the standing surface:	R 2,50

Dimensions "Restricted areas":

above the standing surface:	$R 2,5 + d$
beside the standing surface:	$R 1,4 + d$
below the standing surface:	$R 1,5 + d$

The dimension " d " is determined as described in the Elektrotechnikverordnung 1993.

4.1.3.2.2 (Employee Protection Law, BGBl. Nr. 450/1994)

With standing surfaces adjacent to live parts on the outside of vehicles or adjacent to live parts of a contact line system meshed screens with a mesh size " e " of $20 \text{ mm} < e \leq 30 \text{ mm}$ with a lateral clearance of 0,3 m to 1,3 m having a height of at least 1,8 m and with a lateral clearance of more than 1,3 m having a height of at least 1,4 m may be utilized as obstacles, where the live parts are situated above the standing surface (see Figure 5, example a).

If the live parts are situated at the same height or lower than the standing surface, the height " h " of the obstacle shall be such that the clearance of " $1,40 \text{ m} + d$ " from the top of the obstacle is maintained as shown in Figure 2 (see Figure 5, example b).

The clearance between the obstacle and the live parts shall be at least 0,3 m. If this clearance is not maintained, the obstacles shall be designed in accordance with clause 4.1.3.2.1 or be of solid-wall design and leave no gap between the obstacle and the standing surface (see Figure 5, examples c and d).

Figure 5 (Employee Protection Law, BGBl. Nr. 450/1994)

The distance measurement " $1,35$ " shall be " $1,40 + d$ ".

4.1.3.2.3 (Employee Protection Law, BGBl. Nr. 450/1994)

Obstacles of meshed screen with mesh size " e " of $20 \text{ mm} < e \leq 30 \text{ mm}$ shall be at least provided at the sides of such standing surfaces. The height " h " of the obstacles shall be such that a clearance of " $1,40 \text{ m} + d$ " from the top of the obstacles is maintained as depicted in Figure 2 (see Figure 6, examples a and b). These obstacles shall be at least as long as the length of the standing surface above live parts.

NOTE The height of the side obstacles, if necessary, will generally correspond to the height of the necessary safety railing but must have a height of at least 1,0 m.

Figure 6 (Employee Protection Law, BGBl. Nr. 450/1994)

The distance measurement “1,35” shall be “1,40 + *d*”.

4.1.3.3.1 (Employee Protection Law, BGBl. Nr. 450/1994)

With standing surfaces adjacent to live parts on the outside of vehicles or adjacent to active parts of a contact line system, solid-walled obstacles of minimum height 1,0 m shall be provided and shall leave no gap with respect to the standing surface. Over this, mesh of mesh size “*e*” of 20 mm < *e* ≤ 30 mm with a lateral clearance of 1,0 m to 1,5 m having a height of at least 1,8 m and with a lateral clearance of more than 1,5 m having a height of at least 1,4 m shall be provided (see Figure 12, example a).

If the clearance of the obstacles of 1,0 m to live parts is not maintained, the obstacles shall be of similar solid-wall or Class IP2X as defined in EN 60529 design over their height (see Figure 12, example b).

4.1.3.3.2 (Employee Protection Law, BGBl. Nr. 450/1994)

Any vertical obstacle shall be of solid-wall design or Class IP2X as defined in EN 60529 at least to a height of 1,0 m (see Annex A, Figure A.1). The height of the obstacle shall be designed with a distance of the standing surface from the live part up to 1,5 m by obstacle 1,8 m high in total, up to a distance of more than 1,5 m to 2,0 m by obstacle 1,4 m high in total and at a distance of more than 2,0 m by obstacle 1,0 m high in total.

Figure 14 (Electrical Engineering Regulation 1993, BGBl. No. 47/1994)

(Employee Protection Law, BGBl. Nr. 450/1994)

Dimensions “Public areas”:

above the standing surface:	R 3,50
beside the standing surface:	R 2,25
below the standing surface:	R 3,00

Dimensions “Restricted areas”:

above the standing surface:	R 2,5 + <i>d</i>
beside the standing surface:	R 1,4 + <i>d</i>
below the standing surface:	R 1,5 + <i>d</i>

The dimension “*d*” is determined as described in the Elektrotechnikverordnung 1993.

5.1.2.2 (Electrical Engineering Regulation 1993, BGBl. No. 47/1994)

The clearances to be observed for persons working nearby energized overhead contact line systems shall be defined in national specifications.

If national specifications do not exist, at least the clearances shown in Figure 14 or the clearances according to 5.1.3 shall be used.

5.1.3.2.1 (Employee Protection Law, BGBl. Nr. 450/1994)

With standing surfaces adjacent to live parts on the outside of vehicles or adjacent to live parts of an overhead contact line system meshed screens with a mesh size “*e*” of 20 mm < *e* ≤ 30 mm with a lateral clearance of 0,6 m to 1,3 m having a height “*h*” of at least 1,8 m and with a lateral clearance of more than 1,3 m having a height of at least 1,4 m shall be utilized as obstacles, where the live parts are situated above standing surface (see Figure 16, example a).

If the live parts are situated at same height or lower than the standing surface, the height “*h*” of the obstacle shall be such that a clearance of “1,40 m + *d*” from the top of the obstacle is maintained as shown in Figure 14 (see Figure 16, example b).

The clearance between the obstacle and live parts shall be at least 0,6 m. If this clearance is not maintained, the obstacles shall be of solid-wall design and leave no gap between the obstacle and the standing surface (see Figure 16, examples c and d).

NOTE In addition to the requirements of electrical safety there may be additional requirements for obstacles to provide for security according to national regulations.

5.1.3.2.2 (Employee Protection Law, BGBl. Nr. 450/1994)

Obstacles of meshed screen with a mesh size " e " of $20 \text{ mm} < e \leq 30 \text{ mm}$ shall be provided at the sides of such standing surfaces. The height " h " of the obstacle shall be such that a clearance of " $1,40 \text{ m} + d$ " from the top of the obstacle is maintained as shown in Figure 14 (see Figure 17). These obstacles shall be at least as long as the length of the standing surface above live parts.

NOTE The height of the side obstacles, if necessary, will generally correspond to the height of the necessary safety railing, but must have a height of at least 1,0 m.

Figure 16 (Employee Protection Law, BGBl. Nr. 450/1994)

The distance measurement " $1,5$ " shall be " $1,40 + d$ ".

Figure 17 (Employee Protection Law, BGBl. Nr. 450/1994)

The distance measurement " $1,5$ " shall be " $1,40 + d$ ".

5.1.3.3.1 (Employee Protection Law, BGBl. Nr. 450/1994)

With standing surfaces adjacent to live parts on the outside of vehicles or adjacent to active parts of overhead contact line system, solid-wall obstacles of minimum height 1,0 m shall be provided and shall leave no gap with respect to the standing surface. Above the solid-wall part mesh of mesh size " e " of $20 \text{ mm} < e \leq 30 \text{ mm}$, with a lateral clearance of 1,0 m to 1,5 m having a height of at least 1,8 m in total, with a lateral distance of more than 1,5 m to 2,0 m with a height of 1,4 m in total and with a lateral distance of more than 2,0 m with a height of at least 1,0 m shall be provided. Such mesh shall be arranged so as to ensure that the wall remains unclimbable (see Figure 18, example a).

If the clearance between the obstacles and the live parts of 1,0 m is not maintained, the obstacles shall be of solid-wall design over their full length (see Figure 18, example b).

5.1.3.3.2 (Employee Protection Law, BGBl. Nr. 450/1994)

Any vertical obstacle shall be of solid-wall design at least to a height of 1,0 m (see Annex A, Figure A.2). The height of the obstacle shall be designed at a distance of the standing surface from the live part up to 1,5 m by obstacle 1,8 m high in total, up to a distance of more than 1,5 m to 2,0 m by obstacle 1,4 m high in total and at a distance of more than 2,0 m by obstacle 1,0 m high in total.

Annex A, (Employee Protection Law, BGBl. Nr. 450/1994)

Figure A.1 The distance measurement " $1,45$ " shall be " $1,75$ ".

Finland (Application of electrical safety instructions to fixed installations of electrical tracks; National requirements A1:1993 § 2" and A4:1993 § 1.1 confirmed by resolution 657/96)

5.1.3.3.1 As protection against direct contact on the bridge a solid-wall design or a horizontal obstacle is used. They must be of metal plate or a plate framed with metal and hard to break.

The height of the solid-wall design from the standing surface must be at least 2,00 m and the wall must fulfil a 3,50 m minimum clearance from the middle line of the track (a requirement due to the pantograph) as well as the horizontal minimum clearances from the conductors shown in Figure 1 and provided the operating and climatic conditions allow.

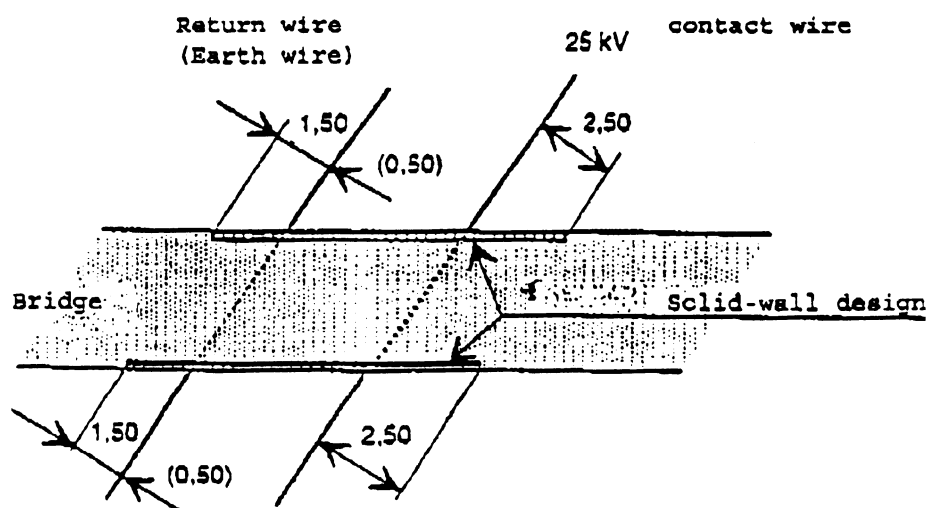


Figure 1

The minimum clearance of the live parts from the solid-wall design must be according to Figure 2 provided the operating and climatic conditions allow.

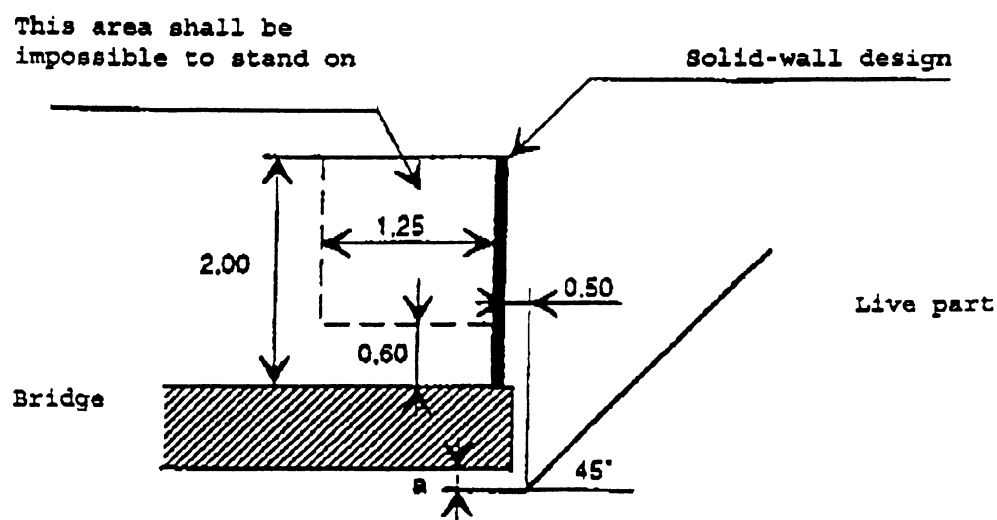
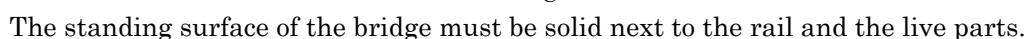


Figure 2

- 5.1.3.3.2 The live part, provided the operating and climatic conditions allow, must fulfil the minimum clearances from the horizontal obstacle shown in Figure 3.



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